International Students and the U.S. Skilled Workforce: Evidence from Optional Practical Training*

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Abstract

Skilled immigration to the U.S. increasingly runs through universities rather than employers: nearly 200,000 international graduates a year now stay to work through Optional Practical Training (OPT), and the program has become the main route into an H-1B visa. Yet whether OPT expands the skilled workforce, or merely re-routes graduates who would have stayed anyway, is unknown. I answer this question using new data that link records on every international student in the U.S. to LinkedIn career histories, and by leveraging the staggered reclassification of degree programs as STEM, which extends graduates' OPT eligibility. Only about half of the added authorization becomes new U.S. employment, about five months per graduate over three years: the rest replaces work graduates would have done anyway, often while enrolled in another degree, and the gain fades once the authorization expires. Even so, the workers the program adds are real for firms: employers with greater access to these graduates hire 29% more OPT workers and sponsor more of them for long-term visas, without cutting other hiring. Post-graduation work authorization thus adds skilled workers to the U.S. economy rather than reshuffling them, but its gains are concentrated in the years the authorization lasts.

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1 Introduction

Over the past twenty years, U.S. employer demand for skilled foreign workers has grown rapidly, outpacing supply. Wait times for employer-sponsored green cards can exceed a decade for applicants from India and China, and applications for employer-sponsored skilled worker (H-1B) visas have grown substantially while the number of cap-subject visas has remained fixed (Donovan, Kandel and Wilson, 2024; U.S. Citizenship and Immigration Services, 2023). As constraints on employer-sponsored channels for high-skilled immigration have continued to bind, a point of entry into the U.S. labor market has played an increasingly important role for prospective skilled immigrants: U.S. higher education.

This pathway operates through a program called Optional Practical Training (OPT), which grants temporary work authorization following graduation to international students in the U.S. on F-1 visas, the primary visa for international students.[1] Over the past two decades, a rapid rise in the international student population, accompanied by a series of policy changes that extended OPT from one year to three years of work authorization for a large share of graduates, has meant that workers and firms alike have increasingly made use of this channel. In 2024, about 195,000 graduates began a period of OPT, and OPT is now the main route into H-1B status: a third of graduates who use OPT go on to request an H-1B, against one in twenty of those who do not.

In recent years, this pathway has attracted substantial political scrutiny. Critics argue that OPT allows employers to circumvent protections for U.S. workers, while proponents contend that it helps the U.S. retain graduates trained at its universities and allows firms to recruit skilled workers who would otherwise leave the country.[2] Despite this debate, empirical evidence on how OPT shapes graduates' subsequent careers and employers' hiring decisions remains limited.

In this paper, I ask whether and how access to more years of OPT expands the U.S. skilled workforce. The answer depends on both the employment of the graduates who gain access to work authorization and the responses of the firms that hire them. Additional authorization need not translate one-for-one into additional U.S. employment: some graduates might otherwise remain by pursuing another degree or obtaining a different visa, while others might leave the U.S. Similarly, firms that hire more OPT workers may expand employment, substitute these graduates for other workers, or recruit workers who would otherwise have

[1] I use 'international student' and 'F-1 student' interchangeably throughout the paper to refer to students in the U.S. on an F-1 visa.

[2] For examples of these opposing views, see Schmitt (2025), who urges DHS to reform or end OPT, arguing that it circumvents employment-based visa restrictions and reduces job opportunities for U.S. graduates, and FWD.us (2026), which argues that restricting OPT would cause U.S.-educated international graduates to take their skills elsewhere and limit employers' access to talent.

worked elsewhere in the U.S. Understanding OPT's contribution therefore requires following international students from U.S. higher education into their subsequent careers, beyond the initial grant of work authorization.

A central obstacle in measuring the effects of OPT on U.S. skilled labor supply is that individual-level administrative immigration records, when available, are generally deidentified, making them difficult to link to subsequent labor market outcomes. I overcome this challenge by constructing a novel dataset that links administrative records covering the universe of F-1 visa holders from 2006–2022 to individual work and education histories from LinkedIn. The administrative records contain detailed information on students' education, OPT employment, and nationality, but no names. Through a combination of imputation, probabilistic linkage, and LLM-assisted review, I link a subset of these records to professional profiles using the school, field, timing and employers they have in common, allowing me to follow international students from school into the labor force. A key advantage of my data is that, for linked graduates, I can observe education, short-term work (such as internships), and even employment outside the U.S., distinguishing continued presence in the U.S. through study or part-time work from return migration. To study employers' responses, I aggregate LinkedIn data to construct a firm-level panel and link it to administrative records of OPT employment and employer-sponsored immigration. These data allow me to examine firms' hiring and employment alongside their demand for skilled foreign workers through other immigration pathways.

I begin by studying the direct effects of OPT authorization on graduates' own U.S. employment, using variation in eligibility for the STEM OPT extension. Introduced in 2008 and subsequently expanded, the extension grants an additional two years of work authorization to international graduates of eligible programs, beyond the initial year of OPT available to graduates more generally.[3] I study a particular response to the extension by universities, which I term 'relabeling'. Flexibility in how universities classify their majors has allowed schools to relabel non-STEM programs, such as Economics and Business, under related STEM-eligible codes, such as Econometrics and Management Science. These changes confer eligibility without changing the name or content of the program: as one school put it, the change "better reflects the content of the curriculum" and was "not a result of a curricular change" (USC Marshall School of Business, 2023). Relabeling occurred gradually across programs and schools, generating variation in eligibility across successive graduating cohorts within the same program. I use a difference-in-differences design, comparing graduates of

[3]Existing studies compare STEM and non-STEM graduates before and after national policy changes. Such comparisons, however, may combine the effects of additional work authorization with differential labor demand trends across fields and changes in students' program choices or universities' program offerings.

relabeled programs with graduates of programs in the same non-STEM fields that have not yet relabeled or never relabel, reweighted on pre-treatment characteristics.

First, I find that relabeling has a large effect on international students' work authorization through OPT. Prior to relabeling, about half of international students in treated programs used OPT, but because they were not yet eligible, virtually none used the STEM extension. After relabeling, graduates in treated programs became 10.2 percentage points (20%) more likely to use OPT at all and 23.6 percentage points more likely to use the STEM extension, relative to graduates in comparison programs. For the graduates I follow into the labor market, these take-up responses translate into 9.7 additional months of potential OPT authorization per graduate over the three years after graduation.[4]

How does this additional authorization translate to actual U.S. employment? Ex ante, the effect of relabeling on the time graduates spend working in the U.S. is unclear and depends on the alternative routes graduates could use to obtain U.S. work authorization. At one extreme, if graduates would have worked in the U.S. *even absent the STEM extension*, through a different work visa or while enrolled in another degree, additional OPT authorization would have no net effect on U.S. employment. At the other extreme, if graduates had no alternative forms of work authorization, the effect of relabeling on U.S. work should correspond one-for-one with the OPT authorization they gain, and there would be no effect on employment after OPT expires. Any long-run response in employment would suggest that longer OPT authorization affects graduates' ability to transition to other forms of work authorization (such as H-1B).

Using the individual linkages from administrative data to LinkedIn, I estimate the change in graduates' recorded U.S. employment. I find that relabeling increases U.S. employment by approximately five months per graduate in the first three years after graduation, so that each additional month of potential OPT authorization raises U.S. employment by only about half a month. After OPT authorization ends, the employment effect declines: for graduates I can follow for five years, employment remains 5.0 percentage points higher in the fourth year and 3.5 percentage points higher in the fifth, but neither effect is statistically distinguishable from zero. These results are robust to alternative estimators, linkage weights and comparison rules, and as a placebo I find that domestic graduates of the same programs, whose employment would also change if relabeling altered the programs themselves, show no change in U.S. employment.

Why does a month of authorization translate into only half a month of U.S. employment?

[4]Graduates who can be linked to a profile respond somewhat more strongly on the extension margin than F-1 graduates as a whole, so this figure exceeds what the population take-up rates imply; Section 4.3 compares authorization and employment for the same linked graduates throughout.

Two factors account for the gap. First, graduates work on OPT for about three-quarters of the authorization they gain. Part of the remainder reflects the timing of the measure, since initial OPT often begins some months after the program ends, and part reflects time between jobs and transitions to an H-1B before the authorization expires. Second, not all work on OPT is additional: about three in ten months of the added OPT work replace U.S. work that graduates would otherwise have done. Half of this replaced work took place during further study. Without the extension, some graduates would have remained in the U.S. past their first year of OPT by enrolling in another degree and working while enrolled, mostly on full-time Curricular Practical Training (CPT); relabeling reduces the time graduates spend in such programs by two-thirds in the second and third years after graduation. The other half was performed under statuses that the student-visa records do not capture, such as a visa obtained through a spouse or an employer, and is imprecisely estimated. For a minority of graduates, the extension instead serves as a bridge to employer sponsorship: recorded requests to change status to H-1B rise by 3.6 percentage points, from 13.7%, and among graduates observed for five years, H-1B status accounts for most of the employment that persists after the authorization ends. OPT therefore adds to graduates' U.S. employment primarily while the authorization lasts, and most of the work it authorizes would not otherwise have taken place in the U.S.

In the second part of the paper, I study employers' responses to increased access to graduates eligible for the STEM OPT extension. Whereas the individual design uses changes in eligibility within existing degree programs, the firm design uses a broader source of variation: the creation and expansion of STEM-eligible programs across schools and over time. I measure firms' exposure to these expansions using historical recruiting ties between firms and schools. A firm that previously hired more graduates from an expanding school experiences a larger increase in exposure than a firm with weaker recruiting ties to that school. I compare firms with large increases in exposure with reweighted firms experiencing smaller increases in a stacked, inverse-probability-weighted difference-in-differences design. Fixed recruiting weights keep firms' subsequent hiring decisions from mechanically determining exposure, while pre-treatment histories account for differences in firms' prior growth, hiring, and recruiting patterns. Because the expansions add graduates of all nationalities, this design measures how firms respond to a larger pool of eligible graduates, including native ones, and I use it to ask whether that recruitment displaces other hiring.

Firms exposed to these expansions recruit more through U.S. education: they hire 29.1% more OPT workers in the administrative records and 26.5% more foreign recent graduates and student interns in LinkedIn, and the additional OPT hires come entirely from STEM-designated programs, so the response follows eligibility for the extension rather than a general

expansion of recruiting from those schools.[5] Whether this recruitment displaces other workers is the central question for the firm design, and I find no evidence that it does. For each additional OPT hire, firms hire 0.05 fewer workers educated abroad, with a 95% confidence interval that rules out replacement of more than about a quarter of a worker; hiring of native workers other than recent graduates does not fall either, and its interval rules out more than about 0.5 displaced native hires per OPT hire; native recent-graduate hiring rises, although expanding programs also supply more native graduates. Exposed firms also sponsor more H-1B workers, with approved petitions rising by 13.8%, which suggests that OPT begins employment relationships that firms then seek to retain. Total employment at these firms is imprecisely estimated, a 0.5% increase with a 95% confidence interval from −0.7 to 1.7%, and because the design follows incumbent firms it does not capture firm creation or spillovers to other local employers. For these margins I draw on the commuting-zone estimates of Kwon, Lee and Monras (2026), who find that the 2016 expansion of the extension increased the number of firms and native high-skilled employment in exposed areas; since those local effects include workers moving between areas, they bound the national response from above.

Finally, I use these estimates to assess a proposed $100,000 fee on OPT, of the size imposed on new H-1B petitions in 2025, which Clemens (2026) models as $70,000 on initial OPT and $30,000 on the extension. The employment a fee removes depends on how many graduates forgo OPT, how much they would have worked, and how much of that work would have occurred in the U.S. anyway. Even though each graduate's additional employment is concentrated in a few years, successive cohorts sustain a larger stock of workers. I take participation responses from Clemens (2026) and the work associated with each OPT path from a duration benchmark, and use this paper's estimates for the share of OPT work that is additional and for replacement by other workers. If universities bear the fee, the U.S. skilled workforce would be about 174,000 workers smaller in steady state, and output about $24 billion lower each year; across the confidence interval for the additional share, the loss ranges from 121,000 to 221,000 workers. If students bear the fee, the loss is at least about 77,000 workers. Most of this loss, about four-fifths, is work during the OPT period itself rather than employment that persists afterwards, so the cost of the fee falls mainly on the years in which graduates would have worked on OPT. The estimate nets out the work graduates would have done through other routes; treating all lost OPT work as additional would raise it to about 241,000 workers. These figures count only graduates' own work net of the replacement measured at incumbent employers. They omit hiring away from other U.S. employers, which would make the loss smaller, and the local firm creation and native employment responses

[5]OPT employment relationships are assigned to graduates' program-end cohorts in the administrative records; these counts are not a measure of the contemporaneous stock of OPT workers.

estimated by Kwon, Lee and Monras (2026), which would make it larger, up to about 480,000 workers.

Related Literature. This paper contributes to two main strands of literature. First, a growing body of literature studies international students in the U.S. Much of this work focuses on international students as inputs into the higher education system, studying how international enrollment impacts university revenue and domestic student outcomes (Shih, 2017; Bound et al., 2020; Chen, 2021; Anelli, Shih and Williams, 2023; Avdeev, (Forthcoming).[6] Several papers in this literature focus specifically on OPT and its effects on U.S. schools. This work finds that the introduction of an extension to OPT for STEM graduates (the STEM OPT extension) induced an increase in international enrollment, particularly at selective schools and in STEM majors, while also increasing the number of domestic STEM graduates (Amuedo-Dorantes, Furtado and Xu, 2019; Amuedo-Dorantes, Shih and Xu, 2023; Khoo, 2025; Amuedo-Dorantes and Shih, 2026). Most closely related to my first empirical design, other papers study the phenomenon of program 'relabeling' in Economics or Business in response to STEM OPT eligibility.[7] I build on this literature by systematically documenting these relabeling events across fields and showing what they do to graduates' authorization, employment and immigration status.

A smaller share of this literature is dedicated to studying international students as potential labor market participants. At the individual level, several papers examine how the STEM OPT extension affected labor market outcomes by comparing graduates of different majors, or workers in different occupations, before and after national policy changes (Demirci, 2019, 2020; Guo, Gong and Pang, 2024).[8] Other papers study how OPT and changes in the international student population spill over into the local labor markets surrounding a school (Peri and Basso, 2016; Beine, Peri and Raux, 2023; Kwon, Lee and Monras, 2026). Most relevant to my work are Demirci (2019) and Kwon, Lee and Monras (2026). Demirci (2019) uses administrative data to show that the 2008 STEM OPT extension induced a small increase in eligible graduates' likelihood of using OPT at all and a large increase in their

[6]Relatedly, several papers study the upstream determinants of international enrollment (Shih, 2016; Chen, Howell and Smith, 2023; Khanna et al., 2025; Jia et al., 2025). See Bound et al. (2021) for a full review of work on international students in the context of the U.S. higher education system.

[7]Kim (2022) finds an increase in the number of STEM extension-eligible 'Management Science' programs following the major's addition to the DHS list in 2012, while Hu and Winters (2025) and Underwood, Marshall and Bilen (2026) find that relabeling Economics programs as STEM extension-eligible increased international students' and women's shares of degrees. Mahon and Asarta (2024) and Kim (2025) find that schools with higher baseline shares of foreign students and large, selective, research-intensive schools relabel Economics programs more often.

[8]Relatedly, Beine, Peri and Raux (2024) use more granular shocks to tuition to study how a larger international student share affects a cohort's entrepreneurship, focusing on master's programs.

likelihood of using the extension, consistent with my first stage results in the first part of the paper. Kwon, Lee and Monras (2026) studies the impacts of the 2016 expansion of the STEM OPT extension on differentially exposed commuting zones (based on international student concentration) and finds that the policy change, in addition to increasing the local supply of high-skilled immigrant workers, also increased firm creation and the employment and wages of native workers. I complement this work by examining recruitment and employment adjustment within incumbent firms, a margin that a commuting-zone design cannot isolate; conversely, their estimates capture the firm creation and local spillovers that my firm design does not, and I use them to bound those margins.

This work establishes that policy changes extending OPT increased initial transition into the U.S. labor market for certain students, with positive effects in some local labor market settings. It also motivates questions that require following students beyond their initial transition into employment: how much time graduates spend working in the U.S., what alternatives that work replaces, and how individual firms adjust to changes in access to this workforce. My main contribution to this literature is to answer these questions: I show that about half of the authorization the extension adds becomes U.S. work, that the rest is lost mainly to late starts, time between jobs and work graduates would have done while studying, and that incumbent firms recruit these graduates without a measurable fall in other hiring. These answers rest on a longitudinal panel linking students to their subsequent employment, including employment after OPT and outside the U.S., which, unlike repeated cross-sectional surveys such as the ACS, allows me to trace careers beyond OPT and distinguish employment changes from moves across local labor markets or abroad.

Second, this paper contributes to a large literature on the effects of high-skilled immigration on domestic labor markets. Much of this research points to skilled immigrants having a positive effect on firm productivity and innovation, with limited employment crowd-out of native workers. For instance, skilled immigration has been linked to increased patenting, knowledge spillovers, and firm expansion and entry across a variety of settings (Hunt and Gauthier-Loiselle, 2010; Moser, Voena and Waldinger, 2014; Beerli et al., 2021).[9] Other work, however, suggests that firm productivity gains may depend on the size and scope of the immigration shock (see, e.g., Paserman, 2013), or that skilled immigration may negatively impact select groups of incumbent workers (see, e.g., Borjas and Doran, 2012).

In particular, much of the evidence on the effects of contemporary skilled immigration on the U.S. labor market comes from the H-1B program. At an aggregate level, this literature documents gains in innovation, productivity, and native welfare (Kerr and Lincoln, 2010;

[9]Related work documents immigrants' disproportionate contributions to innovation and entrepreneurship (Azoulay et al., 2022; Bernstein et al., Forthcoming).

Peri, Shih and Sparber, 2015; Bound, Khanna and Morales, 2018; Abramitzky et al., 2026). Evidence on individual firms is more mixed: although several studies find that access to H-1B workers supports firm growth, survival, and start-up performance (Chen, Hshieh and Zhang, 2021; Dimmock, Huang and Weisbenner, 2022; Mahajan et al., 2024), other work finds displacement of other workers and limited innovation gains as a result of H-1B (Bound et al., 2015; Doran, Gelber and Isen, 2022). Additional work studies how firms adjust when H-1B hiring is restrained, finding that firms respond by retaining incumbent H-1B workers, moving work abroad through offshoring, and acquiring other firms to obtain skilled employees (Mayda et al., 2023; Glennon, 2024; Chen, Hshieh and Zhang, 2024). This work primarily identifies responses to changes in H-1B access, leaving less understood how firms respond to an expanded supply of U.S.-educated foreign graduates. I study this margin by examining recruitment and employment at firms exposed to growth in STEM-eligible degree programs, and I find that these firms recruit more through U.S. education and sponsor more H-1B workers without a measurable decline in hiring through other channels. OPT is also a distinctive setting for the literature on how immigration rules shape workers' mobility (Kahn and MacGarvie, 2020; Gupta, 2023; Kim, 2022; Townsend and Allan, 2025): it offers temporary work authorization without employer sponsorship while providing a route into employer-sponsored employment through the H-1B, and my evidence on the lottery shows graduates using the extension to wait out that route.

The remainder of the paper proceeds as follows. Section 2 outlines key context on international students in the U.S. and work authorization through Optional Practical Training. Section 3 gives an overview of the primary sources of data I use in this paper and how I harmonize and link data across different sources. Section 4 describes how I identify program relabeling events and my empirical methods, and estimates the direct effect of the extension on graduates' potential authorization and U.S. employment. Section 5 describes the school-expansion design and its effects on firms' hiring and employment. Section 6 combines the two sets of estimates into a workforce calculation and applies it to a fee on OPT. Section 7 concludes.

2 International Students and Optional Practical Training

While international students at U.S. postsecondary institutions have a long history, it was not until the 1952 Immigration and Nationality Act that international student visas were

officially codified.[10] International student visas were divided into three categories, a distinction that remains today: J visas, for exchange students or temporary researchers, M visas, for vocational training or study at nonacademic institutions, and F visas, for full-time students enrolled in degree-granting educational programs or language training at authorized academic institutions (United States Congress, n.d.). In this paper, I focus exclusively on F-1 visas, which are the primary F class visa and constitute well over 90% of student visas in the U.S. today (U.S. Department of State, Bureau of Consular Affairs, 2024).[11] Today, a U.S. degree can also provide access to temporary employment and, potentially, to employer-sponsored immigration. In this section I discuss the recent increase in the population of international students in the U.S. as well as several key features of work authorization on the F-1 visa that are relevant for the remainder of this paper.

2.1 The Growth of International Students in the U.S.

International enrollment in the U.S. has grown rapidly over the past twenty years. Between 2004 and 2024, international students doubled both in number (from 570,000 to 1.2 million) and as a share of total higher-education enrollment (from 3.3 to 6.1%). India and China account for much of this growth, together representing more than half of all international students in 2024 (Institute of International Education, n.d., 2025*b*).[12] Recent growth has also been unevenly distributed across different types of postsecondary programs. Bachelor's and master's programs make up the bulk of international graduates, and recent growth has been particularly concentrated in master's programs. Although the absolute number of international students in research-doctorate programs is smaller, international students represent more than a quarter of all U.S. research doctorates. Professional doctorates and associate's degrees have little international representation, so I exclude both from my main analysis.[13]

Growth has also been concentrated in a relatively small set of majors. The ten fields

[10]Historically, foreign students enrolled in U.S. universities served as a type of diplomatic envoy, drawn from elite backgrounds and expected to return to their home country to share what they had learned. This became particularly important in the early 1900s as immigration became more heavily restricted (Kramer, 2009).

[11]The other two F-class visas are the F-2 visa, for dependents of F-1 visa holders, and the F-3 visa, for Mexican and Canadian 'border commuters' who reside in their home country and commute to their program of study.

[12]This data is obtained from Open Doors, which includes graduates on OPT in its international student total. The reported percentage therefore compares enrolled international students plus OPT participants with total U.S. higher-education enrollment (Institute of International Education, 2025*a*). Lengthening OPT can therefore raise the reported total without increasing the number of students who enroll.

[13]International students pursuing associate's degrees may also enter under the M-1 rather than the F-1 visa, complicating analysis based on F-1 records (U.S. Department of State, Bureau of Consular Affairs, n.d.).

with the largest increases account for 60% of the overall rise in international bachelor's, master's, and research-doctorate completions between 2005 and 2024. Computer science, related information fields, and quantitative business programs account for the largest shares, followed by programs in data analytics, economics, applied mathematics, and engineering.

2.2 Optional Practical Training

Even prior to the creation of the F-1 visa, foreign students were permitted to remain in the U.S. for a limited time following graduation for "employment for practical training" (U.S. Immigration and Naturalization Service, 1947). This was formalized in 1992 through the creation of the Optional Practical Training (OPT) program, which allows all students on F-1 visas to work in the U.S. for up to 12 months to complement their education (U.S. Immigration and Naturalization Service, 1992). Standard OPT can be used during the program (e.g. summer internships) or after the program for part or full-time employment, provided that the student has first completed at least one academic year of study and that the employment is related to the student's field of study.[14] Unlike most forms of temporary work authorization in the U.S., such as the H-1B visa, standard OPT does not begin with an employer petition. Instead, a school official recommends the student for OPT, and the student applies to the federal government for work authorization. Once approved, the student can move between qualifying employers and may also be self-employed, as long as the work remains related to the degree. Post-completion OPT must begin within 60 days of the end of the program, and students must report changes in employment and may spend no more than 90 days unemployed during post-completion OPT, or 150 days including the STEM extension (U.S. Citizenship and Immigration Services, n.d.; U.S. Immigration and Customs Enforcement, 2024*a*). In 2024, about 195,000 graduates began a period of post-completion OPT, roughly half of them with a STEM extension (U.S. Immigration and Customs Enforcement, n.d.).

Having OPT authorization is not the same as working. A graduate may not take up the authorization available to them, may begin work after its start date, or may spend part of the authorized period between jobs. Conversely, a graduate without OPT may study further, obtain another work authorization, or leave the U.S. I therefore distinguish eligibility, use of OPT, the length of authorization, and recorded employment throughout the paper.

In 2008, the Department of Homeland Security (DHS) introduced an optional 17-month extension for students graduating with STEM degrees; in 2016, this extension was expanded to 24 months, allowing eligible graduates to work for up to three years in total (U.S. Department of Homeland Security, 2008, 2016). Whether a degree qualifies as 'STEM' depends on its

[14] Pre-completion OPT is limited to 20 hours per week while school is in session (U.S. Citizenship and Immigration Services, n.d.).

6-digit CIP (Classification of Instructional Programs) code, a taxonomy constructed by NCES for reporting purposes. Schools assign and report CIP codes to their programs, and DHS determines which codes appear on the STEM Designated Degree Program List (National Center for Education Statistics, 2026*b*; U.S. Department of Homeland Security, 2024*a*; Student and Exchange Visitor Program, 2022).[15] A school's decision about how to classify a program can therefore change the work opportunities available to its graduates. Section 4 studies this margin within existing programs. Section 5 studies a broader margin: the creation and expansion of eligible programs at schools from which employers recruit.

The STEM extension is less flexible than standard OPT, requiring employers to employ the student for at least 20 hours per week in a bona fide employer–employee relationship, complete and submit a formal training plan, and enroll in E-Verify (an electronic system that checks employment eligibility against DHS and Social Security Administration records (U.S. Citizenship and Immigration Services, n.d.)). Additionally, the duties, hours, and compensation for jobs using the STEM OPT extension are required to be commensurate with those of similarly situated U.S. workers, although in practice enforcement has been historically limited (U.S. Department of Homeland Security, 2016, n.d.). These requirements mean that additional legal eligibility does not automatically translate into extension use.

F-1 students can also work while enrolled. They may work on campus for up to 20 hours a week during the academic term without separate authorization. Off-campus work requires practical-training authorization: pre-completion OPT, which draws down the OPT time available after completion, or Curricular Practical Training (CPT), part-time or full-time. Unlike OPT, CPT must be "integral to [a student's] major and the experience must be part of [the student's] program of study" and generally occurs before program completion. CPT is generally available after one academic year of enrollment, but graduate programs that require practical training may authorize it from the start of the program; a number of schools have built master's programs around this provision, so that students can work full time from their first semester (so-called 'day-one CPT' programs). Twelve months or more of full-time CPT makes a student ineligible for OPT at that education level, and OPT is available once per degree level (U.S. Immigration and Customs Enforcement, 2024*a*).

These rules make further study an alternative to additional OPT. A graduate whose OPT is ending can enroll in a new program, retain F-1 status and, where the curriculum allows, work on CPT; completing a program at a higher degree level also brings a new period

[15] DHS published the first list in 2008 and expanded it in May 2011 and May 2012; the 2012 expansion added, among others, Econometrics and Quantitative Economics and Management Science, the codes that economics and business programs later relabeled into. The list then changed little until January 2022, when 22 fields were added, followed by eight in July 2023 and one in July 2024. The relabelings I study (2014–2019) therefore occur in a period when the list itself was stable.

of OPT. Section 4.5 shows that this is one of the main routes the extension replaces.

2.3 H-1B and Other Post-OPT Pathways

Not only is OPT an important point of entry into the U.S. labor market in its own right, it is also a key bridge to the H-1B visa and other longer-term U.S. employment authorization pathways. The H-1B visa is an employer-sponsored temporary work visa for workers in "specialty occupations," created by the Immigration Act of 1990 as a way for firms to fill skill shortages in the U.S. labor market (Donovan, Kandel and Wilson, 2024). Access to H-1B status is limited and uncertain. For most private employers, the total number of H-1B visas that can be approved in a given year is limited to 85,000, of which 20,000 are reserved for workers with a master's or higher degree from a U.S. institution; universities and certain affiliated nonprofit and research employers are exempt from these caps (Donovan, Kandel and Wilson, 2024; U.S. Department of Homeland Security, 2019). In recent decades, when demand has vastly exceeded this cap, H-1B visas have been allocated through a lottery held each spring for visas that begin the following October. Selection rates have varied considerably. Before 2020, employers filed full petitions for the lottery, and USCIS received between 190,000 and 236,000 cap-subject petitions a year for fiscal years 2017–2020. Since fiscal year 2021, employers have registered electronically and file petitions only for selected registrations: about 45% of eligible registrations were selected in the lotteries held in 2020 and 2021 (for fiscal years 2021 and 2022), about a quarter in those held in 2022 and 2023, when registrations reached 474,000 and 759,000, and 29% and 35% in 2024 and 2025 (U.S. Citizenship and Immigration Services, 2026).

OPT allows graduates to begin working before obtaining H-1B status, and the STEM extension can provide additional annual H-1B lottery opportunities before their student work authorization expires.[16] A recorded request need not result in approval, and an additional H-1B worker at one employer need not increase the national number of H-1B workers under a binding cap.

Graduates use these routes extensively. Among F-1 graduates of bachelor's, master's and doctoral programs completing in 2010–2017, half use OPT: 59% of graduates of programs eligible for the extension and 39% of graduates of other programs (Appendix Table A.1). U.S. work does not end when initial OPT ends. Figure 1 follows these graduates for five years: graduates of ineligible programs work in the U.S. in about 40% of months in each year, and by the third year about a fifth of that work is on an H-1B, a quarter is while enrolled in another program, and half occurs with no status recorded in SEVIS. Much of

[16] Cap-gap provisions can further extend qualifying students' work authorization while an H-1B petition is pending or awaiting its start date (U.S. Department of Homeland Security, 2008, 2016, 2024*b*).

this unrecorded work appears to reflect real stays: for 2012–2019 graduates, stay rates in the National Survey of College Graduates are one and a half to three times the shares in a SEVIS-recorded status (Appendix Table A.1), and many of these stayers have become permanent residents or citizens, often through family rather than employment. OPT is also the main route to an H-1B: 32% of graduates who use OPT request H-1B status, against 6% of those who do not.

For many graduates and their employers, the longer-term objective is employment-based permanent residence. Most employer-sponsored green cards first require a permanent labor certification (PERM) from the Department of Labor, followed by an immigrant petition with USCIS; annual numerical limits and per-country limits can create long waits, particularly for applicants from India and China (U.S. Department of Labor, n.d.; Donovan, Kandel and Wilson, 2024). I use H-1B and PERM filings as measures of employers' demand for these channels, not as counts of workers who obtain them.

2.4 What Additional OPT Adds to the Workforce

These features suggest a simple way to organize what additional OPT authorization contributes to U.S. employment. Suppose a policy gives a graduate P additional months of potential OPT authorization, of which the graduate spends O working on OPT. Some of that work replaces work the graduate would have done in the U.S. anyway, through further study, another visa, or a route that student records do not capture. Let a denote the share of OPT work that is additional, so that the graduate's own U.S. employment rises by

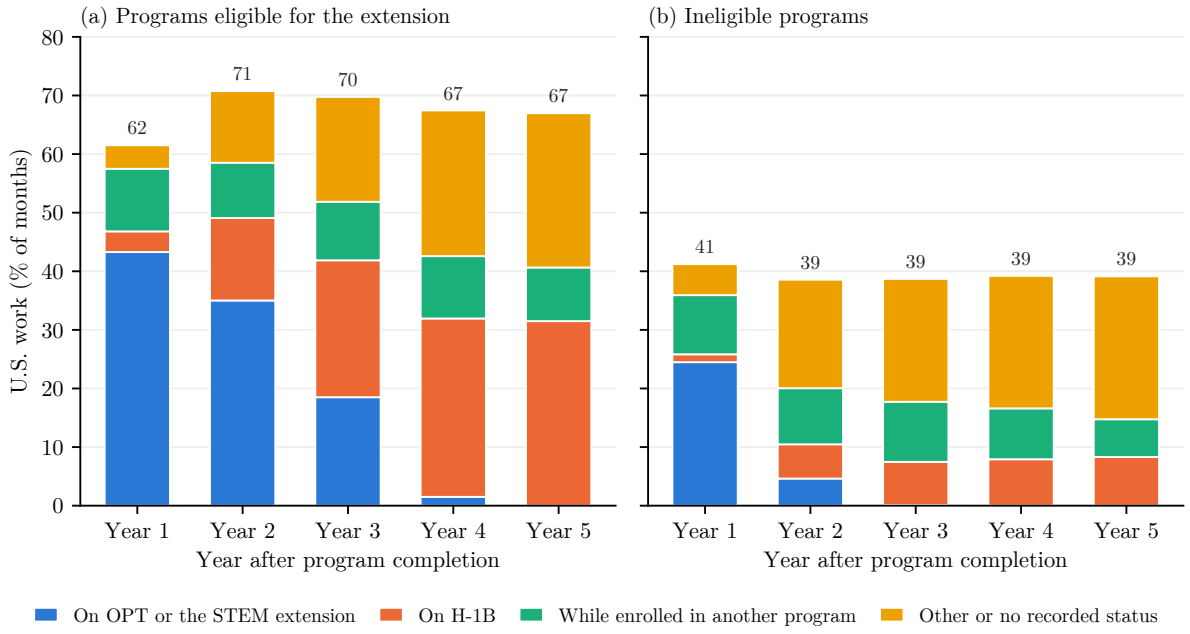
$$\Delta L^{\text{grad}} = P \times \frac{O}{P} \times a. \tag{1}$$

The two natural benchmarks are $a = 0$, when OPT only changes the route through which graduates would have worked, and $a = 1$ with $O = P$, when graduates have no other route and work throughout the added authorization. At employers, each additional OPT worker may replace r workers whom the firm would otherwise have hired, and may support s jobs for other workers through expansion and new firms, so that national employment changes by

$$\Delta L^{\text{US}} = (1 - r + s)\, \Delta L^{\text{grad}}. \tag{2}$$

Section 4 estimates O/P and a using program relabeling, Section 5 bounds r using firms' recruiting networks, and Section 6 combines them, taking s from existing commuting-zone evidence.

Figure 1: What international graduates do after graduation: U.S. work by immigration status, by whether the program qualifies for the STEM extension



Notes: The figure shows how much international graduates work in the U.S. in each of the five years after finishing their degree, and under which immigration status, for graduates of programs that qualify for the STEM OPT extension (panel a) and programs that do not (panel b). The height of each bar is the share of months with a recorded U.S. job; the segments split it by the graduate's status in that month, and the number above each bar is the total. Graduates of ineligible programs keep working in the U.S. well after their single year of OPT ends, through H-1B status, further study and routes that student-visa records do not capture, which is why the extension need not add to U.S. employment one for one. Sample: F-1 graduates of bachelor's, master's and doctoral programs in all fields who completed in 2010–2017 and are linked to a professional profile (152,314 graduates: 81,160 from eligible and 71,154 from ineligible programs), weighted so that they represent all F-1 graduates of their program and cohort; programs are classified by whether their code was on the DHS STEM list in the year the program ended. A month of work while enrolled as an F-1 student counts as half a month unless a full-time work authorization is active. H-1B status is dated from a change-of-status request, since approvals are not observed; "enrolled in another program" covers enrollment recorded in SEVIS or in the profile; "other or no recorded status" includes other requested statuses and months with no SEVIS record. Sources: SEVIS and Revelio.

3 Data and Linkage

To study the usage and impacts of OPT, I construct two analysis panels that link administrative data on schools, F-1 students, and OPT, to downstream outcome data: one panel is constructed at the school-program-year level, while the other is constructed at the employer-year level. In this section I give an overview of the data sources and linkage methods that I use to construct these panels, and of the outcome measures they provide. See Appendices B.1 and B.2 for further details.

3.1 Data Sources

F-1 Administrative Data (SEVIS). The backbone of both panels is an individual-level administrative database on F-1 visa holders: the Student and Exchange Visitor Information System (SEVIS). This is a centralized administrative database used by schools to report details about F-1 students' programs, funding, and authorization for off-campus employment (including OPT) to Immigration and Customs Enforcement (ICE). I use a snapshot of this database from 2023, obtained by the Institute for Progress from ICE via FOIA request (U.S. Immigration and Customs Enforcement, 2024*b*; Institute for Progress, n.d.). The data cover the universe of international students in the U.S. on F-1 visas from 2006–2022, spanning 68 million rows and an estimated 4.6 million individuals, and include information about students' characteristics, programs, OPT employers, and later visa transitions.[17] The annual files run through 2022, but because the snapshot was taken in October 2023, the authorization and status dates they contain extend to October 2023, which is the end of the administrative follow-up in Section 4. The records also report tuition and fees and funding from students' personal funds, schools and other sources. They contain no names: repeated records allow me to connect a student's programs across years even though the released identifiers are not stable across annual files, but linking a student to any outside source has to rely on the school, program, dates, nationality and employers in the record.

School-Program Administrative Data (IPEDS). For analysis at the school-program level, I use data from the Integrated Postsecondary Education Data System (IPEDS) (National Center for Education Statistics, 2026*a*,*b*). These data form my basic school-program analysis unit: I use IPEDS institutions to define 'schools' and I use combinations of IPEDS award level

[17] From 2015 onward, a problem in the released files separates employer names from the dates attached to particular jobs. I can therefore observe the employers a student worked for and the total period spent on OPT, but I cannot reliably attach each set of dates to a particular employer. I therefore use authorization dates to measure authorized time and employer names to identify employment relationships. Appendix B.1 provides more detail on this error, as well as other issues with the raw data.

definitions (subsetting to Bachelor's, Master's, and Research Doctorate degrees) and 4-digit Classification of Instructional Programs (CIP) major definition codes to define 'programs'.[18] Within these programs, six-digit CIP codes identify changes in STEM eligibility. In addition to using the IPEDS taxonomy to form my basic analysis unit, I also use IPEDS degree completions and cost data as supplementary data sources. IPEDS counts awards and classifies international students by nonresident status, while SEVIS records F-1 students and their programs, so I use each source for its own measures and do not treat their counts as interchangeable.

Individual- and Employer-Level Outcomes Data (Revelio). My primary source of outcome data is Revelio Labs' individual profile dataset (Revelio Labs, 2026*b*,*a*). These data list detailed work and education histories ("profiles") for the universe of LinkedIn users as of September 2026. Education histories report institutions, fields, degree levels, and dates, and work histories report employers, positions, locations, and employment dates. Linking these histories to SEVIS allows me to follow recorded work after a student's F-1 activity ends, including work outside the U.S. I use Revelio 'companies' as my primary unit of employer-level analysis.

LinkedIn coverage is not universal, but it is comparatively high in my setting. International comparisons and survey evidence indicate high coverage among the college-educated population and in the Information/Communication, Professional/Scientific/Technical, and Financial/Insurance sectors (Gottfried, 2024; Amanzadeh, Kermani and McQuade, 2024; Zhu, Fritzler and Orlowski, 2018). Users may still omit jobs or fail to update their histories, so I refer to the outcomes as recorded work. Among linked graduates, profiles show a U.S. job in 77% of the months in which SEVIS records an OPT employer, and in 88% and 84% of the months with full-time and part-time CPT in a later program, so work while enrolled is recorded at least as completely as work on OPT (Appendix B.3).

Other Employer Data. I supplement the above sources with additional data on employers. To measure employer demand for H-1B visas and employer-sponsored green cards, I use administrative data on Labor Condition Applications (LCAs), which are a prerequisite for H-1B visas and are commonly used in the literature as a proxy for demand for H-1B visas (see e.g. Kerr and Lincoln, 2010), and Permanent Employment Certification Applications, which generally precede employer-sponsored EB-2 and EB-3 immigrant petitions, subject to

[18]Note also that IPEDS covers the universe of schools that participate in Title IV federal student aid programs. While this excludes some institutions, such as standalone language training schools, whose students are eligible for F-1 visas, from my analysis, I am able to match 97.3% of eligible SEVIS records to an IPEDS institution. To simplify matching, I aggregate branch locations to one central 'main campus' unit for analysis.

exemptions (U.S. Department of Labor, n.d.; U.S. Department of Labor, Office of Foreign Labor Certification, n.d.). LCAs report the number of positions an employer requests, so H-1B demand counts requested positions rather than approved visas; from 2020, when employers began registering for the lottery electronically, I add lottery registrations to this measure. To measure realized H-1B visa volume, I use data on approved H-1B petitions obtained from USCIS via FOIA request and generously provided by Neil Ruiz. Approved petitions can include continuing employment and do not count distinct new immigrants.

3.2 Linkage and Harmonization

The analysis in this paper requires three kinds of linkage: linking schools and fields from SEVIS and Revelio to IPEDS; linking SEVIS employers to Revelio companies and to the H-1B, LCA and PERM employers; and linking SEVIS students to Revelio users. Each is a large-scale record linkage problem with its own difficulties: Revelio text is entered by users and unstandardized; the sources organize entities differently; many units exist in only one source (most SEVIS employers never file for an H-1B, for instance), so an unmatched record can be a true zero or a failed match; and comparing every pair of records across sources is infeasible. I use the same broad approach throughout: normalize names and text, decide on the unit of matching on each side, match exactly and then approximately, review uncertain cases with a large language model, and calibrate the resulting match scores against a sample of human- and LLM-labeled cases so that they can be read as linkage probabilities. For schools, location, institutional aliases, and campus relationships help distinguish otherwise similar names; for fields, text and semantic comparisons map reported majors to the CIP taxonomy.

For individual linkage, the school assignment restricts the candidate set to plausible education histories. Agreement in graduation timing, field, reported OPT employers, and country evidence then distinguishes candidates within that set, and a rare shared employer is particularly informative. The main individual analysis links each student to a single profile when the calibrated match weight of the best candidate exceeds 0.45. Because the records contain no names, only about one in eight F-1 graduates in the analysis programs can be linked this way (10.3% of treated programs' graduates before relabeling), and the linked graduates are selected: 63% of linked graduates used OPT, against 44% of unlinked graduates, and 23% used the extension, against 12%, partly because OPT employers are one of the main pieces of evidence that identifies a profile. Relabeling does not detectably change the linkage rate (Appendix Figure A.3), and the linkage weights described below account for the observable differences between linked and unlinked graduates.

For employer linkage, I use normalized names, aliases, location, and identifying information to connect administrative employers to Revelio companies. Duplicate company pages are consolidated when the evidence supports a common operating identity; common ownership alone is insufficient. A reviewed absence of an employer from a covered administrative file supports a zero, whereas an unresolved identity remains missing, so firm estimates use outcome-specific coverage and do not code every unmatched company as having no activity.

3.3 Outcomes and Analysis Populations

The individual analysis combines administrative program-end dates with subsequent profile histories. The principal employment outcome is the fraction of months in a post-program interval with recorded U.S. work, which I also sum into months of work; Section 4.2 defines it relative to the program-end month. Revelio also estimates annual compensation for each observed job; these estimates are not administrative earnings, and I use them only as supporting evidence. Administrative OPT use, authorization duration, requests to change status, and return to F-1 study do not require a successful profile link and provide complementary outcomes for the broader administrative population.

For firms, U.S. headcount counts people with recorded U.S. employment at the company during the relevant year. Hiring measures capture observed transitions into company employment, and recent graduates and student interns are identified from the timing of work and education histories. Foreign and native categories in the profile data use imputed country-of-origin evidence, since citizenship and visa status are not observed. By contrast, the SEVIS outcome counts OPT employment relationships attached to graduates' program-end cohorts, so a change in cohort hiring is not a change in the contemporaneous employment stock.

The administrative first stage covers graduates regardless of profile linkage. The main employment analysis follows linked graduates from their program-end month using inverse-linkage weights and the 2014–2019 relabeling cohorts. Appendix Table A.7 reports estimates for alternative populations and estimators.

Adjusting for selective linkage. I average linked outcomes within each program-cohort using inverse predicted linkage probabilities. The observation model uses citizenship, degree, cohort, broad field, own-country share within the program-cohort, and counts of administrative education and employment records, including the number of OPT employers; it is refit in the school-level bootstrap. These weights address which outcomes are observed; they are separate from the weights used to make treated and comparison programs comparable.

4 The STEM Extension and Graduates' U.S. Employment

How does the STEM OPT extension affect international graduates' own U.S. employment? The answer is not obvious, because it depends on what graduates would do when their initial year of OPT ends. If every graduate who uses the extension would otherwise have stayed and worked in the U.S. through another route, such as another degree or another visa, the extension changes only the route and U.S. employment does not change at all. If instead graduates have no other route and work throughout the authorization the extension adds, U.S. employment rises one-for-one with that authorization while it lasts, and not afterwards. Figure 1 shows that the alternatives are real: graduates of programs without the extension keep working in the U.S. after initial OPT ends, through H-1B status, further study and routes that student-visa records do not capture. I use university program relabelings, which make graduates of existing programs newly eligible for the extension, to measure where between these two benchmarks the truth lies. I first measure how much authorization relabeling adds, then estimate its effect on graduates' U.S. employment, and finally trace that effect to the routes through which graduates work.

4.1 Program Relabeling

Eligibility for the STEM OPT extension depends on a program's six-digit CIP code. These codes are maintained by the National Center for Education Statistics for reporting purposes; NCES supplies a title, a definition and illustrative examples for each code, but schools choose how to classify their programs (National Center for Education Statistics, 2022), and related codes can describe similar curricula. An economics program, for example, can be reported as General Economics, which does not confer STEM OPT eligibility, or Econometrics and Quantitative Economics, which does. This flexibility has allowed universities to 'relabel' programs in order to qualify their international graduates for the extension. The relabelings I study span economics, psychology, architecture, finance, business, design, communication and a few other fields (Appendix Table A.2).

A key feature of these relabelings is that they change how a program is reported and nothing else. Schools say so themselves: the University of Southern California explained that its undergraduate business programs' move to Management Science "was made because it better reflects the content of the curriculum, not as a result of a curricular change," and Northwestern's economics department wrote that "the curriculum of our degree and the name of our degree are not changing. Rather this reclassification is a recognition that our program

has been misclassified since the 1990s" (USC Marshall School of Business, 2023; Northwestern University Department of Economics, 2024, n.d.). CIP codes are used almost exclusively for reporting and do not appear on transcripts or diplomas; the one place a student sees the code is the Form I-20, which is also the document that determines eligibility for the extension. Figure 2 shows what a relabeling looks like in the data: a program's completions move from the old code to the new one in a single year, and the new code reaches the SEVIS record that governs eligibility.

Why do otherwise similar programs relabel in different years? Relabelings within a field come in waves, both because a program can only be relabeled once a suitable code is on the STEM list and because schools follow their peers: UC Davis's announcement of its 2019 relabeling of Managerial Economics notes that it "brings UC Davis in line with changes made the past few years by other universities" (UC Davis Department of Agricultural and Resource Economics, 2019). Within a wave, the year a given program is relabeled depends on its university's internal approval process. A change of code may need approval only from the department, or may also require approval from a college-wide faculty body, the provost's office and, at some public institutions, a state higher education board; because approved changes typically take effect at the start of the following academic year, differences in the number and length of these steps translate into differences of one or more years in when otherwise similar programs are relabeled (National Center for Education Statistics, 2022).[19] This timing reflects list additions, peer behavior and administrative procedure rather than changes in the program or in its graduates' prospects, which is what the design needs.

I identify relabelings systematically from IPEDS completions, supplementing the administrative patterns with program-specific web evidence. I first assemble an inventory of documented changes and map each to its non-STEM source and STEM-eligible target codes, then search across all schools and degree levels for a large, persistent drop in source-code completions accompanied by a corresponding rise under a previously unused target code. Ambiguous dates, competing code mappings and potentially overlapping events are reviewed using archived catalogs, department announcements and other program-specific evidence; unresolved cases are excluded from both the treatment and comparison groups. Appendix C.1 describes this construction.

The main treatment cohorts are programs first reported under their new codes in IPEDS between 2014 and 2019. I require coverage in each of four complete pre-treatment IPEDS years, at least five total completions across those years, a positive F-1 cohort in the reference

[19] A 2018 survey of universities offering STEM OPT-eligible economics degrees found that while most required department approval for reclassifying the major (80%), some also needed all-college faculty approval (39%) or provost or dean approval (61%), and 18% required approval from a state or federal agency (Marshall and Underwood, 2022).

year, and an observed institutional classification. This leaves 335 treated programs at 169 schools in the administrative analysis (Appendix Table A.2). Comparison programs are drawn from the same non-STEM source codes at other schools: programs that could have relabeled along the same routes, satisfy the same coverage requirements and have no unresolved relabeling. Both never-relabeled programs and programs that relabel sufficiently later serve as comparisons.

The year a program first appears under its new IPEDS code need not coincide exactly with the year eligibility changes in SEVIS. Degree conferral and program completion can fall in different calendar years, and some schools allow recent graduates still on initial OPT to update their code. I therefore date potential exposure from the year before the first relabeled IPEDS cohort, g: event time is $e = t - (g - 1)$, where t is the calendar year of the student's SEVIS program end date, the reference cohort is $e = -1$ (cohort $g - 2$), and the partly exposed cohort is $e = 0$. The main average covers the first four cohorts reported after relabeling, $e = 1, \ldots, 4$.

Relabeling in IPEDS carries through to the administrative records that determine eligibility. Before relabeling, only 2.3% of F-1 graduates of treated programs had a STEM-eligible code recorded in SEVIS; after relabeling, this share rises by 78.2 percentage points (Figure 2 and Appendix Table A.4). Relabeling therefore makes most international graduates of these programs newly eligible for the extension.

4.2 Empirical Strategy

I use a difference-in-differences design, comparing cohorts graduating before and after relabeling with the corresponding cohorts in comparison programs. Two features of the setting shape the design. First, treated programs are larger and much more international than the average program in the same fields (Appendix Table A.3), and outcomes such as U.S. retention plausibly trended differently for heavily international programs over this period, for instance in response to changes in the H-1B lottery or to the pandemic. I therefore assume that parallel trends holds conditional on a small set of pre-treatment program characteristics that plausibly predict both when a program relabels and how its graduates' outcomes evolve: program size, the international share, the shares of international students from India and China, pre-period OPT take-up, recent growth in program size and in the international share, and tuition, together with field-by-degree and institution-type (Carnegie) indicators. Second, because relabelings are staggered, a standard two-way fixed effects regression would use already-relabeled programs as controls for later ones and can assign negative weights to some treatment effects (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020).

I therefore estimate effects separately for each treatment cohort, using only programs that are not yet relabeled as controls, and then combine them (Sun and Abraham, 2021; Cengiz et al., 2019; Wing, Freedman and Hollingsworth, 2024).

Stacked, reweighted event study. For each treatment cohort c, I build a 'stack' containing the programs first relabeled in c and the comparison programs eligible for that stack. Within each stack, I estimate each program's probability of treatment with a logit on the characteristics above, measured over the four pre-treatment years, and reweight comparison programs by their estimated odds of treatment, $\hat{p}/(1-\hat{p})$, so that they resemble that cohort's treated programs (Abadie, 2005). The unit of observation is a program $\times$ graduating cohort. For program p in stack c and graduation year t, I estimate

$$y_{pct} = \sum_{e \neq -1} \theta_{c,e} D_{pc} \mathbf{1}\{t - (c-1) = e\} + \alpha_{pc} + \delta_{c,k(p),t} + \varepsilon_{pct}, \tag{3}$$

where D_{pc} identifies programs treated in stack c, α_{pc} are program $\times$ stack fixed effects, and $\delta_{c,k(p),t}$ are stack $\times$ field-group $\times$ degree-level $\times$ graduation-year fixed effects, so that the comparison is within broad fields and degree levels in a given year. For student outcomes, I weight program-cohorts by their program's reference-cohort number of F-1 graduates, which fixes each program's weight before treatment instead of letting later enrollment responses raise it; program outcomes weight programs equally. Appendix C.2 gives the covariate construction and overlap rules, and Appendix Table A.3 compares treated and comparison programs before and after reweighting.

I aggregate the cohort-specific estimates using the reference-cohort F-1 mass represented at each event time,

$$\theta_e = \sum_c \omega_{c,e} \theta_{c,e}, \qquad \theta = \frac{1}{4} \sum_{e=1}^{4} \theta_e. \tag{4}$$

A program that relabels later is used as a comparison only while its own change is sufficiently far away: a cohort followed for three years cannot serve as a comparison if its program relabels during that follow-up or in the year after. Appendix C.1 states the rule exactly, and I show below how the estimates change with a longer buffer.

Sample and employment measurement. The main specification uses the 2014–2019 treatment cohorts and the linked sample, weighting linked graduates by the inverse of their estimated probability of linkage before forming program-cohort averages (Section 3.2). Of the 335 treated programs, 248 have linked graduates in both the reference cohort and the follow-up window and so contribute to the employment estimates. I measure outcomes month

by month relative to the program-end date recorded in SEVIS: employment in a year is the share of the fifteenth-of-the-month snapshots in that 12-month window on which the graduate has a recorded U.S. job, which I also sum into months of work. Because F-1 students may work at most 20 hours a week during term and profiles do not record hours, a month of work during F-1 study counts as one-half unless SEVIS records a full-time authorization (Appendix B.4).

To compare years after completion, I hold the graduates fixed across years. The first three years use graduates whose third year after completion ends by October 2023, when the SEVIS records end, so that every month of their work can be assigned to an immigration status; the fourth and fifth years use graduates whose fifth year ends by September 2026, when the profile records end. For the first group, I decompose employment by the graduate's SEVIS status in the same month: OPT or the STEM extension; an H-1B period dated from a recorded change-of-status request, since approvals are not observed; enrollment in a later F-1 program, or in other U.S. education recorded in the profile; and any other recorded status or none (Appendix B.4). These components add up to employment by construction.

Identification and inference. Identification requires that, absent relabeling, treated cohorts' outcomes would have evolved like those of reweighted comparisons within field, degree and year, and that the reference cohort is unaffected by the change. Figures 3 and 4 show the pre-treatment estimates alongside the post-treatment effects. Take-up is flat before relabeling apart from a bump of about one percentage point, and potential authorization shows no pre-treatment change. For U.S. work in the first three years, the two pre-treatment cohorts are about one month below the reference cohort, with confidence intervals that include zero. I take this seriously rather than dismiss it. If the difference reflects a linear trend, extrapolating it through the post-treatment cohorts lowers the effect on U.S. work from 5.3 to 4.0 months, or 0.41 months per month of potential authorization, and the 95% confidence interval for the effect widens to -2.1 to 10.5 months; under the relative-magnitude restrictions of Rambachan and Roth (2023), the interval includes zero once post-treatment violations of parallel trends are allowed to be two-thirds as large as the largest change between pre-treatment cohorts. The take-up and authorization results do not depend on this, but the precise size of the employment effect does, and I report the trend-adjusted figure alongside the main one where it matters.

Because employment is observed only for linked graduates, the estimates also require that relabeling not change who is linked, conditional on the characteristics in the linkage model. What the data show is that the share of graduates linked to a profile does not change (0.8 percentage points, standard error 1.0, from 10.3%; Appendix Figure A.3), and that the

linkage rates within OPT users (about 14%) and non-users (about 7%) do not change either; the composition of the linked sample shifts toward OPT users only because relabeling raises OPT use. The linkage weights, which include the number of administrative employers, absorb this shift. What they cannot absorb is a change in which OPT users are linked, so the main assumption is that linkage is as good as random within these groups.

Interpreting the relabeling effect as an effect of eligibility also requires that the code change not independently alter employment prospects through curriculum, reputation or recruitment; Section 4.4 tests this using domestic graduates of the same programs. Relabeling changes the programs' size and prices little: total completions (-0.6 graduates, standard error 1.7) and the tuition and fees reported for international students do not change detectably, but the international share of completions rises by 4.4 percentage points (Appendix Table A.5). I therefore interpret the estimates as effects on successive graduating cohorts, including any composition responses, rather than on a fixed set of students; the first post-treatment cohort, whose enrollment decisions predate relabeling, gives results close to the pooled estimate (below). Comparison programs at schools containing a treated program may also reflect within-school reallocation or spillovers; I have not yet excluded them.

Confidence intervals use 499 school-level Bayesian bootstrap draws, refitting the treatment and linkage weights in each draw, since programs at the same school share approval processes, international student offices and local labor markets; the administrative take-up outcomes in Figure 3 and Appendix Table A.4 use 199 draws. All employment, authorization and status outcomes share the same draws, so intervals for sums, differences and ratios across outcomes use the corresponding sums, differences and ratios of the draws. They account for estimated weights but not for uncertainty in profile assignment or missing employment histories.

4.3 The First Stage: Relabeling and Potential OPT Authorization

Graduates respond to their new eligibility on two margins (Figure 3). Use of the extension rises by 23.6 percentage points, from 1.2%, and use of OPT at all rises by 10.2 percentage points, from 51.6% (Appendix Table A.4). The prospect of a longer period of work authorization therefore draws graduates into initial OPT as well as keeping them on it after the first year.

I translate these responses into the work authorization they make available, which I call potential OPT authorization. Each OPT user may work for 12 months on initial OPT after completion, and each extension user for a further 24 months, or 17 months for extensions that began before May 10, 2016. Counting both from the program-end month, potential authorization covers the first year after completion for OPT users and the second and third years for extension users, and none of the fourth year or later. I estimate the

effect of relabeling on this measure for linked graduates, with the same weights as the employment outcomes, so that authorization and employment are measured for the same people.[20] Potential authorization rises by 5.7 percentage points of months in the first year, through entry into OPT, and by about 37 percentage points in each of the second and third years, through the extension (gray bars in Figure 5). Summed over the three years, relabeling adds 9.7 months of potential OPT authorization per linked graduate (95% confidence interval 7.9 to 11.3; Table 1, column 1).

This is the yardstick for the employment effect. If graduates who gain authorization would have worked in the U.S. anyway, employment would not change; if the extension leaves other routes unchanged and graduates work throughout their potential authorization, employment would rise by the gray bars in Figure 5 in the first three years and not at all after the third.

4.4 The Effect on U.S. Employment

Relabeling substantially increases graduates' U.S. work, but by much less than the authorization it adds. Figure 4 shows the event study for U.S. work over the first three years after completion: treated cohorts work 5.3 months more (95% confidence interval 2.6 to 7.8; Table 1, column 3). Each month of potential OPT authorization therefore raises U.S. employment by 0.54 months (95% confidence interval 0.29 to 0.81): for every two months of added authorization, graduates work about one more month in the U.S.

Figure 5 compares the employment effect in each year with the added authorization, for graduates observed through the fifth year after completion. Their employment rises by 8.3, 14.7 and 14.0 percentage points in the first three years, against 5.7, 37.0 and 37.1 percentage points of potential authorization. In the fourth and fifth years, after potential authorization has ended for every graduate, employment remains 5.0 and 3.5 percentage points higher (standard errors 3.5 and 3.7). Over all five years, U.S. employment rises by 5.5 months per graduate (95% confidence interval 1.6 to 9.0), or 0.57 months per month of potential authorization (Table 2). For these graduates the effect in the first three years is 4.4 months, somewhat smaller than for the group observed in SEVIS through the third year; the five-year sample admits later graduates (through September 2021 rather than October 2020), for whom the effects are smaller.

The employment effect is concentrated in the years of potential authorization. Over

[20] Among linked graduates, use of the extension responds more strongly to relabeling than among all F-1 graduates (about 38 against 24 percentage points) and entry into OPT less (about 6 against 10 percentage points), which is why the potential authorization below exceeds what the population take-up rates imply. Because relabeling makes 78% of graduates newly eligible, effects per newly eligible graduate are roughly 1.3 times those per graduate.

the fourth and fifth years together, U.S. employment rises by 1.0 month per graduate (95% confidence interval −0.6 to 2.8), about a third of the annual effect during the first three years. Once OPT authorization ends, any remaining effect must come through another route, and the most direct is H-1B status: because cap-subject H-1B visas are allocated by lottery, a longer period of authorization gives graduates more chances to be selected while they can still work. For the graduates I can follow in SEVIS through the fifth year, mostly the 2014–2018 relabeling cohorts, work on an H-1B rises by 11.5 and 12.4 percentage points in the fourth and fifth years (standard errors 3.0 and 3.3), most of their 17.4 and 15.4 point employment gains in those years and about a quarter of the extension take-up in that sample. Across all F-1 graduates, the 3.6 point rise in H-1B requests is about one in seven of the graduates induced to use the extension. The two figures differ because the administrative records end in October 2023: a request is recorded only once it is made, so graduates from the later relabeling cohorts, who make up most of the sample, had been through fewer lotteries by the time the records end, and the population estimate understates how many eventually move to an H-1B. Persistence differs across relabeling cohorts: the 2014–2016 cohorts, a small share of the sample, show no decline through the fifth year, while the 2017–2019 cohorts, which graduated into the pandemic and the lotteries with the lowest selection rates, drive the decline (Appendix A.1).

Robustness. The ratio of employment to potential authorization is 0.62 with a doubly robust estimator and 0.53 with linkage weights calibrated to administrative OPT use, and between 0.46 and 0.65 when any one treatment cohort or field group is left out (Appendix Table A.7). The comparison rule matters most. The main specification drops a later-relabeling program from a year's comparison if it relabels within a year after that year's follow-up ends; requiring two or three years instead lowers the ratio to 0.49 and 0.44, and holding the comparison programs fixed across years (so that a program that relabels within six years is never a comparison) lowers it to 0.42. These rules trade comparison programs for distance from their own relabeling, and I use the one-year buffer because the programs it retains are the ones most similar to the treated programs in timing; the range 0.42 to 0.62 is the honest statement of what the design pins down. Over five years, the ratio is 0.69 with the doubly robust estimator and 0.44 with the same comparison programs every year, against 0.57 in the main specification. The first post-treatment cohort, whose enrollment decisions predate relabeling, gives a ratio of 0.59 (95% confidence interval 0.29 to 1.00), close to the pooled estimate.

The ratio also understates the true response if profiles miss work. Potential authorization comes from SEVIS and is measured without error for linked graduates, whereas work comes

from profiles that record a U.S. job in 77% of the months in which SEVIS records an OPT employer and that are wrongly linked in perhaps 5–15% of cases (Appendix B.2); both errors shrink the numerator of the ratio and push it toward zero. If recording were the only problem and affected all work equally, the ratio would be about $0.54/0.77 \approx 0.7$; the marginal months of OPT work that relabeling adds are recorded almost completely (Section 4.5), which suggests the attenuation is smaller than that, but I cannot rule it out. The share of OPT work that is additional, below, divides one profile-based quantity by another and is less sensitive to this problem.

A placebo. Domestic graduates of the same programs provide a placebo test: they cannot use OPT, but a change in the program itself, through its curriculum, reputation or recruitment, would affect them too. Matching graduates in the profiles to programs by school, degree and field, I find that relabeling does not change domestic graduates' U.S. employment: 1.2, 0.3 and 0.5 percentage points in the first three years after graduation (standard errors 0.8 to 1.0), from a base of about 62% (Appendix Table A.8). Foreign graduates matched the same way, a positive control, gain 6.9, 6.5 and 4.2 percentage points. This profile is flatter than the one in Figure 5 because the matched sample uses calendar years rather than months from program end, includes foreign graduates who were never on an F-1 visa, and is not reweighted for linkage, so its first year mixes the end of the program with the start of OPT.

4.5 Channels: How the Extension Changes Graduates' Routes into U.S. Work

Why does a month of potential authorization add only about half a month of U.S. work? The ratio has two parts (Table 1). First, graduates work on OPT or the extension for three-quarters of their potential authorization (74%; 95% confidence interval 64 to 84%). Of the 2.5 potential months not worked on OPT, 0.4 are spent in OPT status without recorded work and 2.2 are not spent in OPT status at all: 1.1 months because graduates move to H-1B status before their potential authorization ends, 0.3 after other early exits, and most of the rest because initial OPT often starts after the program-end month, which shifts OPT time that graduates would have used anyway outside the potential window (Appendix B.4). Second, not all work on OPT is additional: each month of work on OPT or the extension raises net U.S. employment by 0.73 months (95% confidence interval 0.41 to 1.02), because some graduates would otherwise have worked in the U.S. through another route. Multiplying the two parts gives the 0.54.

The status histories show which routes the extension replaces (Figure 6). Work on OPT

and the extension, together with 0.4 months of additional work on an H-1B, adds 7.6 months; of these, 2.3 replace U.S. work that would have occurred anyway, so that net U.S. employment rises by 5.3 months. The replaced work falls into two halves: work while enrolled in a later F-1 program, and work under another status or none recorded in SEVIS.

Further study. The first half, 1.2 months, is work while enrolled in a later F-1 program, and it is the most precisely estimated component. Without the extension, some graduates would have remained in the U.S. after initial OPT by starting another degree and working while they studied. Relabeling reduces time in a later F-1 program by 8.1 and 8.3 percentage points in the second and third years after completion, two-thirds of the pre-treatment level of 12%, and reduces work in such programs by 1.2 months over the three years. SEVIS records how this work was authorized (Appendix Table A.9): more than half of it, 0.65 months, was full-time CPT in the new program, which allows full-time off-campus work while enrolled; 0.17 months was on the new program's own post-completion OPT, 0.08 on part-time CPT, and 0.26 without a recorded off-campus authorization, which the part-time convention counts at one-half. The administrative records for all F-1 graduates show the same pattern: relabeling reduces the time spent in a later F-1 program over the first three years by 1.0 month, from 1.8, including 0.36 months on full-time CPT (Appendix Table A.10). The timing follows the end of initial OPT: time in a later F-1 program barely changes in the first year, and return to F-1 study within a year of completion falls by 1.5 percentage points, within two years by 4.3 percentage points, from 7.8%, mostly into master's and doctoral programs (Appendix Table A.4). With the extension, these graduates work on OPT instead.

The H-1B visa. For a minority of graduates, the extension is a bridge to employer sponsorship rather than a replacement for another route. Recorded requests to change status to H-1B rise by 3.6 percentage points, from 13.7% (Appendix Table A.4). In the first three years, graduates move to H-1B status before their potential authorization ends and work 0.4 months more on an H-1B; after the third year, H-1B status is the most direct route for the remaining employment effect (Section 4.4). The H-1B lottery is the link. A graduate on initial OPT typically has one lottery attempt before authorization ends, while a graduate on the extension can register in up to three. I examine this directly by linking USCIS lottery registrations and selections for the 2020–2022 lotteries to SEVIS records of each employer's OPT graduates, measuring each firm's luck as its number of selections minus the number expected given its registrations and the master's share of its graduates (Appendix Figure A.1). Among firms with at most five registrations, one additional selection in the first lottery after a cohort graduates lowers that cohort's extension take-up by 4.7 percentage points

(standard error 0.3), shortens its work authorization by 2.2 months (0.1), and raises its H-1B requests by 4.9 percentage points (0.4); luck in the lottery held before most of the cohort has begun OPT has no effect on extension use. Graduates whose employers lose the lottery remain on the extension and try again: the extension insures the match between a graduate and an employer against the lottery. Because cap-subject H-1B visas are fixed in number, some of these transitions come at the expense of other applicants, which Section 6 takes into account.

Other routes. The other half of the replaced work, 1.2 months, is work under another recorded status or with no recorded status, and is imprecisely estimated (95% confidence interval −2.7 to 0.5). Several things can sit in this category. SEVIS records a change of status only when it is requested from inside the U.S. on a student record, so a graduate who obtains an H-1B through consular processing, a cap-exempt employer, or after the student record closes, or a status through a spouse (an H-4 or a family-based green card), works here without a recorded status. Staying longer on the extension also gives graduates more time to form such ties. The data cannot separate these routes, and I treat the component as a whole.

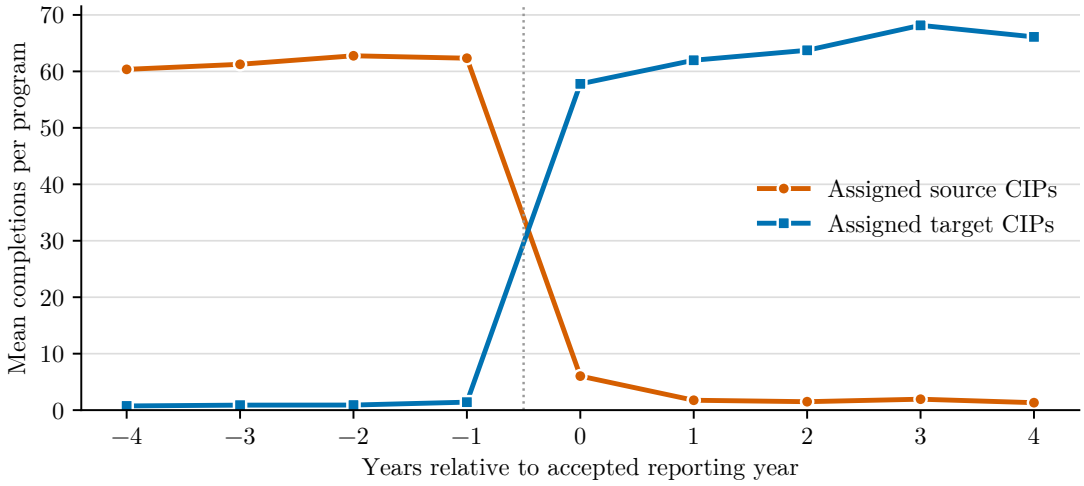
Work before completion and pay. Eligibility also changes work around completion. Employment rises in the final year of the program, before any OPT, by 3.7 percentage points under the part-time convention (Appendix Table A.12). Use of CPT during the degree rises as well, by 5.3 percentage points from 18.7%, and the share of the final program year spent on CPT by 1.5 points from 4.4% (Appendix Table A.11). This is what one would expect if a pre-graduation internship is worth more when the student can stay for three years afterwards: employers that use internships to recruit have more reason to take on a student who can later be retained, and students have more reason to seek them. Most of the rise in final-year employment, however, is in months without a recorded authorization, such as campus jobs and positions that profiles record without a CPT authorization (Appendix Table A.13), so I do not count work during the degree in the employment effect. In the first year after completion, employment rises by 3.4 percentage points more than potential authorization (standard error 4.1), and work on OPT rises by 4.4 points more than time on OPT (standard error 2.6), suggesting that eligibility also raises employment among graduates who would have used initial OPT anyway. Revelio's model-based pay estimates rise with employment (Appendix Table A.14); because Revelio assigns lower pay to positions outside the U.S., this is mostly the mechanical result of more graduates working in the U.S., and I do not read it as a wage effect. Consistent with that, work recorded in graduates' countries of citizenship falls by 8 to 10 percentage points in each of the first three years (Appendix Figure A.6), about

the size of the rise in U.S. presence: the additional U.S. workers come mainly from graduates who would otherwise have gone home. The number of employers and tenure by the fifth year do not change detectably (Appendix Table A.15).

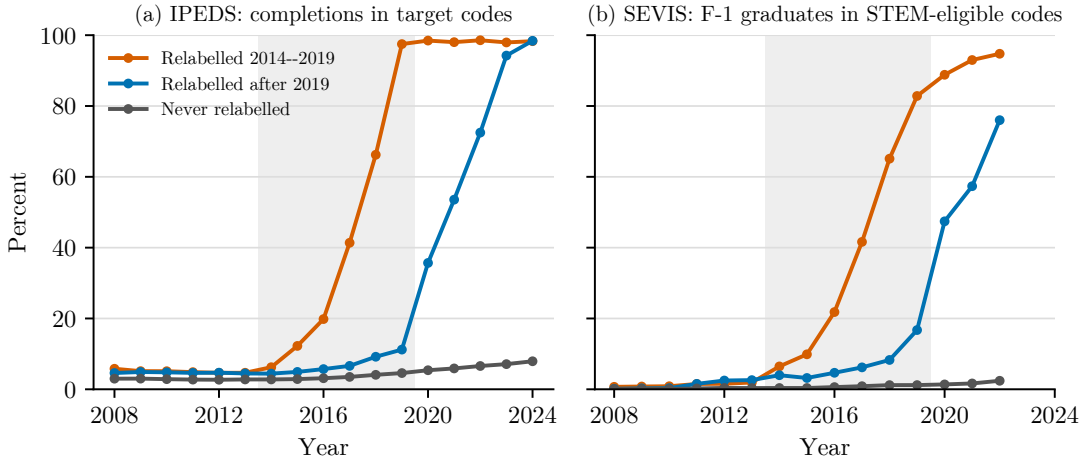
Differences by degree and origin. Master's graduates take up the extension most: their use rises by 32.8 percentage points, against 18.1 for bachelor's and 13.2 for doctoral graduates (Appendix Table A.16). Take-up also differs sharply by origin, rising by 44 percentage points among Indian graduates against 18 among Chinese graduates and about 24 among graduates from English-speaking and other countries, in line with the longer green-card queues Indian graduates face; the linked sample is too selected by origin to estimate employment effects by country. Among master's graduates, a larger part of the added authorization replaces work they would have done while enrolled in another program, and each month of potential authorization raises employment by 0.50 months (95% confidence interval 0.22 to 0.83) over the first three years, against 0.58 (0.18 to 0.99) for bachelor's graduates (Appendix Figures A.7 and A.8); bachelor's graduates' employment also remains higher in the fourth and fifth years, but none of these differences is statistically distinguishable from zero. The next section asks how employers respond when their access to this pool of graduates expands.

Figure 2: What a relabeling looks like in the data

(a) Completions reported under the old and new codes around the relabeling year

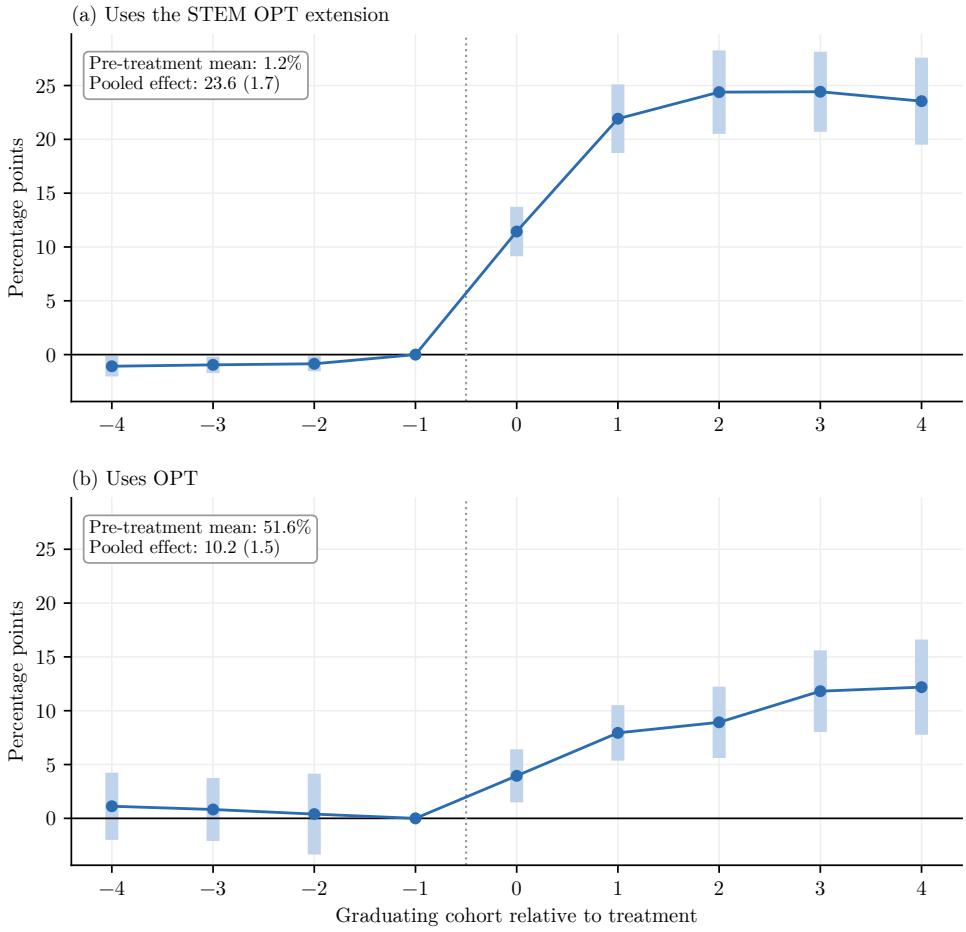


(b) The new code reaches the administrative record that governs eligibility



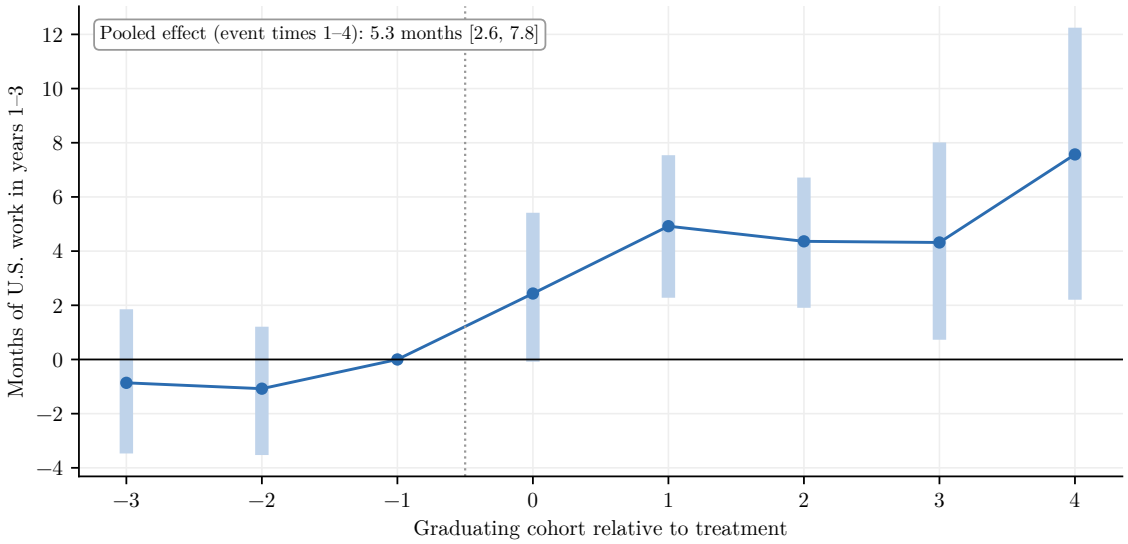
Notes: Panel (a) shows that a relabeling moves a whole program from one reporting code to another at once: mean annual completions per program reported under the program's original (non-STEM) code and under its new STEM-eligible code, in years relative to the first year the program is reported under the new code, for all programs identified as relabeled in 2008–2024. Panel (b) shows that the change reaches the record that determines eligibility: the share of completions reported under STEM-eligible codes in IPEDS (left) and the share of F-1 graduates whose program code in SEVIS is STEM-eligible (right), by calendar year, for programs relabeled in 2014–2019 (the analysis cohorts), programs relabeled after 2019, and comparison programs that never relabel. The shaded band marks the 2014–2019 relabeling years. Sources: IPEDS and SEVIS.

Figure 3: Relabeling raises use of the STEM extension and entry into OPT



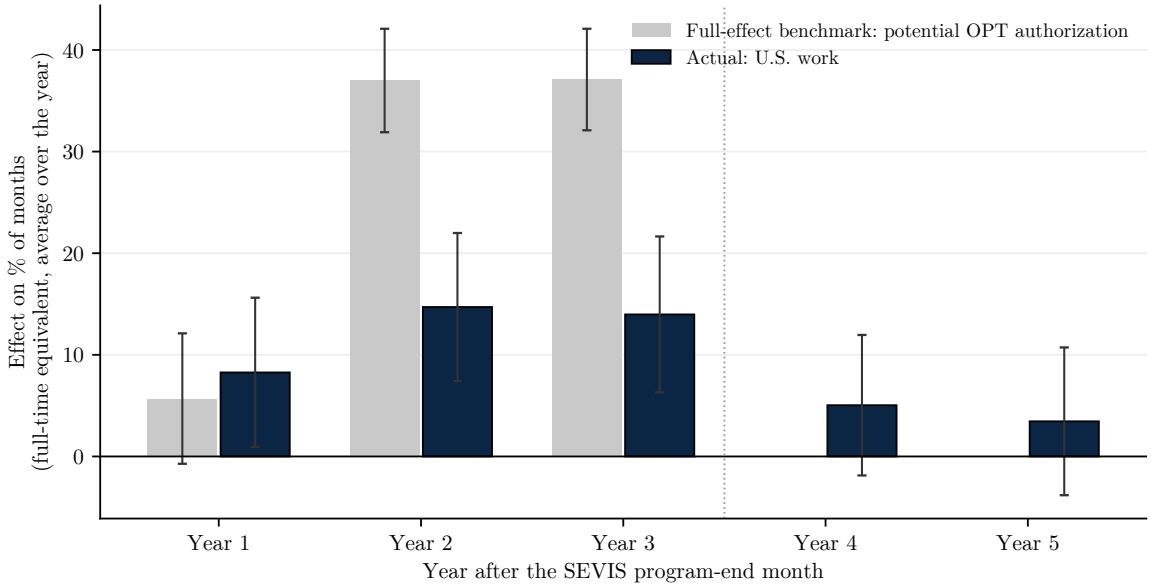
Notes: Effect of relabeling on the share of a program's international graduates who use the STEM extension (panel a) and who use OPT at all (panel b), in percentage points, by graduating cohort relative to relabeling, with 95% confidence intervals. Cohort −1 is the reference cohort, two years before the first cohort reported under the new code; cohort 0 is partly exposed, because some graduates still on OPT can update their code; the pooled effect averages cohorts 1–4. The boxes report the pre-treatment mean among relabeled programs and the pooled effect (standard error). Sample: all F-1 graduates (from SEVIS) of 335 programs relabeled in 2014–2019 and of comparison programs, with comparison programs reweighted to resemble relabeled ones; see Section 4.2. Intervals are from 199 school-level bootstrap draws. Sources: IPEDS and SEVIS.

Figure 4: Relabeling raises U.S. employment in the first three years after graduation



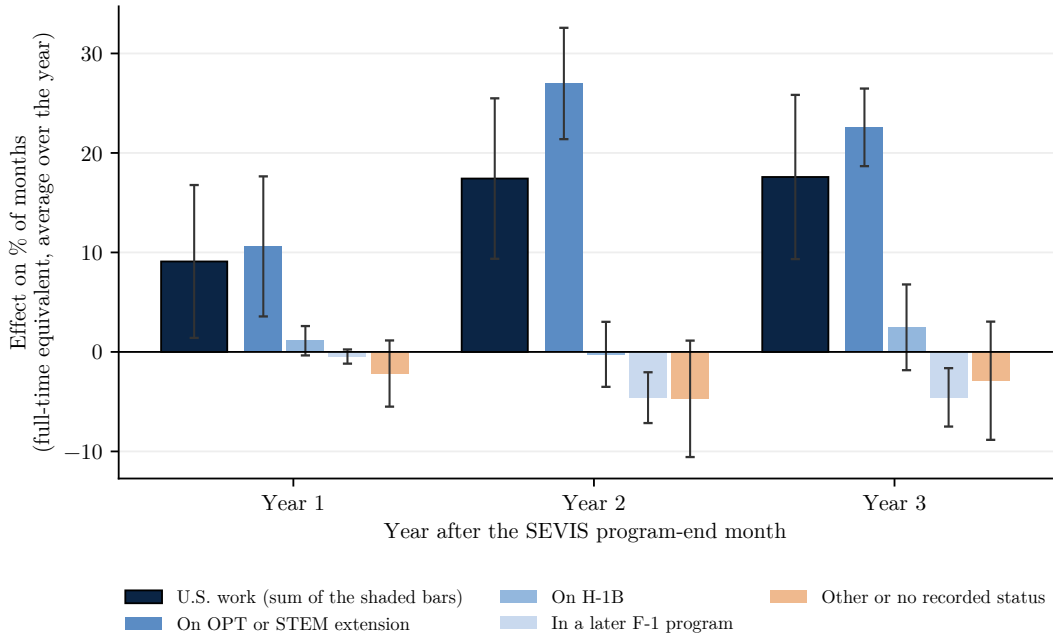
Notes: Effect of relabeling on months of recorded U.S. work in the first three years after graduation, by graduating cohort relative to relabeling, with 95% confidence intervals. Cohort −1 is the reference cohort, two years before the first cohort reported under the new code; cohort 0 is partly exposed; the pooled effect in the box averages cohorts 1–4. The two cohorts before the reference cohort are about one month below it; Section 4.2 discusses what this pre-trend would imply. Sample: graduates of the relabeled and comparison programs who are linked to a professional profile and whose third year after graduation ends by October 2023, the same graduates in every year, weighted to represent all F-1 graduates of their program and cohort (Table 1 gives the specification). A month of work while enrolled as an F-1 student counts as half a month unless a full-time work authorization is active. Sources: IPEDS, SEVIS and Revelio.

Figure 5: Added authorization and added employment over five years



Notes: The gray bars show how much more of each year relabeled cohorts could spend on OPT because of relabeling; the dark bars show how much more of each year they actually spend in a U.S. job. If graduates had no other way to work in the U.S. and worked throughout the added authorization, the dark bars would equal the gray bars in years 1–3 and be zero afterwards; if the added authorization only changed the route through which they would have worked anyway, the dark bars would be zero throughout. Potential authorization counts 12 months of OPT for each graduate who uses OPT and 17 or 24 months of extension (according to the rule in force) for each graduate who uses the extension, from the month the program ends, so it is zero after the third year. Effects are in percentage points of months, with 95% confidence intervals, for graduates linked to a professional profile whose fifth year after graduation ends by September 2026, the same graduates in every year (Table 2 gives the specification). Sources: IPEDS, SEVIS and Revelio.

Figure 6: Which routes the extension replaces: added U.S. work by immigration status



Notes: The dark bar in each year is the effect of relabeling on U.S. employment (share of months with a recorded U.S. job, in percentage points); the lighter bars split it by the graduate's immigration status in the same month and add up to the dark bar. Positive bars for OPT and the extension show the work that the added authorization makes possible; negative bars show work that graduates would otherwise have done while enrolled in a later F-1 program or under another or no recorded status. H-1B status is dated from a change-of-status request, since approvals are not observed; "other or no recorded status" includes education recorded only in the profile. A month of work while enrolled as an F-1 student counts as half a month unless a full-time work authorization is active. Sample and specification as in Table 1; intervals are 95%. Sources: IPEDS, SEVIS and Revelio.

Table 1: U.S. employment in the first three years after completion

				U.S. work by SEVIS status in the same month			
	Potential OPT authorization	Time in OPT status	U.S. work	On OPT or extension	On H-1B	In a later F-1 program	Other or no recorded status
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Effect (months)	9.74***	7.60***	5.29***	7.22***	0.40	−1.16***	−1.17
	(0.86)	(0.81)	(1.38)	(0.89)	(0.48)	(0.34)	(0.80)
Relative to column (1)		0.78	0.54	0.74	0.04	−0.12	−0.12
		[0.73, 0.83]	[0.29, 0.81]	[0.64, 0.84]	[−0.06, 0.14]	[−0.20, −0.06]	[−0.28, 0.06]
Relative to column (4)			0.73				
			[0.41, 1.02]				
Relative to (4), net of H-1B work			0.68				
			[0.43, 0.90]				
Pre-treatment mean	8.0	7.6	15.0	5.1	2.3	1.1	6.4
Treated programs	248	248	248	248	248	248	248
Schools	1,221	1,221	1,221	1,221	1,221	1,221	1,221
Observations	56,593	56,593	56,593	56,593	56,593	56,593	56,593

Notes: The table reports how much the STEM OPT extension changes international graduates' U.S. work in the three years after they finish their degree, in months per graduate, using university program relabelings as the source of eligibility (Section 4). Column (1) is the work authorization that relabeling makes available: 12 months of OPT for each graduate who uses OPT plus 17 or 24 months for each graduate who uses the extension, counted from the month the program ends. Column (2) is the time graduates actually spend in OPT or extension status, and column (3) the months with a recorded U.S. job. Columns (4)–(7) split column (3) by the graduate's immigration status in the same month and add up to it: on OPT or the extension; on an H-1B (dated from a change-of-status request, since approvals are not observed); enrolled in a later F-1 program; and any other status or none, which includes education that appears only in the profile. A month of work while enrolled as an F-1 student counts as half a month unless a full-time work authorization is active, because such students may work at most 20 hours a week. "Relative to column (1)" divides each effect by the added authorization; "relative to column (4)" divides net U.S. work by work on OPT, the share of OPT work that is additional; the next row removes work on an H-1B from the numerator, the version used in the fee calculation (Section 6). Sample: graduates of programs relabeled in 2014–2019 and of comparison programs, linked to a professional profile, whose third year after completion ends by October 2023 (when the SEVIS records end); the same graduates enter every year. Estimates are stacked difference-in-differences effects averaged over the first four relabeled cohorts, with comparison programs reweighted to resemble relabeled ones and linked graduates reweighted for their probability of linkage (Section 4.2). Standard errors in parentheses and 95% intervals in brackets from 499 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Observations are stacked program-cohorts in the first year; later years retain 53,965 to 56,593.

Table 2: U.S. employment over five years after completion

	Potential OPT authorization, years 1–3	U.S. work, years 1–3	U.S. work, years 4–5	U.S. work, years 1–5
	(1)	(2)	(3)	(4)
Effect of relabeling (months)	9.57***	4.43***	1.02	5.45***
	(0.77)	(1.28)	(0.85)	(1.88)
Relative to column (1)		0.46		0.57
		[0.23, 0.68]		[0.18, 0.95]
Annual effect relative to column (2)			0.35	
			[−0.31, 0.91]	
Pre-treatment mean (months)	8.0	15.0	9.5	24.5
Treated programs	248	248	248	248
Schools	1,221	1,221	1,221	1,221
Program-cohort observations	60,781	60,781	60,781	60,781

Notes: The table follows graduates for five years after they finish their degree, two years beyond the longest possible OPT authorization. Effects of relabeling in months per graduate, for graduates linked to a professional profile whose fifth year after completion ends by September 2026 (when the profile records end); the same graduates enter every year. Column (1) is the work authorization relabeling makes available in years 1–3 (defined as in Table 1); it is zero after the third year because no OPT authorization lasts longer. Column (3) sums the fourth and fifth years and column (4) all five years; "annual effect relative to column (2)" divides the average annual effect in years 4–5 by that in years 1–3. Specification and inference as in Table 1; observations are stacked program-cohorts in the first year.

5 Employer Recruitment and Employment

Section 4 shows that most of the work graduates do on OPT would not otherwise have been done by the same graduates. The other margin of OPT's contribution concerns employers: would the jobs that OPT workers fill otherwise have gone to other workers? Firms might hire more graduates and expand employment, replace workers they would otherwise have hired, or draw workers from other U.S. employers. I study recruitment, employment and sponsorship at firms whose historical recruiting schools expand programs eligible for the STEM OPT extension, comparing firms with large increases in exposure with firms experiencing smaller increases, reweighted on their prior histories.

What this design identifies deserves a clear statement up front. The individual design changes the eligibility of a given set of graduates; the firm design changes the number of eligible graduates at the schools a firm recruits from, and those additional graduates include native students and foreign students who would work through other routes. The estimates therefore describe how incumbent firms respond when the pool of eligible graduates they can recruit grows, and they measure employers' responses to OPT through that pool rather than the effect of one more OPT worker with everything else held fixed. This is the right experiment for the question of whether OPT recruitment displaces other hiring, and the ratio of hiring responses it produces is the closest counterpart to r in Section 2.4.

5.1 Eligible Programs and Firms' Recruiting Networks

School expansions. I define a program as STEM-eligible when its six-digit CIP code appears on the DHS eligibility list for that year, and apply this classification to IPEDS bachelor's and master's completions. For each school and year, I measure the growth of eligible programs as the rise in the eligible share of the school's completions that comes from eligible programs expanding, leaving out the part that comes from other programs shrinking, and I cumulate these rises from 2008 into an expansion index H_{kt} for school k in year t (Appendix C.3 gives the decomposition). Because classification follows reported CIP codes and the annual DHS list, an increase can reflect additional graduates, reclassification, or newly eligible fields. I interpret it as growth in the eligible recruiting pool, not as a count of newly created places or work permits.[21]

[21] I retain school–degree combinations with positive total completions in every year from 2008 through 2023, limiting changes caused by reporting coverage. Individual fields can still open or close within this balanced school–degree support. Completions include domestic as well as international graduates.

Recruiting ties. I link these school changes to firms using education and employment histories in Revelio. For people whose first job at a firm began in 2008–2013, I identify their preceding U.S. postsecondary education, and I attribute each such hire to that school. Let N_{fk} be the number of hires attributed to school k at firm f; the fixed recruiting shares and annual firm exposure are

$$w_{fk} = \frac{N_{fk}}{\sum_j N_{fj}}, \qquad E_{ft} = \sum_k w_{fk} H_{kt}. \tag{5}$$

For example, an expansion-index increase of five percentage points at a school supplies three points of exposure to a firm drawing 60% of its attributed recruiting from that school, but only one point to a firm drawing 20%. The recruiting shares are fixed in the 2008–2013 window, so a firm's later hiring cannot raise its own exposure.[22]

Exposure episodes and comparisons. I identify large annual exposure increases in 2015–2019 using a fixed threshold of 2.21 percentage points, the 75th percentile of firms' largest annual increases. An episode begins when the annual increase reaches this threshold after an observed year below it; G denotes the year of the jump, and a firm can contribute more than one episode. Comparison firms have observed exposure from $G-2$ through $G+3$ and annual increases below the median of that distribution, 1.72 percentage points, from G through $G+3$. The thresholds summarize the size of observed jumps; there is no discontinuity in access to work authorization at either.

Both groups must have positive exposure in every year from 2009 through $G-2$, positive employment in each of $G-5$ through $G-2$, complete outcome histories, and valid firm identities and source coverage. Earlier exposure shocks are permitted and enter the adjustment. The core sample contains 40,203 treated episodes at 36,210 distinct firms, with approximately 82,000–101,000 eligible comparison firms per cohort. Appendix Table C.1 reports cohort counts and effective comparison sample sizes, Appendix Table A.17 compares treated and comparison firm-episodes before the exposure jump, and Appendix Figure A.9 shows the exposure paths under the estimated weights.

[22]Firms must have at least three attributed hires from U.S. schools. The denominator includes the firm's total attributed U.S.-school hiring; weights are not renormalized over schools retained in the IPEDS exposure series. Recruiting coverage and concentration enter the pre-event adjustment.

5.2 Empirical Strategy

For each cohort of exposure jumps g, I compare the change in each outcome from $G-2$ to each later year at treated firms with the weighted change at comparison firms,

$$\widehat{\tau}_{g,t} = \frac{1}{N_{1g}} \sum_{f:D_{fg}=1} \Delta Y_{fgt} - \sum_{f:D_{fg}=0} \widehat{\omega}_{fg} \Delta Y_{fgt}, \qquad \Delta Y_{fgt} = Y_{ft} - Y_{f,g-2}, \tag{6}$$

where D_{fg} indicates a treated episode, N_{1g} is the number of treated episodes, and comparison firms receive normalized propensity-score odds $\widehat{\omega}_{fg}$ (Appendix C.3). The propensity score conditions on pre-event outcome histories, firm size and growth, foreign-worker composition, OPT hiring, H-1B demand, exposure histories, recruiting composition, industry, and geography, and is estimated by inverse probability tilting, which balances the means of these covariates between treated and reweighted comparison firms. Each cohort has a balanced outcome window and a fixed comparison pool eligible through $G+3$; the contrasts can equivalently be estimated with firm-by-cohort and cohort-by-year fixed effects and cohort-specific event coefficients. I aggregate cohorts by their shares of treated episodes (Callaway and Sant'Anna, 2021), and the headline is the average annual response during G to $G+2$, relative to $G-2$:

$$\widehat{\tau}_h = \sum_g \frac{N_{1g}}{\sum_j N_{1j}} \widehat{\tau}_{g,g+h}, \qquad \widehat{\tau}^{\text{post}} = \frac{1}{3} \sum_{h=0}^{2} \widehat{\tau}_h. \tag{7}$$

The estimand gives equal weight to treated firm-episodes, not to workers.

Inference uses 499 corporate-cluster multiplier bootstrap draws that re-estimate propensity weights, cohort shares, and outcome contrasts; a corporate group receives the same multiplier across its firms and repeated episodes. I report level effects and percentage effects relative to the estimated untreated counterfactual mean, accounting for their covariance. Firms that recruit from the same schools may share shocks even when they belong to different corporate groups, so I also cluster by each firm's dominant recruiting school; this changes the standard errors little (Appendix Table A.21).

Outcome measurement. Administrative OPT hiring counts employer relationships assigned to each graduate's program-end year. Revelio measures first jobs at the firm among foreign recent U.S. graduates and student interns, as well as other native and foreign hires; native and foreign status is inferred from profiles, since citizenship is not observed. The two sources provide complementary measures of recruitment through U.S. education, with overlapping worker coverage, so their estimates should not be added. Total U.S. headcount counts distinct workers observed at the firm during the calendar year. Appendix B.5 defines

the supplementary hiring groups and sponsorship measures.

Identification and timing. Identification requires that, conditional on the recorded histories, treated firms would have experienced the same outcome changes as reweighted comparisons absent their larger exposure increase. Predetermined recruiting weights remove the mechanical link between current hiring and exposure; they do not eliminate demand shocks shared by firms and their recruiting schools, which I return to below. Because the propensity model conditions on outcome histories that overlap the pre-period, fitted pre-period balance is not an independent test of parallel trends.

OPT hiring is 0.030 workers lower at $G-3$ than at $G-2$ (95% interval -0.054 to -0.006) and 0.074 higher at $G-1$ (0.042 to 0.105), before the measured jump in exposure. Two features of the timing account for this lead. First, IPEDS and SEVIS record graduation at different times. Exposure is dated by IPEDS completions, which count degrees conferred from July to June, while the administrative records date each OPT hire by the calendar year of the graduate's SEVIS program end; 43% of OPT hires come from graduates who finish between July and December, whose degrees IPEDS counts in the following year. Dating OPT hires by the IPEDS academic year lowers the $G-1$ coefficient to 0.056 (standard error 0.014) and the $G-3$ coefficient to -0.004, and raises the average effect over G to $G+2$ from 0.156 to 0.175 (Appendix Table A.20). The main estimates keep calendar-year dating so that the administrative and profile-based outcomes are measured on the same clock, because profiles often record only the year of graduation. Second, employers can respond before the larger eligible cohorts graduate. An expanding eligible program is visible to employers earlier, through students' internships and schools' announcements of new or reclassified programs. Foreign student internships rise by 0.032 at $G-1$ (standard error 0.005), about three-quarters of their average effect over G to $G+2$, and foreign graduate and intern hiring rises by 0.063; both are flat from $G-5$ to $G-3$. Hiring of workers educated abroad, whom the school expansions do not supply, shows no lead (0.004). I therefore treat $G-1$ as partly exposed and measure effects relative to $G-2$, as the individual design does with its partly exposed cohort.

Eligibility. Because exposure is built from the growth of STEM-designated programs, the additional graduates it supplies are, by construction, graduates of those programs. A natural check is therefore whether exposed firms also recruit more OPT workers from the other programs at the same schools, which would indicate that exposure is picking up a general expansion of recruiting from those schools rather than the growth of the eligible pool. They do not: splitting OPT hires by whether the graduate's SEVIS major was on the DHS list in

the program-end year, hiring of STEM-designated graduates rises by 0.175 (standard error 0.021), the full effect, while OPT hiring of other graduates does not change (-0.001, standard error 0.005). Treated firms' recruiting schools also expand ineligible programs slightly more than comparison firms' schools, but by 0.20 percentage points of exposure over G to $G+2$, against 2.36 points for eligible programs (Appendix Table A.20). This check does not separate the longer authorization from the larger number of STEM graduates, which the firm design cannot do; the individual design does that.

The design also follows incumbent firms. Treated and comparison firms must have recruiting histories before the episode and positive employment in each of the four years before it. The estimates therefore measure adjustment within existing employers, and do not capture employment at firms created after the expansion, at firms without prior recruiting ties to the expanding schools, or elsewhere in the local economy. Section 5.4 uses commuting-zone evidence to bound these margins. Appendix Table A.22 reports how an earlier version of the design responds to alternative exposure measures, recruiting windows and comparison pools; the recruitment responses are stable, while the total-employment response is not.

5.3 Recruitment Through U.S. Education

Do expansions in eligible programs translate into greater recruitment through OPT? Table 3 shows that exposed firms hire 0.156 additional OPT workers per annual program-end cohort (95% interval 0.115 to 0.198), a 29.1% increase relative to the counterfactual mean of 0.536. Foreign recent-graduate and student-intern hiring rises by 0.111 workers (0.074 to 0.148), or 26.5%. Figure 7 shows the corresponding paths. Both administrative records and professional histories thus show a substantial increase in recruitment through U.S. education.

To give these numbers a scale: exposed firms' exposure rises by 2.36 percentage points more than comparison firms' over G to $G+2$, so each percentage point of exposure adds about 0.066 OPT hires per firm and year, or 12% of the baseline; across the 40,203 treated episodes, the response amounts to about 6,300 additional OPT hires a year, roughly 3% of annual OPT starts nationally.

The response extends to students who have not yet completed their degrees. Foreign student internships increase by 31.5% (22.4 to 40.6; Appendix Table A.18), and the internship response appears first, a year before the jump in degree completions, as employers see the expanding programs through their students. Employers thus adjust along several stages of the university-to-work pathway: they hire more foreign students while enrolled and more graduates entering the workforce.

5.4 Employment and Other Workers

Increased recruitment through OPT could reflect either an expansion in firms' workforces or a change in whom they hire. An expansion would raise total employment, potentially alongside hiring through other channels; substitution would instead produce declines in other hiring as firms recruit more eligible graduates. I examine total U.S. headcount and the other hiring groups to distinguish these responses.

Figure 8 places the employment response beside native recent-graduate hiring. Total U.S. headcount increases by 0.316 workers per firm, with a 95% interval from −0.481 to 1.113; relative to the counterfactual mean of 62.5 workers, the effect is 0.51% (−0.75 to 1.76). This interval cannot tell expansion from substitution: 0.156 additional OPT hires a year, each staying about two years, would add roughly 0.3 workers to the stock by the third year if none replaced anyone, and zero if all did, and both lie well inside it. The employment evidence therefore has to come from the hiring channels.

Native recent-graduate and student-intern hiring rises by 0.100 workers per year (0.002 to 0.198), or 3.4%. Employers therefore recruit more native graduates alongside the increase in foreign graduate hiring, which is consistent with the expansion of the graduate pool that firms can access, since the expanding programs also graduate more native students. There is no detectable decline in the other measured hiring channels. Other native hiring changes by 1.8% (−0.8 to 4.4), other foreign hiring by −0.1% (−4.2 to 3.9), and hiring of foreign-classified workers with completed foreign postsecondary education and no documented U.S. postsecondary attendance by −1.2% (−6.6 to 4.3; Appendix Table A.18 and Figure A.10). Firms hire more through the U.S.-education pathway without a measured fall in any other hiring group.

These estimates bound how much additional OPT hiring replaces other hiring. The most direct comparison is with hiring from abroad, the alternative channel most similar to OPT: relative to the increase in OPT hiring, hiring of workers educated abroad falls by 0.05 workers per additional OPT hire, with a 95% confidence interval from −0.18 to 0.27.[23] The estimates thus rule out replacement of more than about a quarter of a worker hired from abroad for each additional OPT hire. The debate about OPT is mostly about native workers, and the same calculation for native hiring outside the recent-graduate group gives a point estimate of 1.1 additional native hires per OPT hire and rules out, at the lower end of its interval and ignoring the covariance, a decline of more than about 0.5 native hires per OPT hire. Two qualifications apply. The ratios attribute the whole change in other hiring to the additional

[23]The ratio is the negative of the effect on hiring of workers educated abroad divided by the effect on OPT hiring. Its interval uses the delta method, with the covariance of the two outcomes across the same 499 bootstrap draws (correlation 0.03); ignoring the covariance gives the same interval to two decimal places.

OPT hires, although the same expansions also supply more native graduates, whose hiring rises by 0.100; if those graduates also replaced other workers, the ratios overstate replacement per OPT hire, and if they complemented them, they understate it. And the design does not observe replacement through hiring away from other U.S. employers. Section 6 uses the foreign-educated ratio as the replacement rate r in the fee scenarios.

Because the sample consists of incumbent firms, these estimates also miss firm creation and spillovers to other employers in the local labor market. Kwon, Lee and Monras (2026) measure these margins directly, comparing commuting zones with high and zero exposure to the 2016 expansion of the STEM extension. In exposed areas, college-educated immigrants increase by 0.31 per 100 working-age residents; the number of firms rises by about 1%, concentrated among firms with fewer than 100 employees; and native wage employment rises by 0.45 to 0.51 per 100 residents, or 1.5 to 1.7 native workers per additional high-skilled immigrant, with the increase concentrated among college-educated natives. These local estimates include entry and expansion outside incumbent firms, but they also include natives who move into exposed areas from elsewhere, so they overstate the corresponding national effect. Together, the two designs bound the response of other workers' employment: at incumbent firms, replacement of at most about a quarter of a worker hired from abroad per OPT hire; across local labor markets, including new firms, crowd-in of up to about 1.7 workers per immigrant. Neither design points to substantial displacement of other workers.

5.5 Employer Sponsorship

OPT permits work without immediate employer sponsorship. One possibility is that firms use it in place of sponsoring other foreign workers; another is that access to graduates creates employment matches that firms subsequently seek to retain through another visa. These possibilities predict different responses in employers' demand for sponsorship.

H-1B demand, the number of positions requested on Labor Condition Applications and, from 2020, lottery registrations, rises by 0.179 positions per firm and year (0.072 to 0.287), or 12.0%. Approved petitions rise by 0.089 (0.034 to 0.143), or 13.8%, while the PERM application estimate is −4.9% (−14.8 to 4.9; Appendix Table A.19 and Figure A.11). Relative to the 0.156 additional OPT hires, the 0.089 additional approvals are large, about 0.57 per OPT hire, far above the 3.6 percentage point rise in H-1B requests per 23.6 points of extension take-up in the individual design; the two populations differ, since the firm design's additional graduates are mostly in core STEM fields with better H-1B prospects than the economics and psychology graduates of relabeled programs, and approvals include continuing petitions. Under the binding cap, more approvals at exposed firms need not raise the national

number of H-1B workers, because the capped visas they obtain could otherwise have gone to other employers; Section 6 therefore removes work on an H-1B from the workforce calculation. The increase in H-1B demand alongside OPT hiring shows that recruitment through U.S. education and employer sponsorship expand together: firms use temporary authorization to begin employment relationships and then seek authorization that can support continued employment. The additional OPT hires include graduates both with and without a later recorded request to change to H-1B status (0.081 and 0.075; Appendix Table A.19 and Figure A.12).

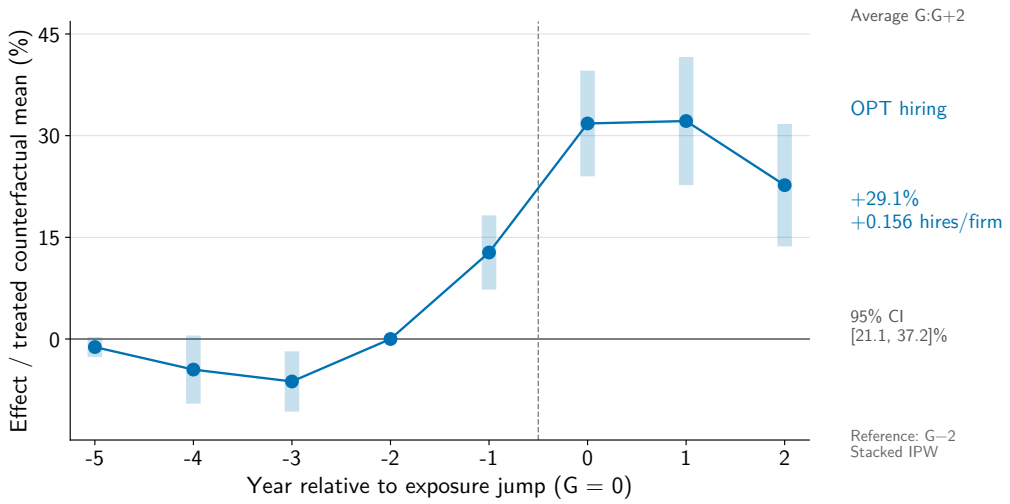
In sum, when access through established school ties expands, incumbent firms hire more OPT workers, more foreign recent graduates and interns, and more native graduates, and they sponsor more H-1B workers, with no detectable displacement of hiring through other channels; total employment is less precisely determined. Section 6 combines these employer responses with graduates' own employment gains to assess the workforce implications.

Table 3: Employer recruitment and employment

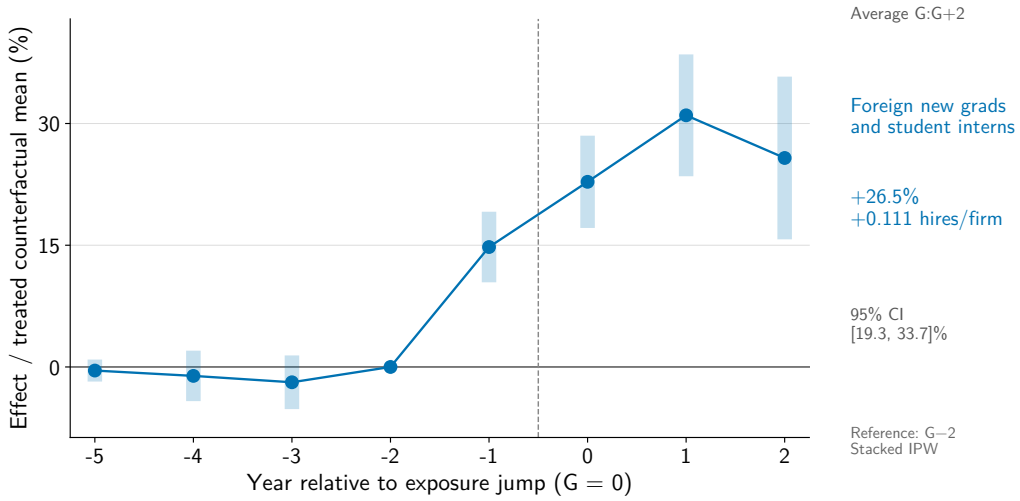
	OPT cohort hiring	Foreign graduate and intern hiring	Native graduate and intern hiring	Total U.S. employment	H-1B demand
	(1)	(2)	(3)	(4)	(5)
Effect per firm and year	0.156***	0.111***	0.100**	0.316	0.179***
	(0.021)	(0.019)	(0.050)	(0.407)	(0.055)
Percent effect	29.1	26.5	3.4	0.5	12.0
	[21.2, 37.0]	[19.3, 33.7]	[0.3, 6.6]	[−0.8, 1.8]	[4.3, 19.7]
Counterfactual mean	0.54	0.42	2.93	62.50	1.50
Treated firm-episodes	40,203	40,203	40,203	40,203	40,203
Treated firms	36,210	36,210	36,210	36,210	36,210
Comparison firm-episodes	451,510	451,510	451,510	451,510	451,510
Corporate clusters	123,437	123,437	123,437	123,437	123,437

Notes: The table reports how recruitment, employment and demand for H-1B visas change at firms whose historical recruiting schools expanded programs eligible for the STEM OPT extension. Each column is a separate outcome; each entry is the average annual effect in the year of the exposure jump and the two following years, relative to two years before the jump, at firms whose historical recruiting schools expanded STEM-eligible programs, compared with firms that experienced smaller increases in exposure, reweighted to match the exposed firms' prior size, growth, hiring and recruiting histories (Section 5). The percent effect divides the effect by the comparison firms' level. Sample: 40,203 exposure jumps at 36,210 firms in 2015–2019 and their comparison firms. Standard errors in parentheses and 95% intervals in brackets from 4499 bootstrap draws that reassign weights by corporate group; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) counts OPT employment relationships in SEVIS, dated by the graduate's program-end year; columns (2)–(4) come from professional profiles, where foreign and native origin is inferred from the profile and total employment counts people with a recorded U.S. job at the firm during the year; column (5) counts positions requested on Labor Condition Applications, the first step of an H-1B petition, capped at the annual 99th percentile.

Figure 7: Exposed firms recruit more through U.S. education



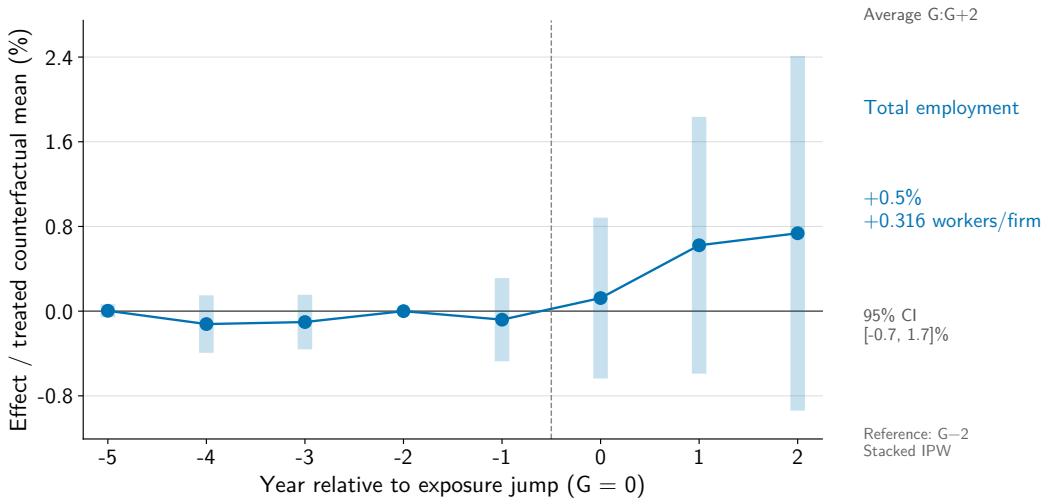
(a) OPT hires (administrative records)



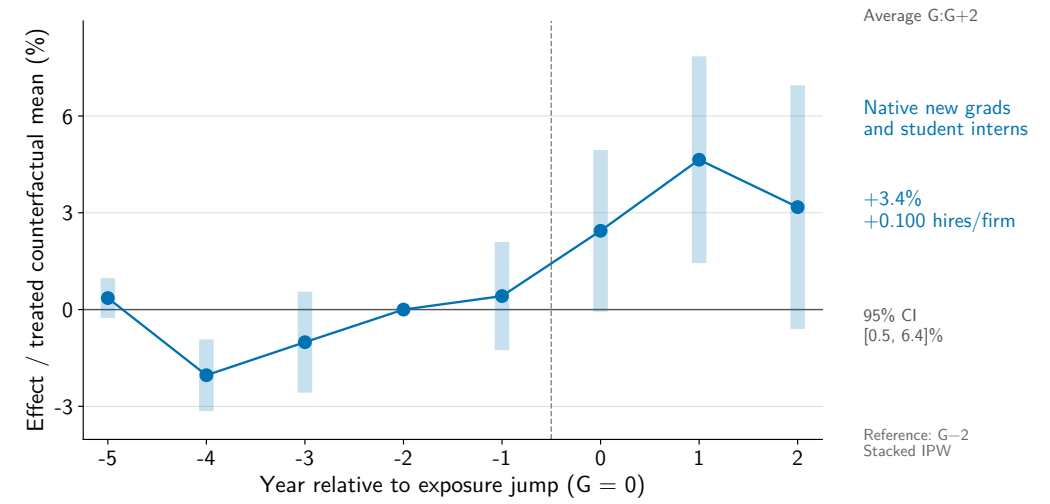
(b) Foreign recent-graduate and student-intern hires (professional profiles)

Notes: Each panel shows how hiring at firms whose recruiting schools expanded STEM-eligible programs changed around the year of the expansion ($G = 0$), relative to two years before and to reweighted comparison firms, as a percent of the comparison firms' level. The right margin reports the average effect over years 0–2, its equivalent in workers per firm, and a 95% interval. Year −1 is partly exposed, because employers see expanding programs through interns and announcements before the larger cohorts graduate, and because IPEDS and SEVIS date graduations differently (Section 5.2). Panel (a) counts OPT employment relationships in SEVIS, dated by the graduate's program-end year; panel (b) counts first jobs at the firm among foreign-classified recent U.S. graduates and student interns in the profiles, where foreign origin is inferred from the profile rather than observed. Sample: 40,203 exposure jumps at 36,210 firms in 2015–2019 and their comparison firms (Table 3 gives the specification); intervals are from 4499 bootstrap draws that reassign weights by corporate group. Sources: SEVIS, Revelio and IPEDS.

Figure 8: Total employment and native graduate hiring at exposed firms



(a) Total U.S. employment



(b) Native recent-graduate and student-intern hires

Notes: As in Figure 7, for the number of people with a recorded U.S. job at the firm during the year (panel a) and first jobs at the firm among native-classified recent U.S. graduates and student interns (panel b). Panel (a) shows that total employment is too imprecisely estimated to tell modest growth from no change. Expanding programs also graduate more native students, so the rise in native graduate hiring in panel (b) partly reflects a larger supply of native graduates rather than a response to OPT hiring. Percent effects are relative to each outcome's own comparison level and cannot be added across panels. Sample, weighting and inference as in Figure 7. Sources: Revelio, SEVIS and IPEDS.

6 Implications for the U.S. Skilled Workforce

The preceding sections measure two margins of OPT's contribution to the U.S. workforce. On the graduates' margin, each month of potential OPT authorization the extension adds raises U.S. employment by about half a month, mostly while the authorization lasts. On the employers' margin, incumbent firms recruit more through U.S. education when their access to eligible graduates expands, without a detectable decline in hiring through other channels. In this section, I combine these estimates following the framework in Section 2.4, and use them to assess what a fee on OPT would remove.

6.1 From Graduates' Employment to the Workforce

Additional employment concentrated in the first years after graduation can still sustain a larger workforce, because each year brings a new graduating cohort. Let d_h be the additional fraction of year h that an affected graduate spends employed in the U.S., and let M_c denote the number of affected graduates in cohort c. Their contribution to employment in calendar year t is

$$\Delta L_t^{\text{grad}} = \sum_{h=1}^{H} M_{t-h+1} d_h. \tag{8}$$

With a constant affected flow M and stable responses, the contribution once all H cohorts are present is $M \sum_h d_h$: the additional employment-years per graduate become additional workers in steady state. In the main linked sample, relabeling adds 5.3 months of U.S. work per graduate over the first three years after completion, or 0.44 employment-years (95% confidence interval 0.21 to 0.65).[24] The national change adds other workers' responses: replacement at the firms that hire the graduates, and the jobs their work supports elsewhere, as in equation (2).

The two designs supply one estimated input for each term. For graduates, the relevant quantity is the share of OPT work that is additional, a. Section 4.5 estimates that each month of work on OPT or the extension raises net U.S. employment by 0.73 months (95% confidence interval 0.41 to 1.02). Part of that net employment, 0.4 months, is work on a cap-subject H-1B that the graduate obtained before the authorization ended, and under a binding cap that visa could otherwise have gone to another applicant. Rather than guess how many of those visas would have been used by someone else, I subtract the measured H-1B work from the numerator and use the remainder: $a = 0.68$ (95% confidence interval 0.43 to 0.90), the share of OPT work that is additional net of work on an H-1B (Table 1).

[24] Appendix Table D.1 reports this effect, the five-year effect, and normalizations by the change in extension use.

For other workers, the firm estimates give the replacement rate r: each additional OPT hire is associated with 0.05 fewer hires of workers educated abroad (95% confidence interval −0.18 to 0.27; Section 5.4). The firm design follows incumbent firms, so it does not measure firm creation, spillovers to other local employers, or how much employment shifts between firms nationally. For s, the commuting-zone estimates of Kwon, Lee and Monras (2026) give an upper bound: after the 2016 expansion of the extension, native employment in exposed areas rose by 1.5 to 1.7 workers per additional high-skilled immigrant, alongside a 1% increase in the number of firms, but because those local estimates include natives who move between areas, they overstate the national effect.

6.2 What a Fee Would Remove

In September 2025 a presidential proclamation imposed a $100,000 fee on new H-1B petitions, and proposals to charge a comparable fee for OPT followed. Clemens (2026) models such a fee as $70,000 on initial OPT and $30,000 on the extension and projects how many graduates would forgo OPT under different assumptions about who ultimately pays. A fee of this kind is a natural application of the estimates: the employment it removes depends on how many graduates forgo OPT, how much they would have worked, and how much of that work would have occurred in the U.S. anyway, and the last two are the quantities this paper measures. For each path p, initial OPT alone or initial OPT followed by the extension, let q_p be the reduction in annual starts, O_p the years of work during OPT per start, and U_p the years of work after OPT that do not depend on a capped visa. With the additional share a, the replacement rate r at incumbent employers, and other workers' employment s through firm creation and local spillovers, each per lost worker, the steady-state employment loss is

$$W^{\text{loss}} = (1 - r + s) \sum_{p} q_p \left(a O_p + U_p \right). \tag{9}$$

I take q_p from the participation responses in Clemens (2026), which differ by who bears the fee. If universities absorb it, he compares the charge with the margin a university earns on a whole degree and assumes that universities stop supplying students whose margin falls below the charge, which gives large declines in OPT starts (63.5% for initial-only OPT and 79.3% for the full path); if students pay, he uses the response of international enrollment to degree prices, which gives lower bounds of 28.8% and 35.0%. The durations O_p and U_p come from a benchmark that assigns 0.85 and 2.20 years of work during OPT to the two paths and 0.12 and 0.35 years of uncapped work afterwards; Appendix D explains where each number comes from. This paper's estimates supply a and r, with a applied to work during OPT, where

the relabeling design measures which routes OPT replaces, and not to the uncapped work after OPT. The central scenarios set $s = 0$; the commuting-zone estimates provide an upper bound of 1.5 to 1.7.

Table 4 reports the results. If universities bear the fee, OPT starts fall by about 139,000 a year, and the U.S. workforce would be about 174,000 workers smaller in steady state, with output about $24 billion lower each year. Across the 95% interval for the additional share, the loss ranges from 121,000 to 221,000 workers; allowing replacement at the upper end of its interval lowers the central loss to 134,000. If students bear the fee, the published participation responses are lower bounds, and the loss is at least about 77,000 workers. Who bears the fee therefore matters more for the size of the loss than the remaining uncertainty about additionality.

Two features of the estimate matter for how a fee would be felt. First, most of the loss, 142,000 of the 174,000 workers, is work during the OPT period itself; the rest is the assumed uncapped work afterwards. This follows from Section 4: the extension's effect on employment is concentrated while it lasts, so a fee removes mainly the years of work that OPT authorizes rather than a permanent stock of workers who would otherwise have settled. Second, the estimate nets out the work graduates would have done through other routes; a calculation that treats every lost year of OPT work as additional gives 241,000 workers under university incidence, so the additional share matters for the level of the loss but not for its order of magnitude.

The central scenarios count only the graduates' own work, net of measured replacement at incumbent employers, so they omit any jobs that OPT workers support at new firms or elsewhere in the local economy. If the local responses estimated by Kwon, Lee and Monras (2026) applied nationally, the loss under university incidence would rise to between 441,000 and 479,000 workers; because those responses include workers moving between areas, these figures are an upper bound. The central scenarios also omit hiring away from other U.S. employers, which the firm design cannot observe and which would make the loss smaller. The two omitted margins point in opposite directions.

These calculations require assumptions beyond the paper's estimates. The additional share is estimated for graduates who respond to extension eligibility in relabeled programs, mainly in economics, business, finance, psychology, architecture, communication and design. Within these programs it varies little across fields or degree levels: leaving out one field gives shares between 0.65 and 0.82, and the shares are 0.66 for master's and 0.82 for bachelor's graduates (Appendix Table A.7 and Appendix Figure A.8). Graduates of core STEM fields may have better H-1B prospects or more employer-sponsored alternatives, which would lower their additional share, and graduates deterred from initial OPT, or from U.S. study altogether,

may have different alternatives again; applying the estimated share to them assumes that their alternatives resemble those of the graduates in relabeled programs. The participation responses depend on assumptions about incidence and behavior, and are projections rather than estimates of the response to an implemented fee. The scenarios assume that graduates who forgo the extension do not switch to initial OPT alone; allowing that switching would reduce the loss. Output is not the cost to other U.S. residents, part of which accrues to the graduates as earnings. Appendix D describes each input, the alternative employer-incidence responses, the treatment of path switching and capped visas, and the output conversion.

Table 4: U.S. employment lost under a $100,000 OPT fee: alternative scenarios

	Universities bear the fee	Students bear the fee
A. Lost OPT starts per year		
Initial OPT only ($70,000)	62,973	28,561
Initial OPT and STEM extension ($100,000)	75,640	33,384
B. Steady-state employment lost, thousands of workers [output lost, $ billions per year]		
Estimated share of OPT work that is additional (0.68)	**174** [23.8]	**77** [10.6]
lower end of its 95% interval (0.43)	121 [17.1]	54 [7.6]
upper end of its 95% interval (0.90)	221 [30.0]	98 [13.3]
with replacement at the upper end of its interval (0.27)	134 [18.3]	59 [8.1]
Every lost OPT work-year additional (no other route)	241 [32.6]	107 [14.5]
C. Upper bound including firm creation and local spillovers		
Estimated share, with local native employment response (1.5–1.7)	441–479	196–213

Notes: The table asks how many fewer people would be working in the U.S., once the change has fully worked through successive graduating cohorts, if a $100,000 fee on OPT ($70,000 for the initial year and $30,000 for the STEM extension) reduced OPT use as projected by Clemens (2026). Panel A applies those projected declines to an annual flow of 99,170 initial-only and 95,384 complete STEM OPT starts, approximated from 2024 authorizations (U.S. Immigration and Customs Enforcement, n.d.): starts fall by 63.5% and 79.3% for the two paths if universities absorb the fee, and by at least 28.8% and 35.0% if students do. Panel B multiplies the lost starts by the net U.S. work each would have produced: assumed years of work during OPT (0.85 for the initial-only path and 2.20 for the complete path) times the share of that work that is additional, plus assumed years of work after OPT that do not depend on a capped visa (0.12 and 0.35), all reduced by the replacement of other workers (0.05 per OPT worker, from Section 5.4, or 0.27 in the fourth row). The additional share is estimated in Section 4.5: net additional U.S. work, excluding work on an H-1B, per month of work on OPT (Table 1). The last row of Panel B treats every lost year of work on OPT as additional. Output values lost employment at assumed annual wages of $80,000 during OPT and $105,000 afterwards, with wages equal to 95% of the marginal product of labor and a labor share of 0.65. Panel C adds the local native employment response to high-skilled immigrants estimated by Kwon, Lee and Monras (2026), 1.46 to 1.67 native workers per immigrant (their Table 4, Panel A); because that response includes natives who move between areas, it is an upper bound on the national response, and no output figure is computed for it. Appendix D describes every input. The scenarios are conditional calculations, not confidence intervals.

7 Conclusion

This paper asks how much Optional Practical Training adds to the U.S. skilled workforce. The answer depends on what graduates and employers would otherwise do: work that OPT authorizes adds to U.S. employment only if graduates would not otherwise have done it, and if employers would not otherwise have filled the same jobs with other workers. Linking administrative student records to professional profiles and employer immigration records, I measure both margins.

When relabeling makes graduates eligible for the extension, use of the extension rises by 23.6 percentage points and use of OPT at all by 10.2 points. Over the first three years after graduation, relabeling adds about ten months of potential OPT authorization and five months of U.S. employment per linked graduate. Graduates work on OPT for three-quarters of the potential authorization, and about three in ten of those months replace work they would otherwise have done in the U.S., half of it while enrolled in another degree program, mostly on full-time curricular practical training, and half under statuses the student-visa records do not capture. Domestic graduates of the same programs, who cannot use OPT, show no change in employment. The gain is concentrated while the authorization lasts; for the minority of graduates who move to an H-1B, the extension is a bridge to employer sponsorship, and the lottery evidence shows them using it to wait out that route.

On the employers' margin, incumbent firms whose recruiting schools expand eligible programs hire more OPT workers, more foreign recent graduates and interns, and more native graduates, and they sponsor more H-1B workers. I find no detectable decline in hiring through other channels: each additional OPT hire replaces at most about a quarter of a worker hired from abroad, and hiring of other native workers does not fall. Total employment at exposed firms is imprecisely estimated, and a design built on incumbent firms cannot measure firm creation or employment shifted between firms. Commuting-zone evidence from Kwon, Lee and Monras (2026) indicates that firm creation and local spillovers add to the employment OPT supports, while hiring away from other employers, which neither design observes, would subtract from it.

These estimates matter for policy because restrictions on OPT remove the work it authorizes only to the extent that the work is additional. Applied to the participation responses projected for a $100,000 fee paid by universities, they imply a U.S. skilled workforce about 174,000 workers smaller in steady state (121,000 to 221,000 across the uncertainty in the additional share) and annual output about $24 billion lower, with most of the loss falling in the years that graduates would have worked on OPT; the loss is smaller, though still substantial, if students rather than universities bear the fee. Evaluating restrictions on

the U.S.-education route therefore requires looking beyond the number of authorizations to the work they make possible and the alternatives they replace.

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A Additional Figures and Tables

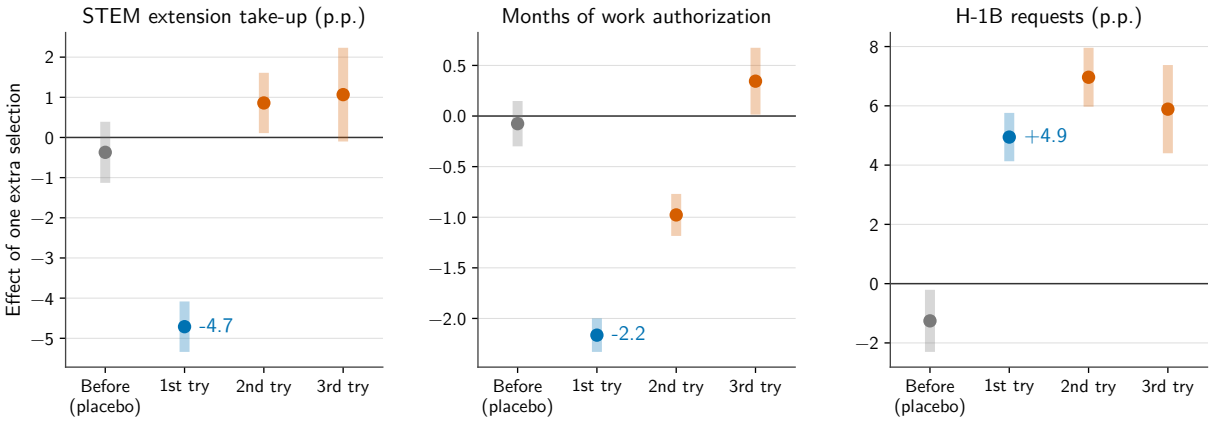
A.1 Graduates

Table A.1: OPT use, H-1B requests and U.S. stay rates among international graduates

			H-1B request		Years 3–5 after completion			NSCG comparison	
	Uses OPT	Uses STEM extension	Never OPT	OPT users	Recorded status	U.S. presence	U.S. work	Recorded status	U.S. stay rate
All graduates	48	14	6	32	36	61	53		
Eligible programs	59	31	12	45	51	75	68		
Ineligible programs	39	0	3	17	21	47	39		
Bachelor's	32	4	3	14	23	49	39	13	37
Master's	60	20	11	38	44	67	60	37	56
Doctoral	60	25	10	44	44	74	73	35	87

Notes: Percentages. F-1 graduates of bachelor's, master's and doctoral programs completing in 2010–2017, all fields. Programs are eligible if their recorded CIP code qualified for the STEM OPT extension under the DHS list in force in the program-end year. OPT use, extension use and H-1B change-of-status requests are from SEVIS records for all graduates. Recorded status is the share of months in years 3–5 after the program-end month spent in a SEVIS-recorded status, U.S. presence is a recorded U.S. job or education spell, and U.S. work counts a month of work during F-1 study as one-half unless a full-time authorization is active; these three columns use linked graduates, weighted by inverse linkage probabilities within program-cohorts and by F-1 graduates across them. The NSCG columns compare 2012–2019 graduates in 2021, two to nine years after graduation: the share in a SEVIS-recorded status, and the share residing in the U.S. in the National Survey of College Graduates among those who first came on a student visa, counting each U.S. degree, relative to IPEDS nonresident completions. SEVIS doctoral completers include some students who left their programs, which understates the doctoral recorded-status rate. Horizons and populations differ across sources, so the comparison is approximate.

Figure A.1: H-1B lottery luck and what graduates do next



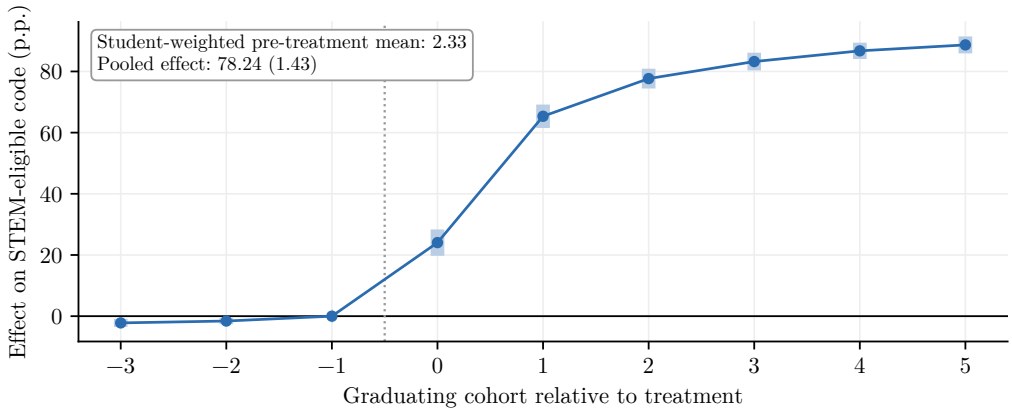
Notes: The figure shows that graduates whose employers lose the H-1B lottery stay on the STEM extension and try again, using a separate design from the rest of Section 4. Each point is the effect of one additional lottery selection at the employer on the subsequent outcomes of the employer's OPT graduates from a given graduating cohort: use of the extension, months of authorization, and recorded H-1B requests. Luck is the employer's number of selections minus the number expected given its registrations and the share of its OPT graduates with a master's degree, pooling the lotteries held in 2020–2022; the sample is employers with at most five registrations and at least two STEM-eligible OPT users. The four lotteries are the one held in the cohort's graduation year ("Before," when most graduates have not yet begun OPT) and the first, second and third lotteries after graduation. One additional selection in the first lottery after graduation lowers extension take-up by 4.7 percentage points (standard error 0.3), shortens authorization by 2.2 months (0.1) and raises H-1B requests by 4.9 percentage points (0.4); luck in the graduation-year lottery has no detectable effect on extension use (−0.4, standard error 0.4) and a small negative association with requests (−1.3, standard error 0.5). Lottery records are linked to employers, not to individual registrants, so these are effects of the employer's luck on its graduates as a group. Intervals are the estimate plus or minus 1.96 standard errors; the analysis uses administrative data only. Sources: SEVIS and USCIS H-1B registration data.

Table A.2: Relabeled programs by first relabeled cohort, degree level and field

	First relabeled IPEDS cohort						
	2014	2015	2016	2017	2018	2019	All
By degree level							
Bachelor's	1	13	12	32	45	52	155
Master's	3	7	7	32	31	41	121
Doctoral	2	6	6	13	14	18	59
By field group							
Economics	1	0	1	15	33	38	88
Architecture	0	5	2	18	16	30	71
Finance	0	1	3	9	2	6	21
Business	0	3	1	2	2	4	12
Psychology	1	7	13	14	17	19	71
Design	0	4	0	7	5	3	19
Communication	0	1	2	5	5	1	14
Other	4	5	3	7	10	10	39
All treated programs	6	26	25	77	90	111	335
Schools	4	18	24	46	63	75	169

Notes: The table counts the relabeled programs in the analysis by the first graduating cohort reported under the new code, by degree level and by the program's original field. Field groups follow the program's original (non-STEM) classification. Examples (the largest pre-treatment F-1 cohort in each group): Economics (bachelor's), CIP 45.0601/45.0605/45.0699 → 45.0603, first relabeled cohort 2019; Finance and quantitative methods (master's), CIP 52.0801 → 27.0305, first relabeled cohort 2019; Architecture and interior design (master's), CIP 04.0201 → 04.0902, first relabeled cohort 2017.

Figure A.2: Relabeling and the eligibility code recorded in SEVIS



Notes: Effect of relabeling on the share of a program's F-1 graduates whose program code in SEVIS, the record that determines eligibility for the extension, is on the DHS STEM list, in percentage points, by graduating cohort relative to relabeling. Cohort −1 is the reference cohort, two years before the first cohort reported under the new code; cohort 0 is partly exposed; the pooled effect averages cohorts 1–4. The figure verifies that a code change reported to IPEDS reaches the administrative record governing eligibility. Sample and specification as in Figure 3; intervals are 95%. Sources: IPEDS and SEVIS.

Table A.3: Pre-treatment characteristics of relabeled and comparison programs

		Comparison programs		Standardized difference	
	Relabeled	Unweighted	Reweighted	Unweighted	Reweighted
Log program completions	4.41	4.33	4.21	0.07	0.17
International share	0.39	0.19	0.32	1.02	0.32
F-1 share from India	0.06	0.06	0.07	−0.02	−0.09
F-1 share from China	0.39	0.31	0.35	0.30	0.15
OPT take-up	0.52	0.37	0.53	0.56	−0.05
Trend in international share	0.022	0.018	0.022	0.07	−0.00
Trend in log completions	0.071	0.035	0.089	0.17	−0.09
Log real tuition	10.70	10.34	10.67	0.91	0.07
Bachelor's program	0.51	0.71	0.56	−0.43	−0.10
Master's program	0.40	0.26	0.40	0.31	−0.00
Doctoral university	0.75	0.64	0.77	0.23	−0.03

Notes: The table compares relabeled programs with their comparison programs before relabeling, before and after comparison programs are reweighted by their estimated odds of relabeling. Program-level means over the four pre-treatment years, weighted by reference-cohort F-1 graduates as in the estimates, computed within each relabeling cohort and averaged across cohorts by their relabeled programs' F-1 graduates. The reweighted column multiplies comparison programs' weights by their estimated odds of treatment. Standardized differences divide the difference from the relabeled mean by the pooled standard deviation across programs. Field-by-degree and Carnegie categories enter the propensity model alongside these characteristics.

Table A.4: Eligibility, OPT use and later status

	Eligibility and take-up				Later status			
	STEM-eligible code (pp)	Uses the STEM extension (pp)	Uses OPT (pp)	Authorization after completion (months)	Requests H-1B status (pp)	Returns to F-1 study within one year (pp)	Returns to F-1 study within two years (pp)	Starts a graduate program within three years (pp)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Effect of relabeling	78.24***	23.57***	10.22***	5.72***	3.57***	−1.45**	−4.29***	−3.02***
	(1.43)	(1.71)	(1.54)	(0.43)	(0.94)	(0.60)	(0.81)	(0.62)
Pre-treatment mean	2.33	1.19	51.63	6.04	13.72	4.24	7.76	6.80

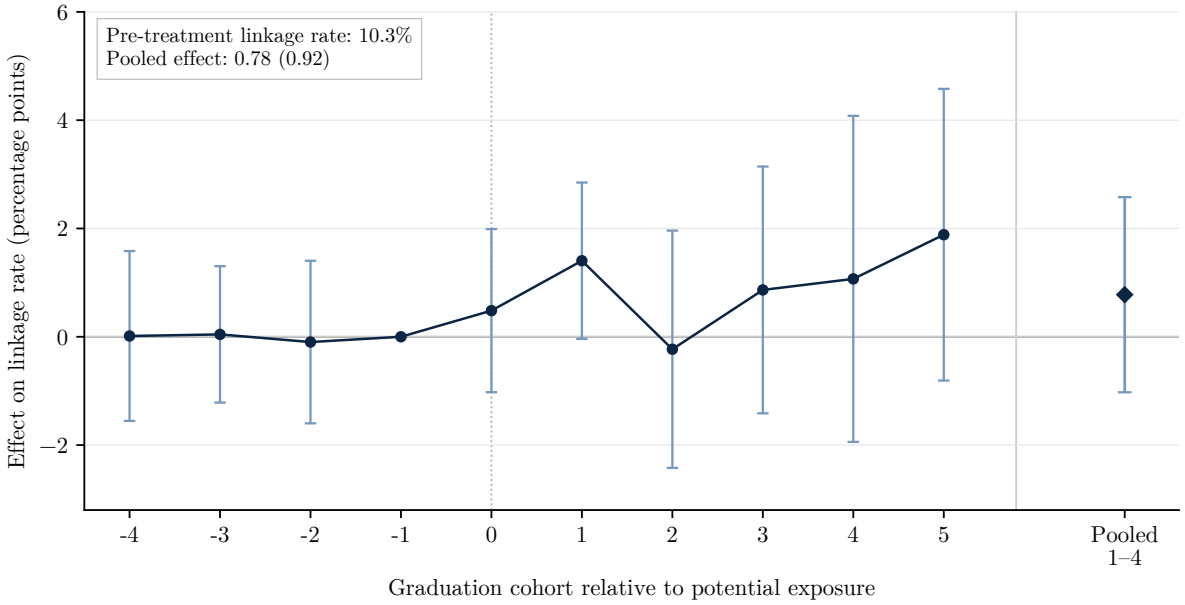
Notes: The table reports how relabeling a program as STEM-eligible changes its international graduates' eligibility, use of OPT and later immigration status, using the SEVIS records of all F-1 graduates (no profile linkage is needed). Each column is a separate outcome; each entry is the effect of relabeling averaged over the first four relabeled graduating cohorts, relative to the cohort two years before relabeling, from the stacked difference-in-differences design of Section 4.2. Column (1) is whether the graduate's program code in SEVIS is on the DHS STEM list; columns (2)–(3) are whether the graduate ever used the extension or OPT; column (4) is months of post-completion OPT authorization from SEVIS authorization dates. H-1B requests are recorded requests to change status, not approvals. Returns to F-1 study within two years and graduate-program starts within three years are unobserved for graduates whose window ends after October 2023. The take-up regressions contain 335 treated programs at 169 schools; other outcomes use the graduates with enough follow-up. Standard errors in parentheses from 199 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; pp denotes percentage points.

Table A.5: Program completions, linkage and reported costs

	IPEDS completions		Linkage	SEVIS-reported costs and funding ($000s)			
	All completions	International share (pp)	Linked to a profile (pp)	Tuition and fees	Students' personal funds	Funds from the school	Funds from other sources
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Effect of relabeling	−0.64	4.40***	0.78	−0.37	1.28	0.88	−0.49
	(1.72)	(0.72)	(0.92)	(0.54)	(0.94)	(1.68)	(2.26)
Pre-treatment mean	64.09	27.70	10.30	43.05	15.01	26.93	60.57

Notes: The table checks whether relabeling changes the programs themselves: their size, the share of international graduates, how often graduates can be linked to a professional profile, and what students report paying and receiving. Each column is a separate outcome; each entry is the effect of relabeling averaged over the first four relabeled cohorts, relative to the cohort two years before relabeling, from the design of Section 4.2. Columns (1)–(2) use IPEDS completions with equal program weights; columns (3)–(7) use SEVIS records for all F-1 graduates. Columns (4)–(7) are in thousands of 2025 dollars and average the nonmissing, nonnegative amounts that schools report to SEVIS; they measure reported amounts, not realized spending. Columns (3)–(5) contain 335 treated programs; columns (6) and (7) contain 257 and 292. Standard errors in parentheses from 199 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; pp denotes percentage points.

Figure A.3: Relabeling does not change the share of graduates linked to a profile



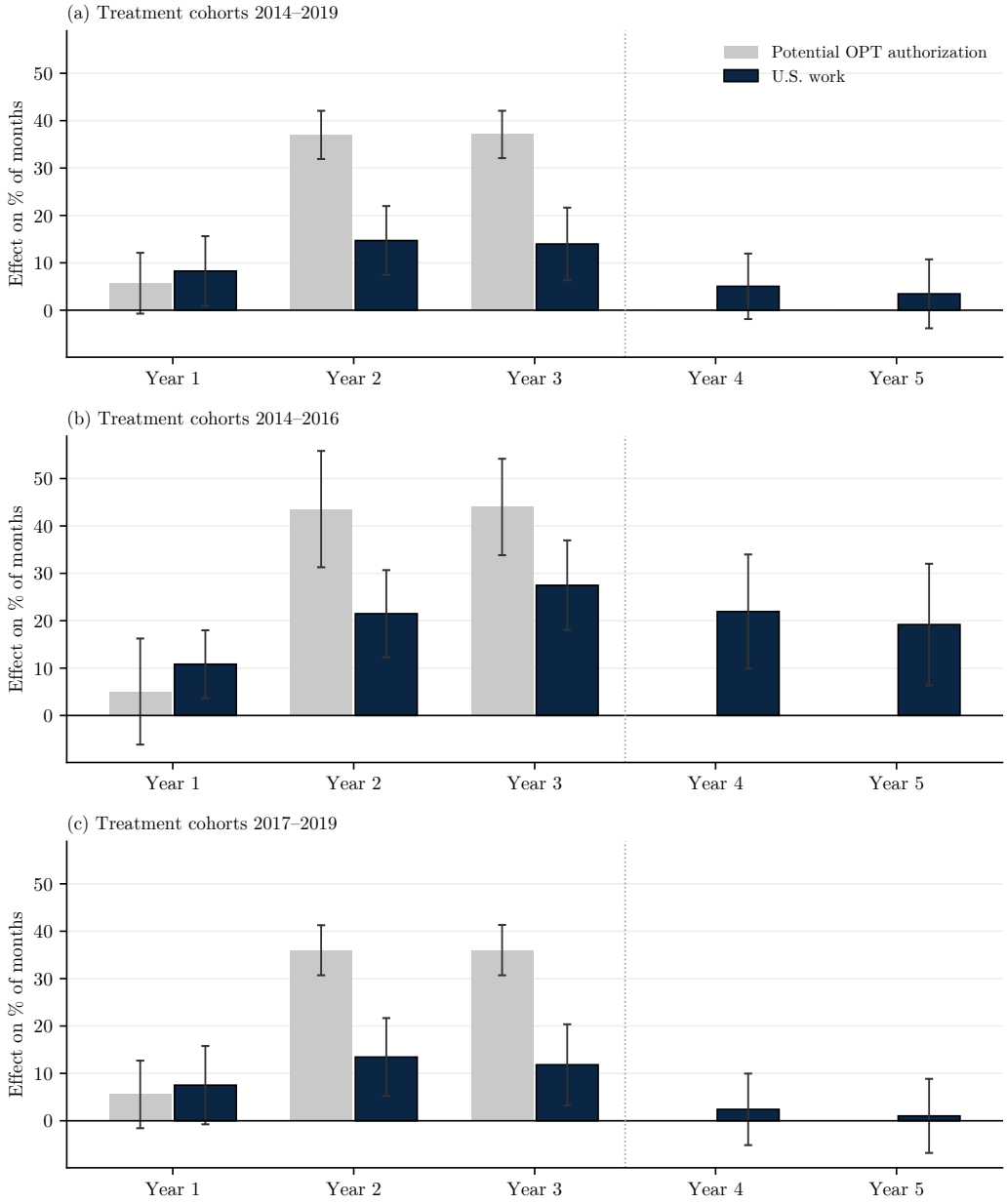
Notes: Effect of relabeling on the share of a program's F-1 graduates who are linked to a professional profile, in percentage points, by graduating cohort relative to relabeling. The employment estimates in Section 4 use only linked graduates, so they require that relabeling not change who is linked; this figure checks the overall linkage rate. Cohort −1 is the reference cohort, two years before the first cohort reported under the new code; cohort 0 is partly exposed; the pooled effect averages cohorts 1–4 (cohort 5 uses the 2014–2018 relabeling cohorts). The box reports the pre-treatment mean among relabeled programs and the pooled effect (standard error). Within OPT users and non-users, linkage rates are about 14% and 7% and do not change; the share of graduates who are both linked and OPT users rises by 1.8 percentage points (standard error 0.8), from 7.1%, and the share linked but not using OPT falls by 1.0 (0.3), from 3.2%, which is what the 10.2 percentage point rise in OPT use implies. Sample and specification as in Figure 3; intervals are 95% from 199 school-level bootstrap draws. Sources: IPEDS, SEVIS and Revelio.

Table A.6: U.S. employment over five years, by treatment cohort and comparison rule

Treatment cohorts	Potential OPT authorization, years 1–3	U.S. work, years 1–3	U.S. work, years 4–5	U.S. work, years 1–5	Column (2) relative to column (1)	Column (4) relative to column (1)	Annual U.S. work, years 4–5 relative to 1–3	Treated programs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Comparison programs vary by year (main specification)								
2014–2019	9.57***	4.43***	1.02	5.45***	0.46	0.57	0.35	248
	(0.77)	(1.28)	(0.85)	(1.88)	[0.23, 0.68]	[0.18, 0.95]	[−0.31, 0.91]	
2014–2016	11.11***	7.18***	4.94***	12.11***	0.65	1.09	1.03	30
	(1.76)	(1.24)	(1.47)	(2.33)	[0.38, 1.07]	[0.64, 1.77]	[0.49, 1.58]	
2017–2019	9.31***	3.93***	0.41	4.35**	0.42	0.47	0.16	218
	(0.82)	(1.44)	(0.92)	(2.09)	[0.16, 0.67]	[0.05, 0.92]	[−1.05, 0.87]	
B. Same comparison programs every year								
2014–2019	9.44***	3.23***	0.92	4.15**	0.34	0.44	0.43	248
	(0.85)	(1.09)	(0.86)	(1.78)	[0.14, 0.56]	[0.09, 0.83]	[−0.65, 1.10]	
2014–2016	11.52***	6.89***	5.09***	11.98***	0.60	1.04	1.11	30
	(2.03)	(1.49)	(1.56)	(2.66)	[0.36, 1.09]	[0.61, 1.83]	[0.47, 1.70]	
2017–2019	9.09***	2.62**	0.29	2.90	0.29	0.32	0.16	218
	(0.90)	(1.21)	(0.92)	(1.94)	[0.05, 0.53]	[−0.07, 0.75]	[−2.28, 1.20]	

Notes: Effects of relabeling in months per linked graduate, for graduates observed in the professional profiles through the fifth year after the SEVIS program-end month; the same graduates enter every year. In Panel A, a comparison program is excluded from a year's comparison if it relabels before the end of that year's follow-up, as in the main estimates; in Panel B, comparison programs that relabel within six years of a cohort's program end are excluded in every year, so that every year uses the same comparison programs. Estimates for the 2014–2016 and 2017–2019 treatment cohorts fit the treatment-propensity and linkage models separately for each group. Columns (5) and (6) divide the effects on U.S. work over years 1–3 and years 1–5 by the effect on potential authorization; column (7) divides the average annual effect in years 4–5 by that in years 1–3. Potential OPT authorization assigns 12 months of initial OPT to each OPT user and 17 or 24 months of extension to each extension user, from the program-end month; U.S. work counts a month of work during F-1 study as one-half unless a full-time F-1 authorization is active. Standard errors in parentheses and 95% intervals in brackets from 499 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.4: Added authorization and added employment over five years, by relabeling cohort



Notes: As in Figure 5, estimated separately for programs relabeled in 2014–2016 and in 2017–2019, each with its own treatment-propensity and linkage models. The employment gain persists through the fifth year for the earlier cohorts and fades after the third year for the later cohorts, which graduated into the pandemic and the H-1B lotteries with the lowest selection rates; the data do not separate these explanations. The 2014–2016 cohorts contain 57 of the 335 treated programs in the administrative sample and 30 of the 248 in the linked sample. Graduates are observed through the fifth year after completion and held fixed across years; intervals are 95%. Sources: IPEDS, SEVIS and Revelio.

Table A.7: Robustness of the main estimates

	Years 1–3				Years 1–5	
	Potential OPT authorization (months) (1)	U.S. work (months) (2)	U.S. work relative to potential authorization (3)	U.S. work relative to work on OPT (4)	U.S. work (months) (5)	U.S. work relative to potential authorization (6)
Main specification	9.74***	5.29***	0.54	0.73	5.45***	0.57
	(0.86)	(1.38)	[0.29, 0.81]	[0.41, 1.02]	(1.88)	[0.18, 0.95]
Doubly robust estimator	9.17***	5.72***	0.62	0.83	6.26**	0.69
	(1.10)	(1.99)	[0.28, 1.11]	[0.41, 1.40]	(2.55)	[0.27, 1.36]
Calibrated linkage weights	8.10***	4.32***	0.53	0.71		
	(0.81)	(1.47)	[0.18, 0.87]	[0.28, 1.08]		
Two-year comparison buffer	9.86***	4.79***	0.49	0.66		
	(0.87)	(1.31)	[0.24, 0.74]	[0.34, 0.96]		
Three-year comparison buffer	9.94***	4.38***	0.44	0.61		
	(0.89)	(1.25)	[0.21, 0.69]	[0.31, 0.89]		
Same comparison programs every year	9.68***	4.03***	0.42	0.58	4.15**	0.44
	(0.96)	(1.28)	[0.18, 0.67]	[0.27, 0.87]	(1.78)	[0.09, 0.83]
Leaving out one treatment cohort			0.46 to 0.65	0.63 to 0.84		
Leaving out one field group			0.47 to 0.60	0.65 to 0.82		

Notes: The table shows how the main estimates change under alternative estimators, linkage weights and comparison-group rules; each row is a specification. Columns (1)–(4) sum effects of relabeling over the first three years after completion, for linked graduates observed in SEVIS through year 3, as in Table 1; columns (3) and (4) divide the effect on U.S. work by the effects on potential OPT authorization and on work on OPT in the same specification. Columns (5) and (6) sum effects over the first five years for linked graduates observed in the profiles through year 5, as in Table 2, column (4), and divide by the same graduates' effect on potential authorization; five-year outcomes were not estimated with calibrated weights or comparison buffers. The doubly robust estimator adds a weighted outcome model for comparison programs' changes to the treatment-propensity weights. Calibrated weights match each arm's administrative shares of graduates using no OPT, OPT only and the STEM extension within stack and event time, dropping cells without linked graduates in a category, so that the pooled support is the 2016–2019 treatment cohorts. A comparison buffer of b years excludes a comparison program from a year's comparison if it relabels within b years after the end of that year's follow-up; the main specification uses one year. The same-comparison row instead excludes, in every year, comparison programs that relabel within six years of a cohort's program end. The last two rows report the range of estimates when one treatment cohort (2014–2019) or one field group (economics, finance, architecture, business, psychology, communication or design) is left out at a time. Standard errors in parentheses and 95% intervals in brackets from 499 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Domestic and foreign graduates matched to programs in the profiles

	Domestic graduates			Foreign graduates			Graduates matched	
	Year 1	Year 2	Year 3	Year 1	Year 2	Year 3	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Effect of relabeling	1.21	0.25	0.47	6.88***	6.52***	4.20*	−3.91	0.97
	(0.80)	(0.93)	(1.01)	(2.38)	(2.41)	(2.40)	(2.45)	(0.93)
Pre-treatment mean	61.6	64.1	64.0	41.1	36.6	33.3	61.5	8.0
Treated programs	271	271	271	218	218	218	271	218
All programs	24,843	24,843	24,512	12,323	12,323	12,067	24,843	12,323

Notes: The table is a placebo test: domestic graduates cannot use OPT, so if relabeling changed only eligibility their employment should not move, while a change in the program itself (its curriculum, reputation or recruiting) would affect them too. Columns (1)–(6) report effects on the percentage of months worked in the U.S. in each calendar year after graduation for graduates of relabeled and comparison programs found in the professional profiles, matched to programs by school, degree level and four-digit CIP code (either the program's former or its new code) and classified as domestic or foreign by the profiles' predicted probability of foreign origin; foreign graduates matched the same way serve as a positive control. U.S. work is the share of monthly snapshots with a spell at a U.S. employer in the firm panel. Columns (7)–(8) report the number of matched graduates per program and cohort. Each cell is a separate stacked difference-in-differences regression with treatment-propensity weights, averaging event times 1–4 relative to event time −1, with programs weighted equally; programs enter only in cohorts with at least one matched graduate in the group. Standard errors in parentheses from 499 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: U.S. work in a later F-1 program by authorization, linked graduates (months, full-time equivalent)

	All later-program months	Full-time CPT	Part-time CPT	Pre-completion OPT	Later program's own OPT	No recorded authorization
	(1)	(2)	(3)	(4)	(5)	(6)
Effect of relabeling	−1.16***	−0.65***	−0.08**	0.00	−0.17	−0.26*
	(0.34)	(0.22)	(0.04)	(0.00)	(0.15)	(0.15)
Pre-treatment mean	1.14	0.30	0.05	0.00	0.27	0.53
Treated programs	248	248	248	248	248	248
Schools	1,221	1,221	1,221	1,221	1,221	1,221

Notes: Effects of relabeling summed over the first three years after the SEVIS program-end month; each column is a separate outcome. A month is in a later F-1 program if it falls within a subsequent F-1 program's dates or under an authorization recorded for that program, and is classified by the authorization active in that month, in the order of columns (5), (2), (4) and (3); column (6) covers months enrolled without a recorded off-campus authorization, which include on-campus work. Linked graduates whose third year after completion ends by October 15, 2023, as in Table 1; column (1) equals column (6) of that table, and work counts one-half of a month unless a full-time authorization is active. Standard errors in parentheses from 499 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Time in a later F-1 program by authorization, all F-1 graduates (months)

	All later-program months	Full-time CPT	Part-time CPT	Pre-completion OPT	Later program's own OPT	No recorded authorization
	(1)	(2)	(3)	(4)	(5)	(6)
Effect of relabeling	−1.02***	−0.36***	−0.04***	−0.00	−0.12***	−0.50***
	(0.21)	(0.06)	(0.01)	(0.01)	(0.04)	(0.16)
Pre-treatment mean	1.81	0.24	0.05	0.00	0.25	1.27
Treated programs	335	335	335	335	335	335
Schools	1,415	1,415	1,415	1,415	1,415	1,415

Notes: Effects of relabeling summed over the first three years after the SEVIS program-end month; each column is a separate outcome. A month is in a later F-1 program if it falls within a subsequent F-1 program's dates or under an authorization recorded for that program, and is classified by the authorization active in that month, in the order of columns (5), (2), (4) and (3); column (6) covers months enrolled without a recorded off-campus authorization, which include on-campus work. SEVIS records for all F-1 graduates of the analysis programs with reference-cohort F-1 weights, for graduates whose third year after completion ends by October 15, 2023. Standard errors in parentheses from 499 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.11: CPT and pre-completion OPT before program completion

	Ever used CPT (pp)	CPT authorization (months)	Ever used pre-completion OPT (pp)	Final program year on CPT (pp)
	(1)	(2)	(3)	(4)
Effect of relabeling	5.3***	0.20**	−0.2	1.5***
	(1.4)	(0.08)	(0.3)	(0.4)
Pre-treatment mean	18.7		2.3	4.4

Notes: The table shows whether relabeling changes work authorization during the degree: Curricular Practical Training (CPT), which covers internships and other work that is part of the curriculum, and pre-completion OPT. Each column reports the effect of relabeling on a separate outcome, averaging the first four relabeled cohorts relative to the cohort two years before relabeling, from the design of Section 4.2. All F-1 graduates in SEVIS records; outcomes refer to authorizations before the program-end date. Standard errors in parentheses from 199 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; pp denotes percentage points.

Table A.12: U.S. work in the final twelve months of the program

	All months, counted fully	Months without recorded authorization	Months with CPT	Months with pre-completion OPT	All months, part-time adjusted	Authorized time, part-time adjusted
	(1)	(2)	(3)	(4)	(5)	(6)
Effect of relabeling	7.0**	6.3**	0.5	−0.1	3.7**	0.4
	(3.2)	(3.1)	(0.8)	(0.1)	(1.6)	(0.8)
Pre-treatment mean	29.4	24.1	4.9	0.2	16.3	6.4

Notes: The table shows whether relabeled cohorts work more in the U.S. during the last year of their degree, before any OPT, and under which authorization. Each column reports the effect of relabeling on a separate outcome, averaging the first four relabeled cohorts relative to the cohort two years before relabeling, from the design of Section 4.2, for linked graduates weighted to represent all F-1 graduates of their program and cohort. Each month of work in the final twelve months before the SEVIS program-end month is assigned to the authorization active in that month; the part-time adjustment counts a month of work during F-1 study as one-half unless a full-time authorization is active, and counts part-time CPT and pre-completion OPT as one-half in authorized time. Standard errors in parentheses from 199 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; pp denotes percentage points.

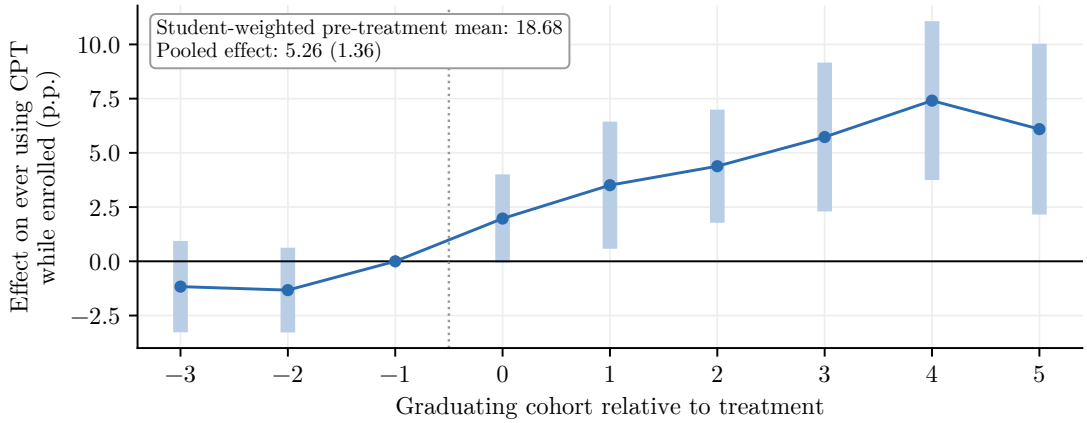
Table A.13: Employment during the program, by position

	All positions	With CPT or pre-completion OPT	Internship title, no authorization	Campus job, no authorization	Other position, ending by completion	Other position, continuing after completion
	(1)	(2)	(3)	(4)	(5)	(6)
Effect of relabeling	0.17**	0.01	0.03	0.05	0.06**	0.03
	(0.08)	(0.01)	(0.02)	(0.07)	(0.03)	(0.02)
Pre-treatment mean	0.66	0.08	0.06	0.38	0.11	0.04

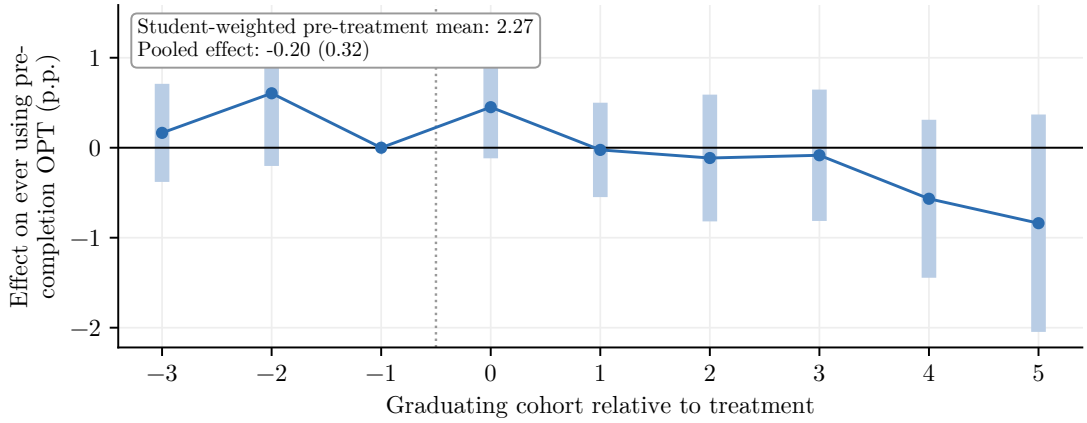
Notes: The table splits the work during the degree in Appendix Table A.12 by the kind of position recorded in the profile. Each column reports the effect of relabeling on a separate outcome, averaging the first four relabeled cohorts relative to the cohort two years before relabeling, from the design of Section 4.2, for linked graduates weighted to represent all F-1 graduates of their program and cohort. Employment-years from program start to completion, classified by the authorization and job title of each position; a position recorded only with a start year in the program-end year that continues past completion is dated from the day after completion. Standard errors in parentheses from 199 school-level bootstrap draws; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; pp denotes percentage points.

Figure A.5: Relabeling and work authorization during the degree

(a) Ever used Curricular Practical Training before completion



(b) Ever used pre-completion OPT



Notes: Effect of relabeling on the share of a program's F-1 graduates who used each kind of work authorization before finishing their degree, in percentage points, by graduating cohort relative to relabeling. Curricular Practical Training covers internships and other work that is part of the curriculum; pre-completion OPT draws down the OPT available after graduation. Cohort −1 is the reference cohort; cohort 0 is partly exposed; the pooled effect averages cohorts 1–4; boxes report the pre-treatment mean and the pooled effect (standard error). Sample and specification as in Figure 3; intervals are 95% from 199 school-level bootstrap draws. Sources: IPEDS and SEVIS.

Table A.14: Estimated compensation after graduation

	Worldwide compensation				U.S. compensation proxy			
	Year 1	Year 2	Year 3	Year 5	Year 1	Year 2	Year 3	Year 5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Effect of relabeling	11.99*	17.73**	12.22	5.25	16.72**	25.26***	20.16***	5.84
	(6.30)	(7.22)	(7.46)	(9.42)	(7.07)	(6.63)	(6.47)	(7.71)
Pre-treatment mean	76.5	80.4	86.9	110.0	53.9	51.4	51.6	64.8

Notes: The table reports effects of relabeling on Revelio's model-based estimates of annual pay, in thousands of dollars, in the given calendar year after graduation, for linked graduates (the calendar-year sample of Appendix Table A.8). These are estimated pay for the jobs recorded in profiles, not earnings: Revelio imputes pay from the position, employer and location, and assigns lower pay to positions outside the U.S., so the worldwide measure rises mechanically when graduates move from jobs abroad to U.S. jobs. The U.S. measure includes defined zeros for graduates without U.S. work; unavailable pay for an observed job is not thereby measured. The two measures use different outcome samples, and changes in location and job composition contribute to the estimates. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.15: Career histories five years after graduation

	Cumulative employers (count)	Cumulative positions (count)	Average employer tenure (months)	Average position tenure (months)
	(1)	(2)	(3)	(4)
Effect of relabeling	0.09	0.17	0.45	1.19
	(0.16)	(0.19)	(1.03)	(0.91)
Pre-treatment mean	3.41	4.90	23.16	20.01

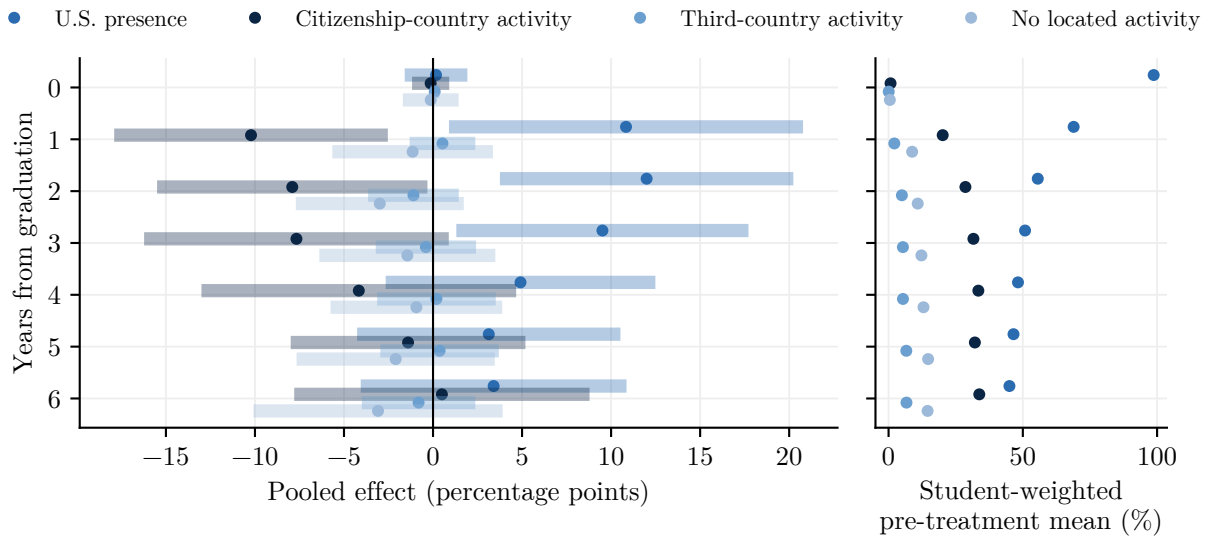
Notes: The table asks whether relabeled graduates' careers look different five years out: each column is the effect of relabeling on a feature of the recorded career history in the fifth calendar year after graduation, for linked graduates (the calendar-year sample of Appendix Table A.8). Selection into observed work qualifies comparisons of tenure and other conditional outcomes. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.16: Take-up of the extension by degree level and country of origin

	Uses the STEM extension (pp) (1)	U.S. work per month of potential authorization, years 1–3 (2)
All graduates	23.6	0.54
	[20.2, 26.9]	[0.30, 0.81]
A. By degree level		
Bachelor's	18.1	0.58
	[14.3, 21.9]	[0.09, 0.99]
Master's	32.8	0.50
	[28.2, 37.5]	[0.25, 0.82]
Doctoral	13.2	
	[9.3, 17.0]	
B. By country of citizenship		
India	44.4	
	[37.3, 51.4]	
China	18.0	
	[13.6, 22.4]	
English-speaking countries	24.1	
	[18.7, 29.6]	
Other countries	24.9	
	[21.5, 28.2]	

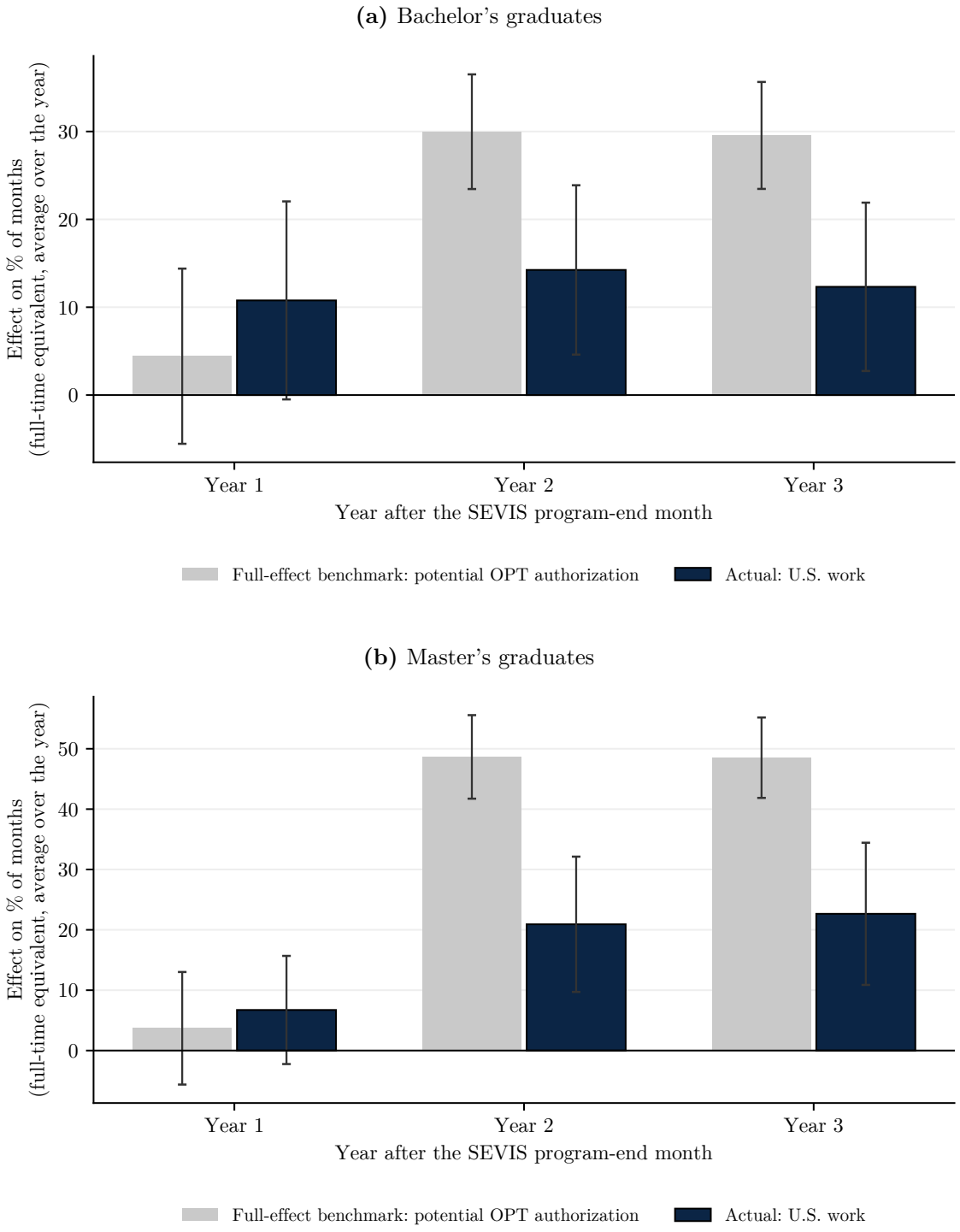
Notes: Each entry is a separate estimate of the effect of relabeling within the indicated group of graduates, with 95% confidence intervals in brackets from 199 school-level bootstrap draws. Column (1) is the percentage-point increase in use of the STEM extension, from the SEVIS records of all F-1 graduates (as in Appendix Table A.4). Column (2) is the ratio of the effect on months of recorded U.S. work to the effect on months of potential OPT authorization over the first three years after completion, for linked graduates (as in Table 1); doctoral graduates are too few in the linked sample for a separate estimate, and estimates by country of citizenship are not reported because the share of graduates who can be linked to a profile falls sharply with the number of classmates from the same country, which makes the origin-specific employment estimates depend on a handful of programs. English-speaking countries are Australia, Canada, Ireland, New Zealand and the United Kingdom. Subgroup estimates are separate fits, not tests of differences between groups.

Figure A.6: Where the additional U.S. workers come from: recorded activity by location



Notes: The left panel shows the effect of relabeling on the share of linked graduates with recorded work or education in each location in each calendar year after graduation, in percentage points with 95% confidence intervals; the right panel shows the pre-treatment means. The rise in U.S. presence is matched by a fall in activity recorded in the graduate's country of citizenship, with no change in third countries, so the additional U.S. workers come mainly from graduates who would otherwise have returned home. Locations are those of recorded jobs and education, so "no located activity" means that the profile records nothing in that year, not that the graduate has left. Sample: the calendar-year linked sample (as in Appendix Table A.8), with inverse-linkage weights; effects average the first four relabeled cohorts. Sources: IPEDS, SEVIS and Revelio.

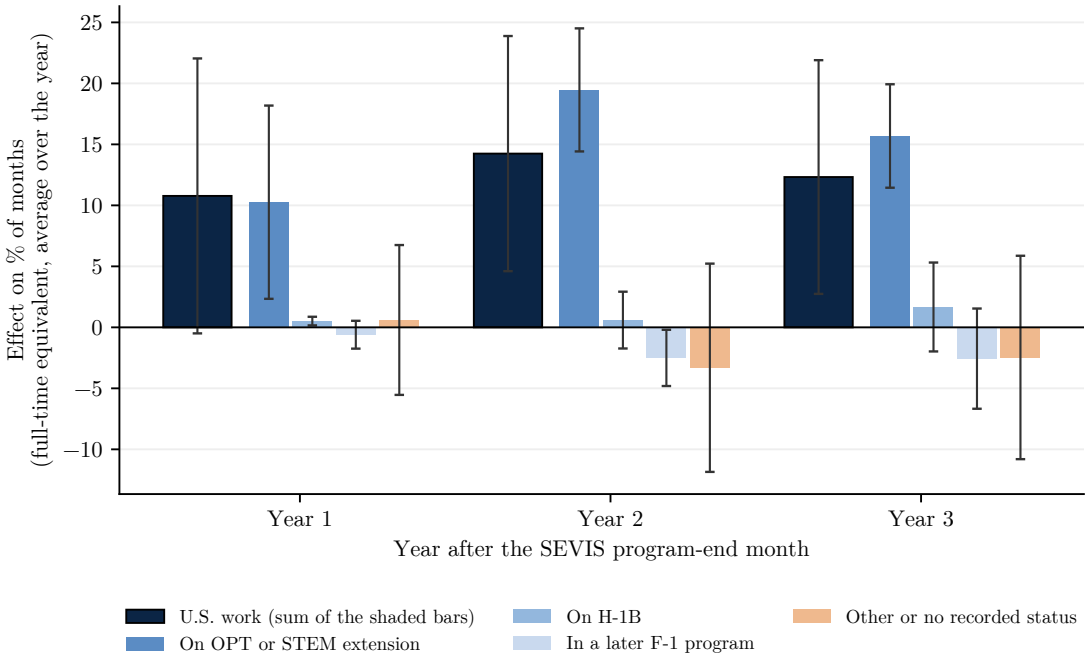
Figure A.7: Added authorization and added employment, by degree level



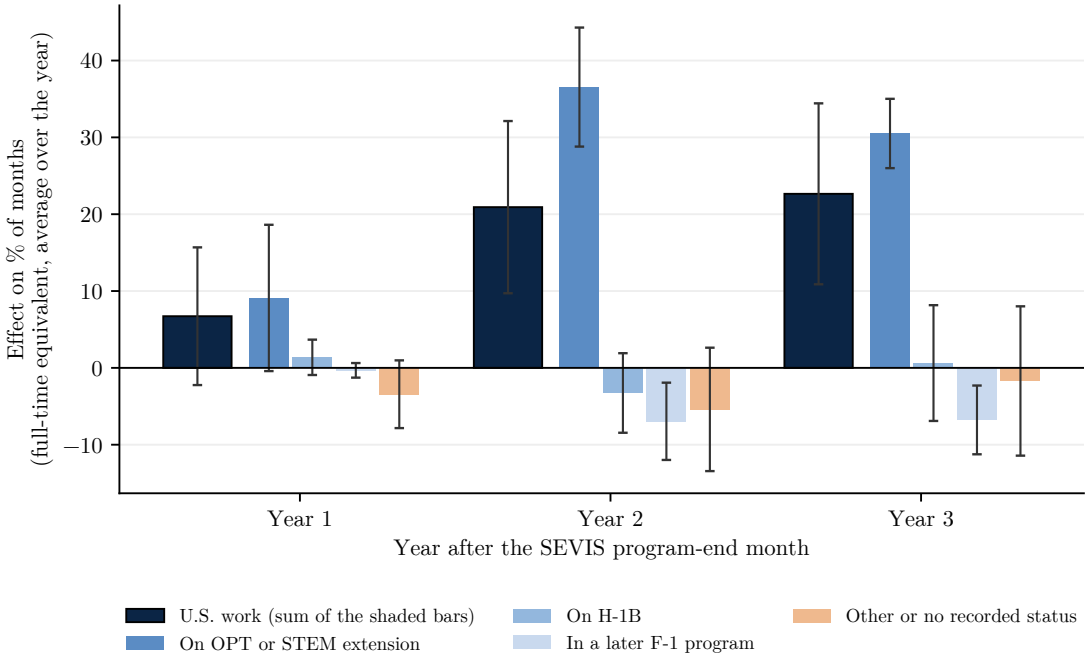
Notes: As in Figure 5 for the first three years after completion, estimated separately by degree level: gray bars are the effect of relabeling on potential OPT authorization and dark bars the effect on recorded U.S. work, in percentage points of months. Sample as in Table 1, with the common treatment-propensity model and degree-specific weights and means; doctoral graduates are too few in the linked sample for separate estimates. Intervals are 95%. Sources: IPEDS, SEVIS and Revelio.

Figure A.8: Added U.S. work by immigration status, by degree level

(a) Bachelor's graduates



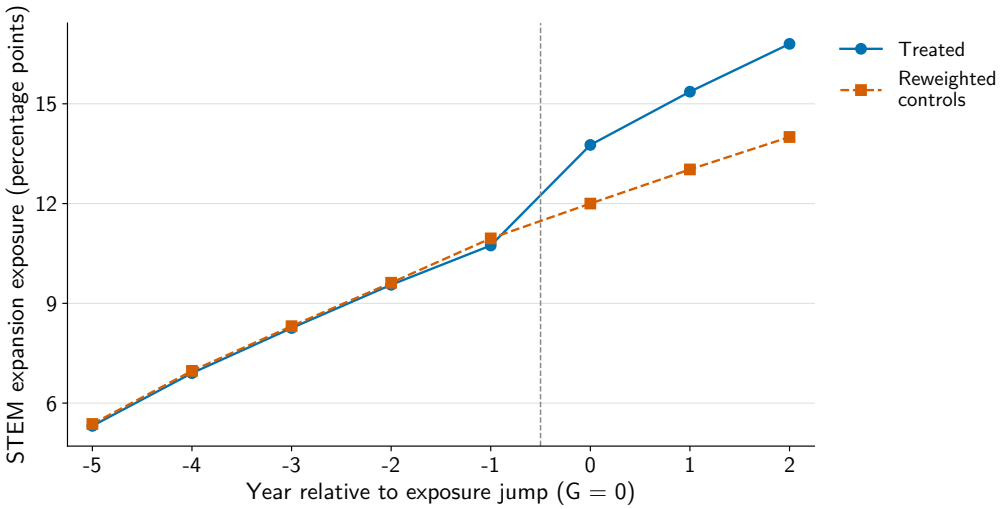
(b) Master's graduates



Notes: As in Figure 6, estimated separately by degree level: the dark bar is the effect of relabeling on U.S. employment and the lighter bars split it by immigration status in the same month. Master's graduates' added authorization replaces more work in a later F-1 program than bachelor's graduates'. Intervals are 95%. Sources: IPEDS, SEVIS and Revelio.

A.2 Employers

Figure A.9: Recruiting exposure of exposed and comparison firms



Notes: The figure shows the treatment in the employer design: mean recruiting exposure (the share of a firm's recruiting network's completions in STEM-eligible programs, in percentage points, using recruiting ties fixed in 2008–2013) for exposed firms and reweighted comparison firms, by year relative to the exposure jump ($G = 0$). The two groups track each other before the jump and diverge at it; comparison firms are required to have small increases through $G + 3$. Exposure is a feature of the schools a firm recruits from; it is not the share of the firm's workers on OPT. Sample: 40,203 exposure jumps at 36,210 firms in 2015–2019, aggregated across cohorts by their numbers of exposed firms; intervals are from 4499 bootstrap draws that reassign weights by corporate group. Sources: IPEDS, SEVIS and Revelio.

Table A.17: Treated and comparison firm-episodes before the exposure jump

	Treated	Comparison episodes		Standardized difference	
	episodes	Unweighted	Reweighted	Unweighted	Reweighted
	(1)	(2)	(3)	(4)	(5)
A. Outcomes in $G-2$, per firm					
Total U.S. employment	50.8	291.4	57.7	−0.18	−0.01
OPT hiring	0.53	0.61	0.49	−0.01	0.01
Foreign graduate and intern hiring	0.43	0.58	0.41	−0.03	0.01
Native graduate and intern hiring	2.60	9.88	2.62	−0.16	−0.00
Other foreign hiring	1.70	4.77	1.83	−0.10	−0.00
Other native hiring	8.13	43.44	9.26	−0.18	−0.01
Foreign-educated hiring	0.51	1.44	0.56	−0.08	−0.00
H-1B demand	1.40	1.52	1.31	−0.01	0.01
Approved H-1B petitions	0.62	0.72	0.61	−0.02	0.00
B. Recruiting exposure (percentage points)					
Exposure in $G-2$	9.56	7.61	9.62	0.84	−0.03
Change from $G-2$ to G	4.21	2.06	2.38	2.02	1.72
Firm-episodes	40,203	451,510			

Notes: The table compares exposed firms with their comparison firms two years before the exposure jump, before and after the comparison firms are reweighted to resemble the exposed firms. Panel A reports the outcomes used in Section 5, in workers or positions per firm; Panel B reports the recruiting-exposure index (the share of a firm's recruiting network's completions in STEM-eligible programs, in percentage points) and its change through the jump year. Means are averages over the 2015–2019 exposure-jump cohorts, weighted by each cohort's number of exposed firms; standardized differences divide the difference in means by the average of the two groups' standard deviations. Because the reweighting matches firms on their pre-event outcome histories, the reweighted outcome levels are close by construction; the exposure change in the last row is the treatment and is not balanced. Comparison firm-episodes count each eligible comparison firm once per cohort.

Table A.18: Hiring through other channels

	Other foreign hiring (1)	Other native hiring (2)	Foreign student internships (3)	Foreign-educated hiring (4)
Effect per firm and year	−0.003	0.173	0.043***	−0.007
	(0.042)	(0.126)	(0.008)	(0.018)
Percent effect	−0.1	1.8	31.5	−1.2
	[−4.2, 3.9]	[−0.8, 4.4]	[22.4, 40.6]	[−6.6, 4.3]
Counterfactual mean	2.04	9.38	0.14	0.63
Treated firm-episodes	40,203	40,203	40,203	40,203
Treated firms	36,210	36,210	36,210	36,210
Comparison firm-episodes	451,510	451,510	451,510	451,510
Corporate clusters	123,437	123,437	123,437	123,437

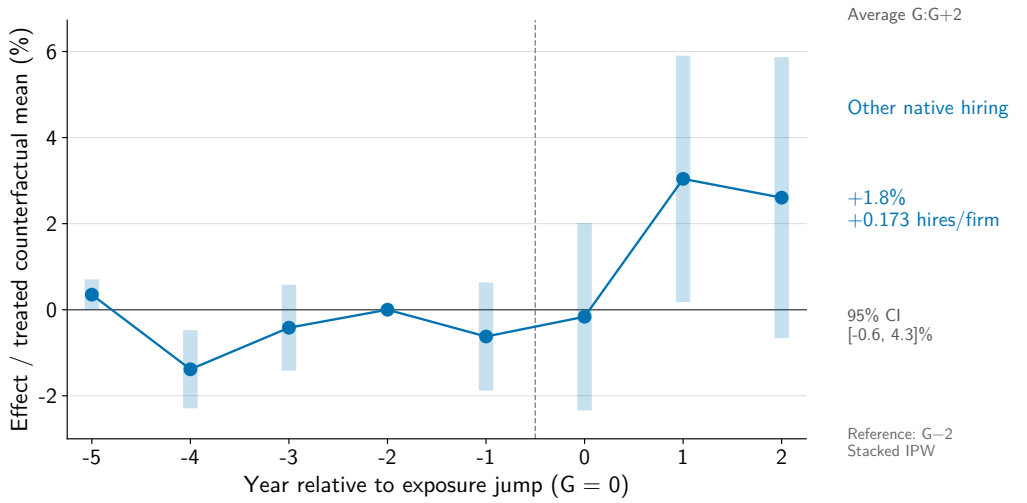
Notes: The table asks whether exposed firms hire less through other channels when they recruit more OPT workers (Table 3). Each column is a separate outcome; each entry is the average annual effect in the year of the exposure jump and the two following years, relative to two years before the jump, at firms whose historical recruiting schools expanded STEM-eligible programs, compared with firms that experienced smaller increases in exposure, reweighted to match the exposed firms' prior size, growth, hiring and recruiting histories (Section 5). The percent effect divides the effect by the comparison firms' level. Sample: 40,203 exposure jumps at 36,210 firms in 2015–2019 and their comparison firms. Standard errors in parentheses and 95% intervals in brackets from 4499 bootstrap draws that reassign weights by corporate group; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All outcomes count first jobs at the firm in the professional profiles, with origin inferred from the profile: other foreign and other native hiring exclude recent U.S. graduates and student interns; foreign student internships are short first jobs that overlap U.S. enrollment; foreign-educated hiring counts foreign-classified hires with a completed foreign degree and no recorded U.S. postsecondary education, the group most directly comparable to OPT hires.

Table A.19: Sponsorship and OPT hiring by later H-1B request

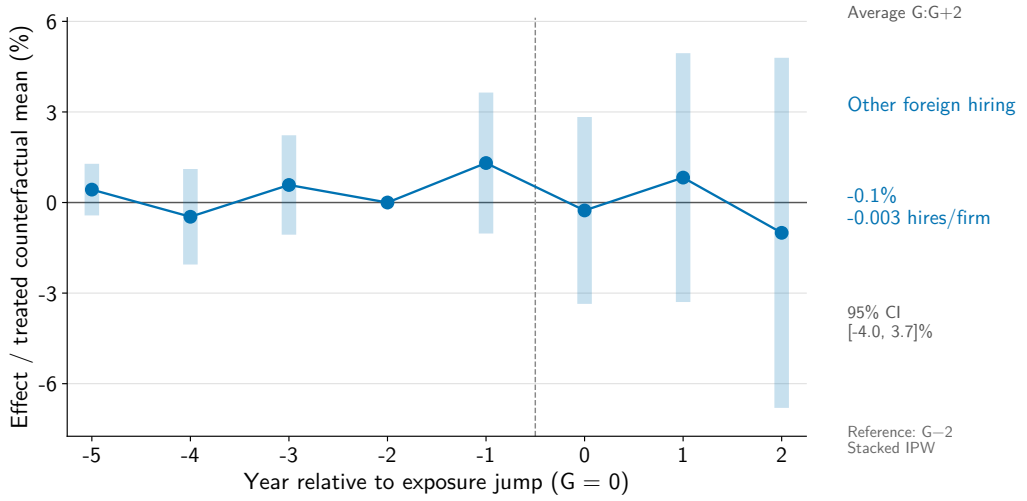
	Approved H-1B petitions	PERM applications	OPT hiring: H-1B request recorded	OPT hiring: no request recorded
	(1)	(2)	(3)	(4)
Effect per firm and year	0.089***	−0.011	0.081***	0.075***
	(0.028)	(0.012)	(0.012)	(0.011)
Percent effect	13.8	−4.9	40.7	22.3
	[4.6, 23.0]	[−14.8, 4.9]	[27.3, 54.0]	[15.8, 28.7]
Counterfactual mean	0.64	0.23	0.20	0.34
Treated firm-episodes	40,203	39,991	40,203	40,203
Treated firms	36,210	36,014	36,210	36,210
Comparison firm-episodes	451,510	448,877	451,510	451,510
Corporate clusters	123,437	122,830	123,437	123,437

Notes: The table reports employer sponsorship outcomes and splits the OPT hiring response by whether the graduate later requested H-1B status. Each column is a separate outcome; each entry is the average annual effect in the year of the exposure jump and the two following years, relative to two years before the jump, at firms whose historical recruiting schools expanded STEM-eligible programs, compared with firms that experienced smaller increases in exposure, reweighted to match the exposed firms' prior size, growth, hiring and recruiting histories (Section 5). The percent effect divides the effect by the comparison firms' level. Sample: 40,203 exposure jumps at 36,210 firms in 2015–2019 and their comparison firms. Standard errors in parentheses and 95% intervals in brackets from 4499 bootstrap draws that reassign weights by corporate group; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Approved H-1B petitions (USCIS records) include new and continuing petitions by fiscal year of decision; PERM applications (Department of Labor) are the first step of employer-sponsored permanent residence. Columns (3) and (4) divide OPT hiring by whether the graduate has any recorded H-1B request in the administrative records, which is not linked to a particular employer or date.

Figure A.10: Hiring of other native and other foreign workers at exposed firms



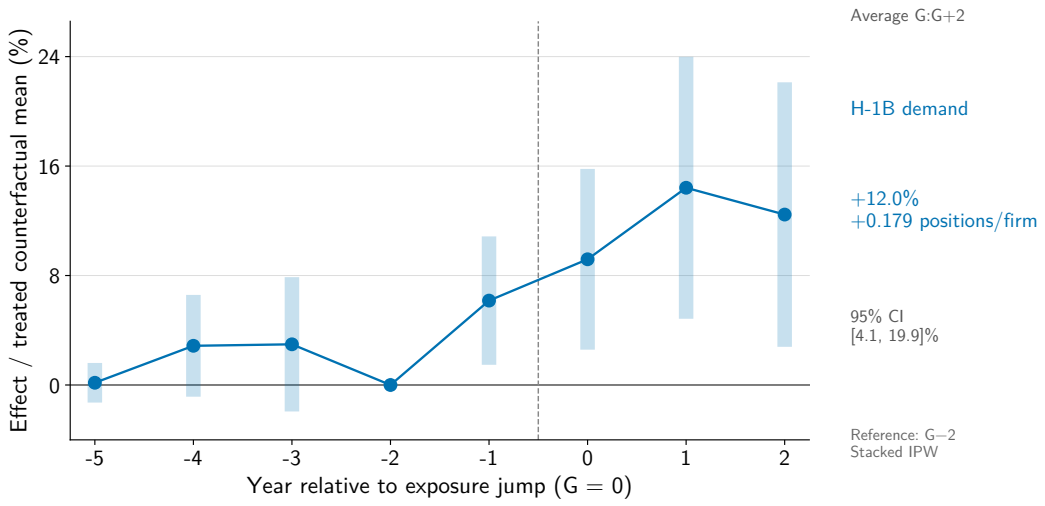
(a) Other native hires



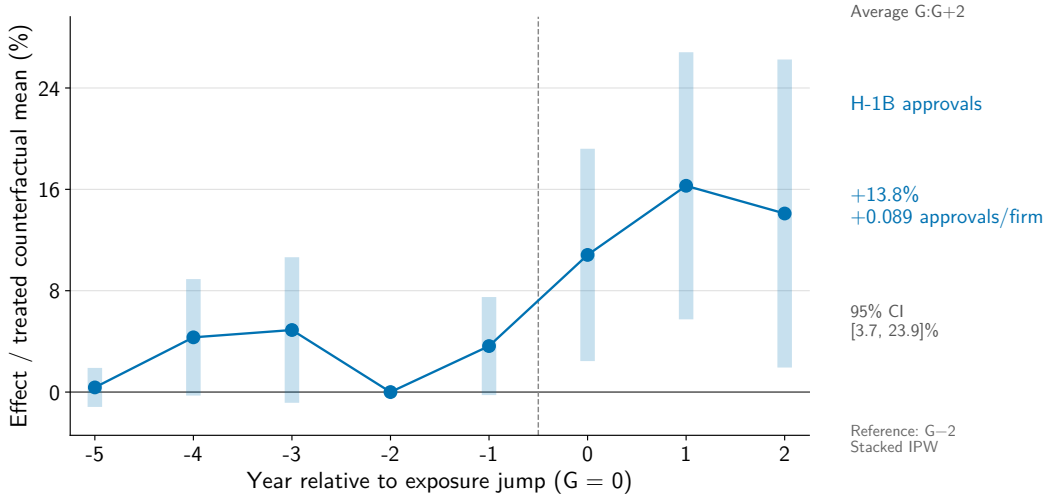
(b) Other foreign hires

Notes: As in Figure 7, for first jobs at the firm among native (panel a) and foreign (panel b) workers other than recent U.S. graduates and student interns; origin is inferred from the profile. Neither group's hiring falls when exposed firms recruit more OPT workers. Sample, weighting and inference as in Figure 7. Sources: Revelio, SEVIS and IPEDS.

Figure A.11: H-1B demand and approved petitions at exposed firms



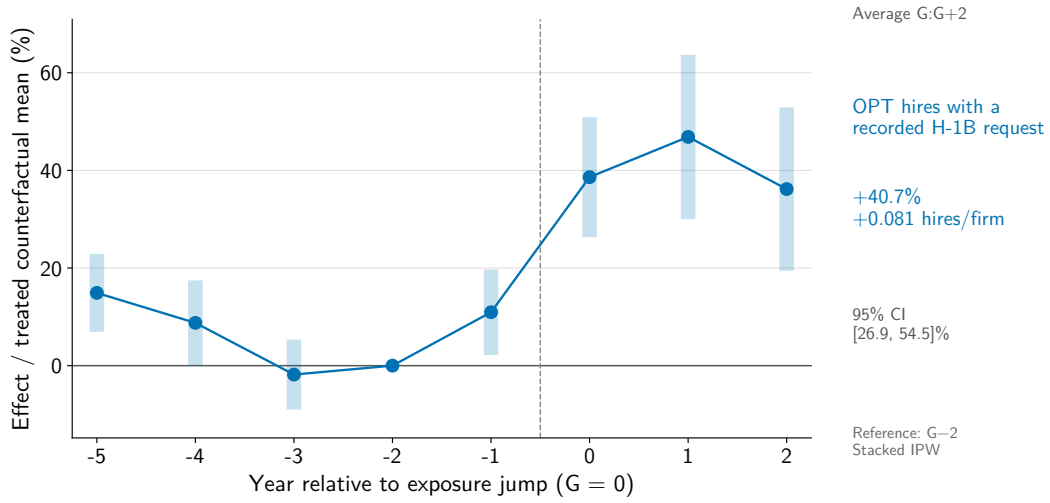
(a) H-1B demand



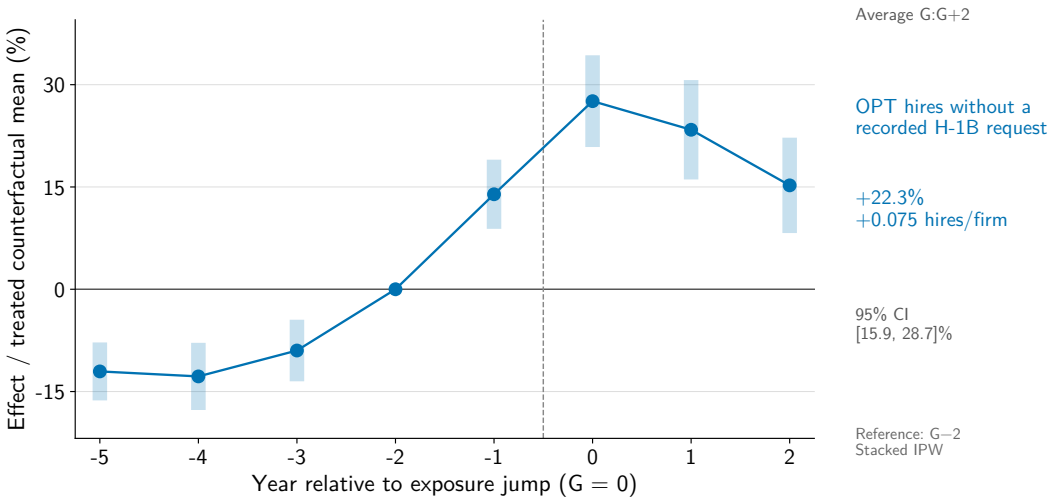
(b) Approved H-1B petitions

Notes: As in Figure 7, for employers' demand for H-1B workers (positions requested on Labor Condition Applications and, from 2020, lottery registrations, without deduplicating workers) and approved H-1B petitions (new and continuing, by fiscal year of decision). Both are capped at the annual 99th percentile among positive values and do not follow the same applicants across stages. Sample, weighting and inference as in Figure 7. Sources: DOL, USCIS, Revelio, SEVIS and IPEDS.

Figure A.12: OPT hiring at exposed firms, by whether the graduate later requested H-1B status



(a) Graduates with a recorded H-1B request



(b) Graduates with no recorded H-1B request

Notes: As in Figure 7, panel (a), splitting each firm's OPT hires by whether the graduate has any recorded request to change to H-1B status in SEVIS. The request is not linked to a particular employer or date, so the split describes the composition of the additional hires: recruitment rises both among graduates who later appear in sponsorship records and among those who do not. The two outcomes use the same observations and weights as total OPT hiring, so the level effects add (0.081 and 0.075, summing to 0.156); the percentage effects do not. Sample, weighting and inference as in Figure 7. Sources: SEVIS, Revelio and IPEDS.

Table A.20: Timing and program eligibility of the OPT hiring response

	OPT hiring				Recruiting exposure (pp)	
	All graduates		By graduate's program		Expansion of	
	Calendar year	Academic year	STEM-designated	Other programs	Eligible programs	Ineligible programs
	(1)	(2)	(3)	(4)	(5)	(6)
$G-3$	-0.030^{**}	-0.004	0.006	-0.010^{**}	0.006^{***}	0.006^{**}
	(0.012)	(0.011)	(0.009)	(0.005)	(0.002)	(0.003)
$G-1$	0.074^{***}	0.056^{***}	0.042^{***}	0.014^{***}	-0.153^{***}	-0.001
	(0.016)	(0.014)	(0.013)	(0.005)	(0.007)	(0.003)
G to $G+2$	0.156^{***}	0.175^{***}	0.175^{***}	-0.001	2.362^{***}	0.201^{***}
	(0.021)	(0.022)	(0.021)	(0.005)	(0.010)	(0.007)
Counterfactual mean	0.536	0.536	0.355	0.179	12.95	3.73

Notes: The table examines the timing of the OPT hiring response and whether it is confined to graduates of STEM-designated programs (Section 5.2). Rows are years relative to the exposure jump (G), with two years before the jump as the reference; the last row averages the jump year and the two following years. Column (1) is the main estimate, which dates each OPT hire by the calendar year in which the graduate's program ended; column (2) dates hires by the July–June academic year that IPEDS uses for completions, the clock on which exposure is measured. Columns (3) and (4) split column (2) by whether the graduate's program code in SEVIS was on the DHS STEM list in the year the program ended. Columns (5) and (6) take the firm's own recruiting exposure as the outcome: the paper's index of eligible-program expansion at its recruiting schools and the corresponding index for ineligible programs, in percentage points. All columns use the 40,203 exposure jumps, weights and bootstrap draws of the main OPT-hiring estimate (Table 3); only the outcome changes. Standard errors in parentheses from 4499 bootstrap draws that reassign weights by corporate group; * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A.21: Inference with clusters at the dominant recruiting school

	OPT hiring	Foreign graduate and intern hiring	Native graduate and intern hiring	Total U.S. employment	H-1B demand	Foreign-educated hiring
	(1)	(2)	(3)	(4)	(5)	(6)
Average effect, G to $G+2$	0.156	0.111	0.100	0.316	0.179	-0.007
SE, corporate groups	(0.021)	(0.019)	(0.050)	(0.407)	(0.055)	(0.018)
SE, recruiting schools	(0.022)	(0.019)	(0.038)	(0.377)	(0.054)	(0.017)
Clusters: corporate groups	123,437	123,437	123,437	123,437	123,437	123,437
Clusters: recruiting schools	3,238	3,238	3,238	3,238	3,238	3,238

Notes: The table checks whether treating firms that recruit from the same school as dependent changes the precision of the main estimates. Point estimates are the main estimates (Table 3 and Appendix Table A.18). The first row of standard errors is the paper's main inference, from 4499 bootstrap draws in which all firms in the same corporate group receive the same weight; the second row instead gives the same weight to all firms whose largest recruiting share is at the same school, refitting the propensity weights in each draw. With school clusters, the 95% interval for the replacement ratio of foreign-educated hiring per additional OPT hire is $[-0.16, 0.25]$ (point estimate 0.05).

Table A.22: Earlier version of the employer design: alternative exposure measures and comparison groups

	Effect per percentage point of exposure, % of pre-period mean					
	OPT hiring	Recent-graduate hiring	Total employment	H-1B demand	Approved H-1B petitions	PERM applications
	(1)	(2)	(3)	(4)	(5)	(6)
Baseline (never-treated comparisons) (29,261)	26.2	3.0	0.6	56.6	5.8	14.0
	(6.7)	(1.0)	(0.4)	(12.5)	(3.4)	(4.7)
Exclude firms with an earlier exposure jump (23,365)	14.3	2.5	1.0	59.0	6.6	19.0
	(3.4)	(1.0)	(0.5)	(16.5)	(4.4)	(6.8)
Recruiting ties measured 2008–2011 only (22,718)	17.7	1.1	-0.3	22.7	3.9	6.1
	(3.8)	(1.0)	(0.5)	(9.7)	(3.5)	(6.7)
Ties from all hires, not only first jobs (29,359)	20.1	2.2	0.3	55.5	4.6	14.3
	(3.7)	(0.9)	(0.4)	(13.2)	(3.3)	(4.7)
Ties from recent-graduate hires only (14,811)	17.6	1.1	-0.4	46.1	6.2	5.4
	(2.7)	(0.9)	(0.4)	(18.3)	(4.0)	(11.1)
Exposure from eligible-program growth only (28,889)	20.4	1.4	-0.5	50.8	8.2	17.2
	(3.4)	(0.9)	(0.4)	(11.0)	(2.8)	(6.2)
Exposure from other-program contraction only (26,360)	45.8	1.0	-0.4	147.8	25.5	-8.8
	(9.8)	(3.1)	(1.4)	(46.9)	(10.1)	(11.6)
Match on exposure history and coverage (15,291)	13.0	-1.0	-1.7	37.5	7.0	9.7
	(4.0)	(1.5)	(0.8)	(17.1)	(3.7)	(5.8)
Later-treated firms as comparisons (24,904)	8.8	0.9	-0.1	41.6	-4.2	10.6
	(3.5)	(0.8)	(0.5)	(11.1)	(3.8)	(4.9)
Later-treated comparisons, matched on exposure history (23,398)	6.5	1.4	-0.1	29.4	-1.3	17.5
	(2.5)	(0.9)	(0.5)	(9.3)	(3.7)	(6.2)

Notes: These estimates come from an earlier version of the employer design (September 2026) that matched each exposed firm to a comparison firm in the same industry and cohort and estimated the effect of a one-percentage-point increase in recruiting exposure, pooled over the year of the exposure jump and the three following years, relative to the year before the jump. Each row changes one feature of that design; the first row is its baseline, and the number in parentheses after each label is the number of treated firm-events. Effects are divided by the treated firms' pre-period mean of the outcome so that rows with different exposure doses and samples can be compared; standard errors (in parentheses) are clustered by matched pair and corporate group. The main estimates in Section 5 use the current stacked, inverse-probability-weighted design, whose sample, growth control and reference year differ, so the rows are not reruns of Table 3; they show which design choices move the recruitment, employment and sponsorship responses. Replacing never-treated comparison firms with firms whose own exposure jump comes later (last two rows) balances exposure histories and lowers the total-employment and recent-graduate estimates while the OPT, H-1B demand and PERM responses remain positive. Sources: IPEDS, SEVIS, Revelio, DOL and USCIS.

B Data Construction, Linkage, and Outcome Measurement

This appendix describes how I turn the raw administrative files and professional profiles into the data used in the paper: recovering people and programs from the SEVIS files, linking schools, fields, employers and people across sources, and measuring outcomes after linkage. Appendix C describes the restrictions specific to each design.

B.1 SEVIS Administrative Records

The disclosed SEVIS files are organized as annual snapshots. A student who remains enrolled or uses practical training can appear in several files, often with the same program and employment information repeated. Within a file, identifiers distinguish people and their educational programs, but these identifiers are not consistent across annual files, so simply appending the files would count many students and authorizations more than once.

I use repetition across files to recover longitudinal identifiers. Exact agreements in record contents, excluding regenerated identifiers, connect observations across annual files; within each file, the reported person identifier connects programs and associated employment information; and connected groups of these records define reconstructed people. Groups that imply inconsistent person identifiers within the same year are excluded. In the 2006–2022 files, this procedure retains 68,063,835 of 68,104,536 source rows and identifies 4,641,311 people; the 40,701 excluded rows reflect ambiguous or inconsistent connections.

The annual files also have date and coverage problems. Several early files require year relabeling, one nominal year duplicates another, and the final 2023 file is incomplete, so the continuous window is 2006–2022 after correcting file years. This window describes administrative coverage; an education or authorization date reported in a retained record can lie outside the year of the file that contains it, and the latest dates run to October 2023. Program cohorts are assigned using actual program-end calendar years, not snapshot years.

Employer sets and authorization intervals. Beginning with the 2015 files, a student's distinct employers can be paired with multiple reported employment intervals in ways that do not preserve the original employer–interval relationship. Suppose a student works first at employer A and then at employer B and has three distinct intervals; the files can attach all three intervals to both employers. The set of employers remains informative, but assigning every interval to each employer would overstate both employment duration and firm-level activity. I therefore keep the distinct employer set and the authorization intervals as separate

objects. Authorization duration uses valid authorization dates, combining overlapping intervals so that a day is counted once. Employer relationships identify which firms hired a student on OPT but do not supply a reliable calendar-year OPT headcount, which is why the firm outcome assigns these relationships to the student's program-end cohort, and why the individual employment outcome comes from the linked work history and not from the administrative employer dates.

Changes of status. SEVIS records requests to change status, including requests associated with H-1B. A request is not evidence of approval, of its effective date, or of how long the resulting status lasted, and the field is person-level and undated. Requests are recorded only when made from inside the U.S. on the student's record, so a graduate who obtains an H-1B through consular processing, through a cap-exempt employer after the student record closes, or a status through a spouse, has no recorded status. The main text reports the request outcome directly; the status assignments used in the decomposition of employment are described in Appendix B.4.

B.2 Linking Schools, Fields, Employers, and Individuals

School matching uses normalized institution names, aliases, location, and campus relationships to link SEVIS sites and profile education entries to IPEDS. Campus identifiers and primary-institution relationships require particular care when schools change names or combine reporting units, and the program analysis harmonizes these identifiers before following degree–field cells over time. A SEVIS school without an IPEDS counterpart can be a valid institution outside the IPEDS reporting universe, so a non-match is not automatically an error.

Profile fields require a separate harmonization step because users supply free text. Candidate retrieval uses lexical, semantic, and translated field descriptions to identify plausible CIP families, and the selected field and institution assignments define education cells; lower-ranked candidates inform review but are not treated as additional degrees. Historical CIP codes are mapped to a common taxonomy before assigning STEM eligibility and connecting programs over time.

Employer matching combines conservative name cleaning with location, aliases, and source identifiers. The target is a common operating company, not a parent group defined only by ownership: multiple company pages can represent the same employer, but a shared parent or web domain does not alone establish identity. Administrative employer outcomes are merged only where identity and source coverage support interpretation, and a reviewed no-match in a covered source is coded as zero whereas an unresolved link is left missing.

Person linkage. The individual procedure begins with students who have a bachelor's, master's, or doctoral program at an IPEDS-linked school. Within school-based candidate sets, agreement in education timing and field identifies plausible profiles, and employer overlap and country evidence provide additional information. Citizenship is observed in SEVIS; the corresponding evidence in profiles is inferred from education, work, location, and available name information, and helps distinguish otherwise similar candidates without itself establishing a user's citizenship.

Candidate evidence is combined into match weights that allow for the possibility that no candidate is correct, and the analysis assigns a student to the leading candidate when its weight is at least 0.45. I do not force a one-to-one assignment between administrative identifiers and profiles, because unresolved duplication of administrative people would otherwise discard valid links; duplicate profiles supported by common name, school, timing, and employer evidence are consolidated before constructing histories, so the number of matched administrative people need not equal the number of distinct profile people.

To validate the linkage, I drew 400 stratified student cases and had them reviewed with the algorithmic scores hidden. The reviewers are LLM agents that inspect the administrative record and the candidate histories and select a candidate, reject all candidates, or mark the case uncertain; their decisions are validation evidence, not an independent record of identity. At the 0.45 threshold, precision among decided cases is 95.6%, and 96.0% in the portion not used to choose the threshold; 13.8% of accepted cases are unresolved in review, and treating every unresolved case as incorrect lowers precision to about 82%. These figures describe the linkage as a whole; accuracy can vary with field, origin, cohort, and employer history, and a stricter threshold would trade coverage for stronger evidence while changing the represented population. Section 4.4 discusses what this error does to the main ratio.

B.3 Selection into Observed Histories

Let R_i indicate that an administrative student has a retained link and let X_i^{obs} denote the characteristics used to model linkage. The fitted linkage probability is $q_i = \Pr(R_i = 1 \mid X_i^{\text{obs}})$, and for linked students in program p and graduating cohort t the outcome mean is

$$\overline{Y}_{pt} = \frac{\sum_{i \in (p,t)} R_i Y_i / \widehat{q}_i}{\sum_{i \in (p,t)} R_i / \widehat{q}_i}. \tag{10}$$

The linkage model is a logit that includes citizenship, degree, cohort, broad field, own-country share and its square, indicators for missing own-country information and singleton cells, and categories of the number of administrative employers, employment spells, and education

spells; sparse country and field categories are pooled. It does not include indicators for OPT use, STEM-extension use, or H-1B requests, and it is refit within the school-level bootstrap, separately from the treatment-comparison weights.

Some of these predictors reflect post-graduation history and can respond to relabeling. Their purpose is to model which outcomes are observed, not to hold a treatment mechanism fixed in the outcome regression; recovering the full cohort's mean requires that linked students' outcomes represent unlinked students' outcomes conditional on these predictors, which cannot be tested using the missing histories. Incorrect links are a separate source of error that inverse-linkage weights do not remove.

Recording by immigration status. The replacement estimates in Section 4.5 would be too small if profiles recorded jobs held on OPT more completely than jobs held while enrolled. SEVIS identifies the months in which a linked graduate has an active OPT employer or an active CPT authorization, which requires an employer, so the share of these months with a recorded U.S. job measures how completely profiles record work in each status. In the first three years after completion, profiles show a U.S. job in 77% of the 641,822 months with an OPT employer, 88% of the 32,919 months with full-time CPT in a later program, and 84% of the 17,794 months with only part-time CPT in a later program; in the final year of the original program, the shares are 76% for full-time and 77% for part-time CPT. Work while enrolled is therefore recorded at least as completely as work on OPT. When SEVIS records OPT authorization without an employer, the share falls to 42%, consistent with time between jobs during OPT.

Calibrated linkage weights. The calibrated alternative in Appendix Table A.7 additionally matches, within each treatment arm, relabeling cohort and event time, the administrative shares of graduates using no OPT, initial OPT only, and the STEM extension; each arm keeps its own distribution, so calibration does not force treated and comparison programs to have equal take-up. A cell with administrative mass but no linked observations in a category cannot be calibrated and is dropped, so the calibrated specification covers the 2016–2019 treatment cohorts.

B.4 Employment and Immigration Status

The principal work measures use reported employment intervals with a U.S. location, counting overlapping jobs once. The main post-completion path counts monthly snapshots on the 15th of each month relative to the SEVIS program-end month: years one to five are months 0–11, 12–23, 24–35, 36–47 and 48–59, and the final year in the program covers the twelve

months before the program-end month. A month counts as employed if any recorded U.S. job is active on the snapshot date, under any authorization and alongside study. A position recorded only with a start year, coded as January 1 of the program-end year, that continues past program completion is dated from the day after completion; without this rule, such positions would appear as work throughout the final months of the program. Calendar-year measures, used in the domestic placebo, compensation and destination analyses, instead refer to calendar year $t + h$, where t is the SEVIS program-end year.

F-1 students may work at most 20 hours a week during the academic term, on campus or through part-time CPT, and profiles do not record hours. A month of work during F-1 study, in the original program or a later one, therefore counts as one-half unless SEVIS records a full-time CPT, pre-completion OPT or OPT authorization for that month; all other months count fully. Within the administrative follow-up, each month of work is classified by the graduate's SEVIS status in the same month: OPT or the STEM extension; an H-1B period; enrollment in a later F-1 program; and any other recorded status or none, which also includes enrollment recorded only in the profile. H-1B periods are assigned from change-of-status requests, which are recorded without a date: a period starts at the latest recorded F-1 program or authorization end and lasts 72 months, and similar rules assign other requested statuses. These assignments do not observe a visa approval, its effective date or the duration of employment under that visa; the total-employment outcome does not depend on them, but its decomposition by status does.

Potential OPT authorization assigns 12 months of initial OPT to each graduate who uses OPT and 17 or 24 months of extension to each graduate who uses the extension, according to the rule in force when the extension began, starting at the program-end month; time in OPT status is measured from the same SEVIS records. The administrative outcome for post-completion authorization in Appendix Table A.4 is instead measured from SEVIS authorization start and end dates.

Potential months outside OPT status. Each potential month outside OPT or STEM OPT status is classified by its timing: before the first OPT authorization starts, between authorizations, or after the last one ends, and in the last case by whether the graduate has requested H-1B status, has started a later F-1 program by that month, or neither. OPT months outside the potential window, mainly initial OPT that starts after the program-end month and continues into the second year, are counted separately.

Location and compensation. Location measures reflect recorded work and education in the U.S., the country of citizenship, or a third country; they do not continuously observe

physical residence, so missing profile employment is not evidence of absence from the U.S., and a recorded U.S. education spell is not itself a job. Compensation is estimated by Revelio from the characteristics of reported positions; it describes jobs with usable pay information, assigns lower pay to positions outside the U.S., and does not measure realized annual earnings. Administrative follow-up ends earlier than the professional histories, so a linked job can be observed after the last SEVIS record without revealing the worker's immigration status at the time.

B.5 Employer Outcomes

Hiring measures count first jobs at the firm in the professional profiles. Recent graduates have a U.S. postsecondary graduation in the year before entry; qualifying student interns have a short first spell that overlaps U.S. enrollment; other native and other foreign hiring exclude these groups. Foreign-educated hiring is a subset of foreign hiring: it requires completed foreign postsecondary education and no documented U.S. postsecondary attendance by entry to the firm, but does not exclude prior U.S. employment or establish a border crossing. Native and foreign status is inferred from profile evidence on country of origin, since citizenship is not observed. The administrative OPT hiring count and the profile-based graduate hiring measures overlap in worker coverage and should not be added.

H-1B demand counts positions requested on Labor Condition Applications and, from 2020, lottery registrations, without deduplicating workers. Approved H-1B petitions include new and continuing petitions by fiscal decision year. PERM applications cover the separate permanent-residence sponsorship process and do not count green cards granted. All three are capped at the annual 99th percentile among positive firm-year values. The split of OPT hiring by recorded H-1B request uses any request to change to H-1B status in a graduate's administrative records; the request may occur outside the focal employer relationship and is undated.

C Program Relabelings and Empirical Methods

C.1 Identifying Program Relabelings

The relabeling design follows programs across changes in their reported CIP codes. Its starting point is an inventory of documented transitions from non-STEM source codes to STEM-eligible target codes, assembled with a large language model with web search and matched to IPEDS; the most common are Economics to Econometrics and Quantitative Economics, General Psychology to Experimental or other quantitative psychology codes,

Business to Management Science, and Architecture to Architectural and Building Sciences. IPEDS completion patterns then identify analogous changes at every institution and degree level. A fall in source-code completions alone can indicate a closure, and a rise in target-code completions alone can indicate a new program; requiring both, and requiring persistence after the change, distinguishes relabeling from these alternatives (Figure 2). Source codes are combined where a documented program previously reported multiple tracks, and related changes at the same institution and degree level are reviewed together so that one event is not counted several times.

Cases with competing target codes, conflicting dates or ambiguous program identities are resolved using program-specific documentary evidence: archived catalogs, department announcements and similar sources. The review can accept a mapping, revise its source codes or date, or leave it unresolved, and unresolved programs are excluded rather than assigned to the untreated group. Master's changes that merely duplicate the corresponding doctoral program's code change are also excluded where they would count the same underlying program twice.

For a first relabeled IPEDS year g, the four complete pre-treatment years used to screen coverage and construct baseline characteristics are $g-5, \ldots, g-2$. The screen requires complete IPEDS coverage in those years, at least five total completions across them, positive F-1 completions in $g-2$, and an observed Carnegie classification. The main administrative take-up fit contains 335 treated programs from the 2014–2019 relabeling cohorts, 207,974 stacked program-cohort observations and 27,671 program-by-stack units; the last two are not counts of distinct programs, because comparison programs can appear in several stacks. Appendix Table A.2 counts the treated programs by first relabeled cohort, degree level and field, with examples.

All relabeling event studies date potential exposure from $g-1$, so that the reference cohort $e=-1$ is $g-2$ and event time zero is the cohort immediately before the first relabeled IPEDS cohort. This convention keeps the partly exposed cohort in the event study without using it as the baseline.

Comparison programs are drawn from the potentially eligible source-code universe. For a comparison program first relabeled in year g_c, a student outcome measured for graduation cohort t at horizon h is included only if

$$g_c > \max\{g-2, t\} + h + b, \tag{11}$$

where the buffer $b=1$ in the main specification; never-relabeled programs have no finite g_c, and administrative ever-use outcomes use $h=0$. Appendix Table A.7 reports $b=2$ and

$b = 3$, and a version that applies the rule with $h = 5$ in every year so that the comparison programs are the same in every year.

C.2 Weights, Estimands and Inference in the Relabeling Design

The treatment propensity is estimated separately within each stack. Its eight continuous pre-treatment predictors are mean log program size, mean international share, the pooled shares of F-1 graduates from India and China, pooled OPT take-up, the trend in international share, the trend in log program size, and mean log real tuition. Means and trends use the four pre-treatment years above, with at least three observed years required where applicable; logarithms are taken of positive values only, and missing or undefined predictors receive missing-value indicators. Field-by-degree and Carnegie categories enter alongside these covariates, and categories without treatment-comparison overlap are removed. The propensity model is an unpenalized logit, and the odds are neither trimmed nor clipped; treated programs receive a weight of one and comparison programs $\widehat{p}/(1-\widehat{p})$. Appendix Table A.3 compares pre-treatment characteristics before and after reweighting: every characteristic is within 0.2 standard deviations except the international share (0.32), where the remaining difference comes from the 2014 and 2015 stacks, which contain 32 of the 335 relabeled programs, and the standardized difference is below 0.12 within each of the 2016–2019 stacks.

Student outcomes additionally weight programs by their reference-cohort F-1 count. The estimand therefore weights programs in proportion to their pre-treatment F-1 cohorts but averages the outcomes of the graduates who actually complete the program in each year, so fixed weights do not remove changes in student composition. Program-level outcomes, including completions and their international share, use equal program weights and a separately fitted treatment propensity.

Linked program-cohort means use the inverse-linkage adjustment described in Appendix B.3. Confidence intervals use school-level exponential multiplier draws that refit the treatment and linkage models, and the calibration when it is used, in each of 499 draws. Doubly robust estimates combine a treatment propensity with an outcome model for comparison-group changes, with school-clustered influence functions that incorporate the estimated nuisance parameters; I validate their standard errors against the bootstrap, requiring at least 190 successful refits and a ratio between 0.65 and 1.60.

Changes in extension use do not uniquely mediate the employment effect, because initial OPT use also responds and cohorts can change composition. Dividing employment by the change in extension take-up therefore produces a response ratio, not a local average treatment effect of extension use on a fixed group, which is why the main text divides by potential

authorization instead.

C.3 The Employer Design

The expansion index. Let A_{kt} and B_{kt} denote eligible and other completions at school k in year t, with $T_{kt} = A_{kt} + B_{kt}$ and eligible share $S_{kt} = A_{kt}/T_{kt}$. The share can rise either because eligible programs expand or because other programs contract, and the change decomposes exactly as

$$\Delta S_{kt} = \frac{(B_{kt} + B_{k,t-1})\Delta A_{kt}}{2T_{kt}T_{k,t-1}} - \frac{(A_{kt} + A_{k,t-1})\Delta B_{kt}}{2T_{kt}T_{k,t-1}}. \tag{12}$$

The expansion index cumulates only the positive eligible-program component,

$$H_{kt} = \sum_{u=2009}^{t} \max\left\{\frac{(B_{ku} + B_{k,u-1})\Delta A_{ku}}{2T_{ku}T_{k,u-1}}, 0\right\}, \qquad H_{k,2008} = 0, \tag{13}$$

so that it increases with expansion in the eligible component and not with contraction elsewhere. Among the exposure jumps in the sample, eligible-program growth supplies about 90% of the gross positive contributions to the eligible share and other-program contraction about 10%.

Weights and sample. Comparison firms receive normalized propensity-score odds,

$$\widehat{\omega}_{fg} = \frac{\widehat{p}_{fg}/(1 - \widehat{p}_{fg})}{\sum_{j:D_{jg}=0} \widehat{p}_{jg}/(1 - \widehat{p}_{jg})}, \tag{14}$$

estimated by inverse probability tilting rather than the maximum-likelihood logit of the individual design. Firm growth is the difference between the logarithms of mean headcount in $G-3$ to $G-2$ and in $G-5$ to $G-4$, which the four-year positive-employment requirement makes well defined for every retained firm. All annual employer estimates use the same exposure definition, treatment cohorts, reference year and weighting construction. The 13 outcomes include the five main outcomes, other native and foreign hiring, foreign-educated hiring, foreign student internships, H-1B approvals, PERM applications, and the two recorded-request components of OPT hiring; all use 40,203 treated episodes at 36,210 distinct firms except PERM, whose coverage supports 39,991 episodes at 36,014 firms. Outcomes retain observed zeros, and unobserved firm-years are not converted to zero. Table C.1 describes the OPT-hiring comparison: the eligible comparison pool is much larger than its effective sample size, reflecting unequal propensity odds. Other nonnegative covariates use positive-value

Table C.1: Cohort support and control weights in the OPT-hiring sample

Cohort	Treated episodes	Controls	Effective controls	Largest share (%)
2015	7,956	82,159	6,520	0.39
2016	8,843	83,484	12,374	0.15
2017	9,816	89,252	9,679	0.18
2018	8,341	95,125	7,699	0.30
2019	5,247	101,490	12,822	0.16

Notes: Control weights are normalized within cohort. Effective controls are the inverse sum of squared normalized weights; the last column is the largest single control weight. Repeated firms may appear in more than one cohort. The original firm roster is preserved when defining growth as the ordinary-log difference of late and early two-year headcount means; no additional observations are excluded for nonpositive or missing headcount. Diagnostics come from the completed propensity fits used for the reported estimates, not from earlier fits.

components and separate positive or missing indicators where applicable; signed changes remain in levels, and the annual outcomes enter the comparison in levels. Administrative demand, approvals and PERM counts are capped at the annual 99th percentile among positive observations.

Doubly robust comparison. Let $\widehat{m}_{0gt}(X)$ be a weighted linear regression of comparison-firm changes on the same covariates used for tilting. The doubly robust estimator

$$\widehat{\tau}^{DR}_{g,t} = \frac{1}{N_{1g}} \sum_{f:D_{fg}=1} \left[\Delta Y_{fgt} - \widehat{m}_{0gt}(X_{fg})\right] - \sum_{f:D_{fg}=0} \widehat{\omega}_{fg} \left[\Delta Y_{fgt} - \widehat{m}_{0gt}(X_{fg})\right] \tag{15}$$

is consistent under conditional parallel trends and overlap if either the propensity or the outcome model is correctly specified (Sant'Anna and Zhao, 2020). Because tilting balances the regressors of the linear outcome model, the two estimators coincide to three decimal places for every main outcome; this checks the implementation, not the identifying assumptions.

Clustering by recruiting school. The main intervals assign the same bootstrap multiplier to all firms in a corporate group. Appendix Table A.21 instead assigns the same multiplier to all firms whose largest recruiting weight is on the same school, refitting the propensity model in each of 499 draws, so that firms recruiting mainly from one school are treated as dependent.

Stocks, flows and alternative outcome models. Let L denote total employment and K the stock of OPT workers, measured over the same period and population. The change in other employment per additional OPT worker would be $(\Delta L - \Delta K)/\Delta K$. Neither an annual

Table C.2: Other workers' employment and OPT hiring under different outcome models

Specification	Episodes	Estimate	95% interval
Paired episode-FE Poisson response ratio	223,202	0.352	[0.200, 0.658]
Log IV: mean annual logs	18,824	0.149	[−0.433, 0.899]
Log IV: log of post mean	49,053	−0.051	[−0.252, 0.104]

Notes: Each row relates the change in other workers' December employment at exposed firms to the change in their OPT cohort hiring under a different model, using the 2015–2019 exposure jumps and their comparison firms, with two years before the jump as the baseline and the jump year and two following years as the post period. The first row is the ratio of two Poisson coefficients (0.036 for other employment and 0.103 for OPT hiring), each the proportional change at exposed firms relative to comparison firms, with a joint corporate-cluster Fieller interval; the next two rows are instrumental-variable elasticities of other employment with respect to OPT hiring, using the exposure jump as the instrument, with weak-instrument-robust Anderson–Rubin intervals (the first-stage Wald statistic for the second row is 6.05). All three are elasticities or ratios of proportional changes, not counts of other workers per additional OPT worker, because the numerator is a stock and the denominator a hiring flow; rows differ in sample because nonpositive values are excluded.

count of workers observed during the year nor the administrative cohort-hiring measure supplies the aligned point-in-time stocks this requires: translating cohort hiring into a stock requires job-start dates, durations and employment carried over from earlier cohorts. Dividing the headcount effect by the hiring effect therefore does not estimate this quantity, and even a correctly measured firm stock response would not distinguish additional national employment from employment reallocated across firms. Table C.2 reports two related comparisons with different targets. A paired Poisson model relates the post-period mean of OPT cohort hiring and of other workers' December employment to the $G-2$ values, with firm-episode effects, cohort-by-period effects and cohort-specific prehistory interactions, using the 223,202 episodes with positive totals in both outcomes. Instrumental-variable regressions of log other employment on log OPT hiring, using the exposure jump as the instrument, require positive OPT hiring and retain only 18,824 or 49,053 episodes, depending on whether logs are taken year by year or of the post-period mean.

Earlier design sensitivities. Before settling on the stacked design, I estimated an earlier matched-pair, continuous-dose version of the employer design under seventeen alternative exposure measures, recruiting windows and comparison pools. Appendix Table A.22 reports the main ones. The recruitment responses are positive throughout; total employment is the least stable outcome, and using firms whose own exposure jump comes later as comparisons, which balances exposure histories much better than never-exposed firms, lowers the employment and recent-graduate estimates. This is one reason the main text does not rest any conclusion on the total-employment estimate.

D Conditional Calculations for an OPT Fee

This appendix explains each input to the fee scenarios in Section 6.2. The employment estimates in the paper describe responses to changes in extension eligibility and in the supply of eligible graduates; they do not identify the effect of charging for initial OPT, an extension, or both. I therefore keep three inputs separate throughout: the empirical employment response, the externally supplied participation response, and the assumptions connecting those responses to national employment and output. The scenarios use assumed employment durations for each OPT path, scale the work during OPT by the share this paper estimates to be additional, and reduce the total by the replacement rate from the firm design.

D.1 Which Employment Path Is Lost?

Consider two paths relative to a common no-OPT counterfactual: initial OPT without an extension, denoted 0, and initial OPT followed by STEM OPT, denoted S. Let q_0 and q_S be the reductions in annual entrants to these paths and T_0 and T_S the net U.S. employment-years generated by each path for the graduates whose choices change, including any change in other workers' employment. With stable annual flows and employment responses, the steady-state employment loss is

$$W^{\text{loss}} = q_0 T_0 + q_S T_S. \tag{16}$$

The signs of the path changes matter. If a graduate gives up the extension but still uses initial OPT, $q_S = 1$ and $q_0 = -1$, and the employment loss is $T_S - T_0$, not T_S; a reduction in full-path starts therefore cannot be read as a reduction in initial OPT starts without specifying this switching. Similarly, the effect of an extension-only fee depends on the incremental value of the extension among graduates deterred by that fee, which need not equal the effect of relabeling among graduates who respond to expanded eligibility.

An enrollment response introduces another counterfactual. A student who does not study in the U.S. might otherwise have entered through a different route, and forgone enrollment can change universities' finances and the supply of other graduates; the relabeling design does not identify these effects. If a participation projection already incorporates reduced enrollment, multiplying it by a second enrollment-loss factor would count the same response twice, so the scenarios use the published OPT participation responses directly and impose no additional enrollment contraction.

D.2 Employment Evidence and Normalizations

Table D.1 reports the employment quantities the scenarios draw on. The reduced forms in Panel A are the most direct empirical inputs: relabeling increases U.S. work in the first three years after completion by 0.44 years per linked graduate, and by 0.45 years over five years for the graduates observed that long. They measure the entire employment response to relabeling, including changes in initial OPT participation, further study, and other work, and their confidence intervals describe sampling uncertainty under the maintained design, not uncertainty about the other assumptions in a fee calculation.

Panel B reports the ratios used in the scenarios. Over the first three years, net additional U.S. work is 0.54 of the added potential OPT authorization and 0.73 of the added work on OPT and the extension (Section 4.5). Part of the net work, 0.4 months, is on a cap-subject H-1B obtained before the authorization ended; because a binding cap means that visa could otherwise have gone to another applicant, the scenarios remove it and use the share net of H-1B work, $a = 0.68$. This is the share in equation (9): it scales assumed work during OPT by the part that graduates would not otherwise have supplied in the U.S., and because both its numerator and denominator are measured for the same linked graduates, it does not depend on a first-stage normalization. Replacement by other workers, r, is the ratio of the firm design's effects on hiring of workers educated abroad and on OPT hiring, with the sign reversed; its interval uses the delta method with the covariance of the two outcomes across the same 499 bootstrap draws, which has little effect on it.

Other workers' employment beyond replacement at incumbent firms is not estimated here. The firm evidence concerns a different supply change, follows incumbent firms, and includes recruitment that may shift across firms, so it does not supply a national value; it bounds replacement of hiring from abroad at about 0.27 per OPT hire. For firm creation and spillovers to other local employers, the commuting-zone estimates of Kwon, Lee and Monras (2026) imply 1.46 to 1.67 additional native wage workers per additional high-skilled immigrant; these are local effects that include natives moving between areas, so they bound the national response from above, and Panel C of Table 4 applies them.

D.3 External Participation Responses

I use the September 2026 version of Clemens (2026), which considers a fee borne by students, universities, or employers. The calculations use a $70,000 initial charge and a $30,000 charge for the two extension years, for a total of $100,000 for the complete STEM path, in the source's 2024-dollar convention. The incidence cases are alternative behavioral assumptions, not confidence limits around one forecast.

Table D.1: Employment quantities used in the fee calculation

Quantity	Estimate	95% interval
A. Reduced forms and normalizations (years)		
Years 1–3: reduced form per graduate	0.44	[0.21, 0.65]
Years 1–3: normalization per induced extension user	1.17	[0.63, 1.73]
Years 1–5: reduced form per graduate	0.45	[0.13, 0.75]
Years 1–5: normalization per induced extension user	1.23	[0.39, 1.99]
B. Ratios used in the fee scenarios		
Net additional work per month of potential OPT authorization	0.54	[0.29, 0.81]
Net additional work per month of work on OPT	0.73	[0.41, 1.02]
excluding work on an H-1B, a	0.68	[0.43, 0.90]
Replacement of hiring from abroad per OPT hire, r	0.05	[−0.18, 0.27]

Notes: The table collects the estimates from Sections 4 and 5 that enter the fee scenarios in Table 4. The first row is the effect of relabeling on U.S. work over the first three years after completion in Table 1, column (3), expressed in years; the third row is the effect over five years in Table 2, column (4). The second and fourth rows divide these effects by the effect on extension use for the same graduates, 37.7 and 37.0 percentage points; these are response ratios, not treatment effects of extension use on a fixed group, because relabeling also changes initial OPT use and cohort composition. Panel B ratios use the first three years and the same linked graduates as Table 1; the third row removes months worked on an assigned H-1B from the numerator and is the share used in the scenarios. Intervals are percentiles across 4499 bootstrap draws (499 for the replacement rate). None of the intervals includes uncertainty about fee incidence, participation, national replacement, or transport to a different population.

Table D.2: Published participation responses to the fee

Assumed incidence and response	$70,000	$100,000
Students: lower bound on the decline in OPT starts	28.8%	35.0%
Universities: decline in OPT starts	63.5%	79.3%

Notes: Projected declines in annual OPT starts under a fee of the indicated size, from Tables 2(b) and 5(b) of Clemens (2026), combining bachelor's and master's graduates. If students pay, the decline is a lower bound derived from the response of international enrollment to degree prices, weighted by the distribution of initial OPT starts across university-degree cells; if universities pay, the decline assumes that universities stop supplying the students whose whole-degree margin (revenue less marginal cost) falls below the charge. The two columns index the total charge attached to a path; they do not describe switching between paths.

Under student incidence, the published lower bounds use the response of international study to degree prices; under university incidence, the calculation compares the charge with universities' estimated revenue less marginal cost over a whole degree, which can give a large response even without assuming that the graduate personally pays the fee. Both require a view of how the charge changes education and OPT choices; neither is an estimate of the response to a previously implemented OPT fee. Applying the combined bachelor's/master's rates to all degrees also assumes that doctoral participation responds in the same way.

An employer-incidence variant uses a different mechanism. Table 6 of Clemens (2026) spreads the $100,000 total charge over three years and applies labor-demand elasticities motivated by H-1B evidence. At an elasticity of −10, the projected decline is 84.4% when the charge is compared with total labor cost, assumed to equal 1.45 times an annual wage of $112,600; elasticities of −4 and −16 produce declines of 52.4% and 94.9% on that basis, and comparing the charge with wages alone gives 64.6%, 92.5%, and 98.4%. These are published conditional demand responses for the complete three-year path; they require assumptions about incidence, wage adjustment, and the relevance of H-1B demand to OPT, and cannot be substituted mechanically for separate initial-only and extension responses. An extension-only version would additionally require the share who move to initial-only OPT rather than leave OPT altogether. Keeping these cases separate is more informative than averaging across them.

D.4 A Duration-Based Numerical Benchmark

The benchmark in Tables D.3 and D.4 uses the university-incidence participation responses and assumes the employment contribution of each lost path, treating all work during OPT as additional; it corresponds to the last row of Panel B in Table 4, and the other rows scale its work during OPT by the estimated additional share.

The annual baseline uses 194,554 initial OPT authorizations and 95,384 STEM OPT authorizations with employment start dates in 2024, as reported by U.S. Immigration and Customs Enforcement (n.d.). I approximate initial-only paths by the difference, 99,170, and complete STEM paths by the 95,384 extension starts. These are authorizations beginning in the same calendar year, not linked entrants followed through both stages, so the approximation requires a stable-flow interpretation and should not be read as the observed division of one graduating cohort into two paths.

Durations. For each lost initial-only start, the benchmark assumes 0.85 years of work during OPT and 0.12 years of subsequent uncapped work; for each lost complete STEM path, 2.20 and 0.35 years. These come from the following considerations. Initial OPT lasts 12

Table D.3: Annual OPT starts and lost starts in the benchmark

Path	Baseline flow	Charge	Decline	Lost starts
Initial-only	99,170	$70,000	63.5%	62,973
Complete STEM	95,384	$100,000	79.3%	75,640
Total	194,554			138,612

Notes: Annual OPT starts by path, approximated from 2024 authorizations (U.S. Immigration and Customs Enforcement, n.d.), and the starts lost under the university-incidence participation responses of Clemens (2026). Counts are rounded; the total uses unrounded calculations. The benchmark applies combined bachelor's/master's response rates to all degrees, assumes no switching between the two paths, and imposes no additional enrollment response.

months, and graduates may spend up to 90 days unemployed, so 0.75 years is the minimum consistent with full use; 0.85 allows for graduates who use most of the permitted gap at the start and work continuously thereafter. For the complete path, the first year is the same 0.85; the two extension years add between 1.1 years, the additional U.S. work per induced extension user in years one to three implied by the relabeling estimates, and about 1.6 years, the 24-month extension less the 150 days of unemployment permitted over the whole period; 2.20 is between these cases (the low and high cases are 1.85 and 2.60). Uncapped work after OPT is the hardest input: for the complete path, 0.35 years is about half of the roughly 0.7 years of additional U.S. presence in years four to six estimated for induced extension users in an earlier version of the relabeling analysis, with the other half assumed to depend on a capped visa; the low and high cases (0.17 and 0.60) bracket it, and the 0.12 years for the initial-only path is an assumption. The post-OPT component already excludes employment through fully refilled capped visas, which is why the scenarios do not scale it by a. These durations specify work relative to the no-OPT counterfactual; they are not the legal maximum authorization periods and are not the relabeling estimates themselves, which concern eligibility for the extension and not the loss of the whole OPT route.

The benchmark sets replacement by other workers at 5% of the remaining employment, the point estimate of replacement per OPT hire from the firm design. That estimate concerns hiring from abroad at exposed firms; applying it to national employment assumes that other replacement margins are small, and the same replacement share across paths and time is also an assumption. Applying it to both duration components gives 0.92 net years per lost initial-only start and 2.42 per lost complete STEM path.

The implied 241,000-worker reduction is a fully specified scale calculation that depends jointly on the large university-incidence response, the annual-flow approximation, no switching, the duration assumptions, and the imposed replacement rate. Scaling the work during OPT by the estimated additional share of 0.68 gives the central scenario of 174,000 workers in Table 4, and the interval for the share gives 121,000 to 221,000. The scenarios do not scale

Table D.4: Assumed durations and implied employment in the benchmark

Input or conditional outcome	Initial-only	Complete STEM	Total
OPT work-years per lost start, before replacement	0.85	2.20	
Uncapped post-OPT work-years, before replacement	0.12	0.35	
Assumed other-worker replacement	5%	5%	
Net work-years per lost start	0.92	2.42	
Steady-state employment loss	58,000	183,000	241,000

Notes: The employment loss is the product of the unrounded lost starts in Table D.3 and the net years per start, treating all work during OPT as additional; the paragraph above explains where each duration comes from. A permanent annual reduction in starts and stable durations are required for the steady-state interpretation. The total consists of about 209,000 workers during OPT and 32,000 during subsequent uncapped work; it is a conditional calculation with no sampling confidence interval.

the 32,000 workers in uncapped post-OPT work by the same share; doing so would give about 164,000. Replacing the assumed complete-path duration with the normalizations of 1.17 or 1.23 years in Table D.1 would mix distinct counterfactuals and populations, so the scenarios scale the assumed durations by the estimated share and do not replace them.

D.5 Output, Welfare, and Uncertainty

A monetary conversion adds assumptions beyond the employment accounting. The scenarios assign annual wages of $80,000 to the missing OPT workers and $105,000 to the missing post-OPT workers. The first is the median estimated pay of recent foreign graduates in the profiles ($93,000) deflated by 10–30% toward the earnings of comparable workers in the American Community Survey; the second is between the median wages on Labor Condition Applications and approved H-1B petitions (about $98,000) and the $112,600 used by Clemens (2026). If wages equal 95% of the marginal product of labor and the labor share is 0.65, the corresponding output conversion is

$$Y^{\text{loss}} = \frac{80{,}000 L^{\text{loss}}_{\text{OPT}} + 105{,}000 L^{\text{loss}}_{\text{post}}}{(1 - 0.05) \times 0.65}, \tag{17}$$

which gives $23.8 billion annually in the central scenario and $32.6 billion under the benchmark. Dividing by the labor share assumes that capital leaves along with the workers, which is a long-run assumption; with capital fixed, the loss is the wage bill divided by 0.95 alone, about 35% smaller. The output value is a conditional conversion of the duration scenario, not an estimated consequence of the fee or an empirical estimate of the wage markdown, which the paper's compensation evidence does not identify.

Output is also different from the cost to other U.S. residents. Part of the output would

accrue to the graduates as earnings; a resident-welfare calculation would need changes in firm profits, other workers' earnings, tax receipts and public costs, and universities' net revenue, and fee receipts would be transfers whose incidence depends on who pays and who ultimately bears the burden. The scenarios do not estimate these components.

Finally, the empirical intervals and the scenario uncertainty should remain separate. The intervals in Table D.1 do not include uncertainty about which graduates a fee deters, transport from relabeling to withdrawal from OPT, the participation projection, future cohort flows, or national replacement, and the duration benchmark has no estimated policy-level confidence interval. Its purpose is to make the consequences of a stated set of assumptions reviewable.